04820KA001 Mono Slitter 903_V12_1106_5

Author: Michou
Created: 11/6/2026 5:11:11 PM
Last Modified: 11/6/2026 5:11:11 PM
Comment: Modified to Connect 2nd Diameter Sens

Sysmac Studio Module Version: 1.60.0.64010

1.PLC

1-8.POUs

1-8-1.Programs

1-8-1-1.TaglMonoTRM

1-8-1-1-1.Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Comment
VAR						
SpdDelta	REAL			False	False	Minimum or maximum speed SpdDeltaerence
UwSlowDiamCalc	REAL			False	False	V7c: speed-dependent unwinder slow-down diameter (mm)
SonarRunMem	bool			False	False	
EndStopReq	BOOL			False	False	V12: end-of-roll stop requested (AutoStopCmd or unwinder diameter)
Buffer_R	BOOL			False	False	Buffer Result
EndRampOK	BOOL			False	False	V12: end-of-roll ramp usable (FinalStopDecel, FF active, running)
EndRampHard	BOOL			False	False	V12: safety net during the ramp -> immediate stop
EndStopNow	BOOL			False	False	V12: immediate end-of-roll stop (Motor_Manager rung 4)
SonarInit	bool			False	False	
AlarmMem	ARRAY[0..32] OF bool			False	False	
UwSlowDiamLatch	BOOL			False	False	V7c: unwinder diameter slow-down reached (frozen)
FFRate	REAL			False	False	V7f: ramp rate (reference per 4 ms scan)
FFRateSet	REAL			False	False	V7f: rate allowed to still stop at the target
FFJerk	REAL			False	False	V7f: max rate change per scan
SpdScanCnt	INT			False	False	V9: scan counter for the 100 ms speed window
HornRun	BOOL			False	False	V9: slow-down / final speed running
HornRunMem	BOOL			False	False	V9: edge memory
HornStartMem	BOOL			False	False	V9: start edge memory
HornTicks	INT			False	False	V9: scans since slow-down start
HornCnt	INT			False	False	V9: scans in the beep period
HornPer	INT			False	False	V9: beep period in scans
HornStartScans	INT			False	False	V9: start beep in scans
HornLenScans	INT			False	False	V9: short beep in scans
HornSlowBeep	BOOL			False	False	V9: slow-down beep
HornStopPend	BOOL			False	False	V9: stopped after a slow-down, waiting for standstill
HornStopTicks	INT			False	False	V9: stop beep counter
LongActive	BOOL			False	False	V9: long-baseline learning running
LongValid	BOOL			False	False	V9: long baseline >= 20 mm growth, replaces the windows
LongStartPos	DINT			False	False	V9: encoder at the start of the long baseline
LongStartD	REAL			False	False	V9: rewinder diameter at the start (mm)
LongGrow_mm	REAL			False	False	V9: growth since the start (mm)

LongPls	REAL			False	False	V9: pulses since the start
LongK	REAL			False	False	V9: K from the long baseline
RevW	REAL			False	False	V9: encoder pulses summed in the current revolution
RevDP	DINT			False	False	V9: encoder pulses this scan
RevPrevPos	DINT			False	False	V9: encoder last scan
RevSum	REAL			False	False	V9: sum of raw DT35 distance x pulses
RevMin	REAL			False	False	V9: min raw distance in the revolution
RevMax	REAL			False	False	V9: max raw distance in the revolution
RwDiamRevOk	BOOL			False	False	V9: one full revolution averaged since the last fault
UwEmptyTicks	INT			False	False	V9: unwinder empty time (100 ms ticks)
UwFullTicks	INT			False	False	V9: new roll seen time (100 ms ticks)
GsmArmed	BOOL			False	False	V9: roll was empty -> next new roll asks the GSM question
T_JogStartBlk	TOF			False	False	V9: START blocked while JOG held + 2 s
StartHeldBlk	BOOL			False	False	V9: START pressed during the block, wait for release
StartPB_OK	BOOL			False	False	V9: START push-button accepted (replaces I_Start_PB in General 7/8/10)
FFSoftMem	BOOL			False	False	V9: soft stop last scan (rung B)
FFSlowAcc	BOOL			False	False	V9: gentle ramp-up after a cancelled soft stop
FFStartEdge	BOOL			False	False	V9: START push-button edge (rung B)
FFStartMem	BOOL			False	False	V9: START last scan (rung B)
SpdDeltaMin	REAL			False	False	Minimum speed difference
FFSqrtArg	REAL			False	False	SQRT argument (REAL)
SpdDeltaMax	REAL			False	False	Maximum speed SpdDeltaerence
SoftStopMem	BOOL			False	False	v6r: soft stop edge memory
SoftStopRatio	REAL			False	False	v6r: InvRef counts per m/min, frozen at the soft stop start
MemPos	DINT			False	False	Previous encoder count
MatChangeCnt	INT			False	False	v6s: consecutive learning windows far from RwMat_K
DeltaPulse	DINT			False	False	Encoder count difference
SlowMem	BOOL			False	False	SlowDown edge memory
DecelDiam	REAL			False	False	Rewinder diameter used by the decel model (mm, min 100)
TeachSwap	REAL			False	False	Temporary for UNDO swap
SlowStartDiam	REAL			False	False	Diameter at slow-down start (mm)
BrakeHold	BOOL			False	False	Keep T-one in auto during a normal stop (O_Brake)
StopNow	BOOL			False	False	Immediate stop request (replaces I_Stop_PB in General rung 9)
StopPB_Mem	BOOL			False	False	STOP button edge memory
T_StopLong	TON			False	False	STOP held 1 s = immediate stop
T_SoftStop	TON			False	False	Soft stop timeout (120 s)
BrakeHoldTicks	INT			False	False	BrakeHold duration after stop (100 ms ticks, max 150)
StopStartDiam	REAL			False	False	Diameter at stop command (mm)
DecelK_Meas	REAL			False	False	Measured decel per mm of diame

						ter
SlowMeasActive	BOOL			False	False	Slow-down measurement running
TeachTicks	INT			False	False	TEACH CORE stable time (100 ms ticks)
SlowStartPos	DINT			False	False	Encoder at slow-down start
SlowStartSpd_Pps	REAL			False	False	Speed at slow-down start (pulses/s)
SlowMeas_Pls	REAL			False	False	Measured slow-down length (pulses)
SlowDecelMeas_mm	REAL			False	False	Measured slow-down decel (mm/s2)
SlowOneRamp	BOOL			False	False	Slow-down uses one ramp (manual or learned)
Velocity	REAL			False	False	Speed in [mm/s]
Alarm_Array	ARRAY[0..32] OF BOOL			False	False	Alarms
T_PlcOn	TON			False	False	PLC active timeout
ControlVoltage	BOOL			False	False	Auxiliaries OK
StartMotor_Cmd	BOOL			False	False	Motor start command
HubExtractEnable	BOOL			False	False	Hub extraction enable command
Brake_Cmd	BOOL			False	False	Brake command
T_StopVent	TOF			False	False	Fans stop timer
TimStopVen	TIME			False	False	Fans stop timeout
MotorCooling	BOOL			False	False	Fans command
OverRider_Auto	BOOL			False	False	OverRider in automatic
OverRider_Hoover	BOOL			False	False	OverRider in Hoover Mode
OverRider_Lock	BOOL			False	False	OverRider lock command
GSM_LearnFault	BOOL			False	False	Paper [Grams / Square Meter] LearnFailCnt ≥ RwLearnFailMax
TickCnt	INT			False	False	100 ms tick scan counter (25 x 4 ms)
RunWasJog	BOOL			False	False	Run used jog / manual / slow button (2nd ramp)
StopLearnValid	BOOL			False	False	This stop may be learned
SpdBand_Pps	REAL			False	False	Speed band of the trim ramp (pulses/s)
TrimDecel_Pps2	REAL			False	False	Trim ramp decel (pulses/s2)
SpdKnee_Pps	REAL			False	False	Speed where coarse ramp hands over to trim ramp (pulses/s)
OverRider_RaiseUp	BOOL			False	False	OverRider raise command
Pot1_Mem	INT			False	False	Work potentiometer value
VelSet_mms	REAL			False	False	Set speed in mm/s
Vfd_SP_Modif	REAL			False	False	Inverter speed modifier
RwAutoLimAct	BOOL			False	False	v6y: rewinder AUTO speed limit active (taken over at standstill)
UwTeachTicks	INT			False	False	Unwinder TEACH CORE samples (100 ms)
T_PosPresOff	TON			False	False	OverRider photocell OFF timer
T_PosPresOn	TON			False	False	OverRider photocell ON timer
TimPosPresOff	TIME			False	False	OverRider photocell OFF timeout
TimPosPresOn	TIME			False	False	OverRider photocell ON timeout
RollEnd_Active	BOOL			False	False	Roll diameter photocell intercepted
Output_Ratio	REAL			False	False	Analog output ratio
Input_Ratio	REAL			False	False	Analog input ratio

T_StartLow	TON			False	False	Slow start timer
T_UwDiamDebounce	TON			False	False	Roll diameter photocell Debounce timer 500mS
UwDiamReached	BOOL			False	False	Roll diameter reached
Index_A	INT			False	False	
AlarmStrobe	BOOL			False	False	Alarm message display strobe
T_Display	TON			False	False	Alarm display timer
RangeOK	BOOL			False	False	Range reached
Vel_Max_mms	REAL			False	False	Maximum speed in mm/s
Slow_Inhibit	BOOL			False	False	Inhibit slow button
Decel_Ramp	REAL			False	False	Deceleration ramp
Accel_Ramp	REAL			False	False	Acceleration ramp
Pot2_Mem	INT			False	False	Final potentiometer value
Current_mm	REAL			False	False	Current millimetres
EncoderPulse_to_mm	REAL			False	False	mm per encoder pulse
Stop	BOOL			False	False	Stop
T_EncoderStill	TON			False	False	(?) Encoder stopped timer
Analog_Res	UINT			True	False	Analog board resolution
Mt_To_mm	REAL			False	False	m/min to mm/s factor
Init_Ramp	BOOL			True	False	(?) Ramp initialisation
SlowByDiam	BOOL			False	False	Slow down by diameter
UwStopByDiam	BOOL			False	False	Stop by diameter
T_Screen_Delay	TON			False	False	(?) Start page timer
T_PulStop	TON			False	False	(?) Stop button timer
Reading_Max	REAL	2000		True	False	Maximum sensor reading (full scale)
DeadZone	REAL	60		True	False	Sensor dead zone
In_UwDiameter	REAL			True	False	UnWinder Diameter Analog input voltage
Tick	BOOL			False	False	100 ms tick
MmPerPulse	REAL			False	False	mm per encoder pulse
RwDistRaw_mm	REAL			False	False	Rewinder Unfiltered distance (mm)
RwFiltInit	BOOL			False	False	Rewinder Filter initialised
SpdInit	BOOL			False	False	Speed init
EncPrev	DINT			False	False	Encoder at previous tick
SpdAct_Pps	REAL			False	False	Actual Speed (pulses/s)
LearnInit	BOOL			False	False	Learn initialised
LearnWinRunning	BOOL			False	False	Motor ran whole window
LearnWinTicks	INT			False	False	Learn window counter
LearnWin_Pls	DINT			False	False	Pulses in window
LearnWinDiamDelta	REAL			False	False	Diameter change in window
MatK_Window	REAL			False	False	Material factor this window
MotorRunMem	BOOL			False	False	Motor start edge memory
StopMeasActive	BOOL			False	False	Stop measurement running
StopStartPos	DINT			False	False	Encoder at stop command
StopStartSpd_Pps	REAL			False	False	Speed at stop command (pulses/s)
StandstillTicks	INT			False	False	Standstill tick counter
LearnWinOk	BOOL			False	False	Learning window gave a valid result
ThickWin_mm	REAL			False	False	Paper thickness this window (mm)
LearnFailCnt	INT			False	False	Consecutive failed learning windows
StopLearnOk	BOOL			False	False	Stop measurement gave a valid re

						sult
StopDecelMeas_mm	REAL			False	False	Measured stop decel (mm/s2)
StopMeas_Pls	REAL			False	False	Measured stop length (pulses)
StopDecelMeas	REAL			False	False	Measured stop decel (pulses/s2)
StopLen_Pls	REAL			False	False	Stop length (pulses)
StopLenFinal_Pls	REAL			False	False	Stop length from slow speed (pulses)
RwSlowLen_Pls	REAL			False	False	Rewinder Slow-down length (pulses)
SlowDecel_Pps2	REAL			False	False	Slow-down decel (pulses/s2)
SpdFinal_Pps	REAL			False	False	Final speed (pulses/s)
RwRemain_Pls	REAL			False	False	Rewinder Remaining length to target (pulses)
LearnWinStartPos	DINT			False	False	Rewinder Position Window 0
LearnWinStartDiam	REAL			False	False	Rewinder Diameter Window 0
T_ByPassProtOff	TOF			False	False	Delay Off Timer
UwEnMem	BOOL			False	False	Edge memory UwStopEnable
RwEnMem	BOOL			False	False	Edge memory RwEnable
SpdRef_Pps	REAL			False	False	Reference speed MAX (actual,setpoint) (pulses/s)
SpdWork_Pps	REAL			False	False	Work speed setpoint (pulses/s)
SlowLenBase_Pls	REAL			False	False	Slow-down length without margin (pulses)
UwSlowLen_Pls	REAL			False	False	Unwinder slow-down length (pulses)
UwMarginUsed_m	REAL			False	False	Unwinder margin at slow speed (m)
SqrtArg	REAL			False	False	Square root argument
AutoSlowCmd	BOOL			False	False	Automatic Slow Command
AutoStopCmd	BOOL			False	False	Automatic Stop Command
RwLastValidDiam	REAL			False	False	Rewinder Last valid diameter (mm)
RwRemainMax_Pls	REAL			False	False	Rewinder Remaining length to max diameter (pulses)
UnWinder_Empty	BOOL			False	False	UnWinder Empty
StopFailCnt	INT			False	False	Consecutive failed stop measurements
UwTeachSum	REAL			False	False	Unwinder TEACH CORE sonar sum (mm)
T_MatReset	TON			False	False	v6s: RwMatReset stays ON 3 s
RwAutoMaxSpdAct	REAL			False	False	v6y: active rewinder AUTO max speed (taken over at standstill)
UwEmptyMem	BOOL			False	False	v6x: UnWinder_Empty edge memory
LearnWinLen_Pls	REAL			False	False	v6w: learning window length in encoder pulses
WebRatio	REAL			False	False	V7: drive counts per m/min of web at the current diameter
FFActive	BOOL			False	False	V7: feed-forward loop active
FFWasActive	BOOL			False	False	V7: edge memory
FFTarget	REAL			False	False	V7: reference for the set speed (mm/s of the 0-10 V scale)
RampRef	REAL			False	False	V7: ramped reference (mm/s of the 0-10 V scale)
StepAcc	REAL			False	False	V7: accel step per scan

	StepDec	REAL			False	False	V7: decel step per scan
	FFSteadyTicks	INT			False	False	V7: scans at target
	FFErr	REAL			False	False	V7: relative speed error
	Clock_100ms	bool			False	False	Clock100ms
	AlarmCnt	INT			False	False	
	AlarmCntMem	INT			False	False	
	RstEdge	BOOL			False	False	
	RstMem	BOOL			False	False	
	SilenceTicks	INT			False	False	
	WarnCnt	INT			False	False	
	WarnCntMem	INT			False	False	
	WarnBeepTicks	INT			False	False	
	TensUnitMem	BOOL			False	False	
	TensIdeal_N	REAL			False	False	
	TensionSeen	BOOL	False		False	False	Tension reached once since START
	T_TensOn	TON			False	False	Tension ON debounce 0.5 s
	T_TensOff	TON			False	False	Tension OFF debounce 2 s
	T_UwSensFlt	TON			False	False	Unwinder sonar fault debounce 1 s
	Buffer_A	BOOL			False	False	
	Buffer_B	BOOL			False	False	
	Buffer_C	BOOL			False	False	
	Buffer_D	BOOL			False	False	
	AS_RunMem	BOOL			False	False	
	AutoStopSeen	BOOL			False	False	
	T_AutoStopDisp	TON			False	False	
	RwArmMem	BOOL			False	False	
	UwArmMem	BOOL			False	False	
	T_AutoEnRef	TON			False	False	
	TensRef	REAL			False	False	
	WebBreakNow	BOOL			False	False	
VAR_EXTERNAL							
	VelSet	REAL				False	Set speed
	OverSpeed	BOOL				False	Speed too high
	SpdDelta_Percent	REAL				False	Speed difference percentage
	Encoder_Pos	DINT				False	Encoder counter
	FinalStopDecel	REAL				False	
	FinalStopRamp	BOOL				False	
	Velocity_M_min	REAL				False	Speed in m/min
	Current_Metre	REAL				False	Current metres
	I_RsetCnt	BOOL				False	Metre counter reset button
	SonarAsk	BOOL				False	
	SonarAskYes	BOOL				False	
	WorkSpd_Max_Set	REAL				False	HMI request: work pot full scale (m/min), applied only with motor stopped
	FinalSpd_Max_Set	REAL				False	HMI request: final pot full scale (m/min), applied only with motor stopped
	SpdMax_Pending	BOOL				False	New pot range waiting for the motor to stop
	RwAxisDist_Prev	REAL				False	Rewinder: RwAxisDist before the last TEACH (for UNDO)
	UwCoreDist_Prev	REAL				False	Unwinder: UwCoreDistance before the last TEACH (for UNDO)
	RwTeachUndo	BOOL				False	HMI: UNDO rewinder TEACH (PLC resets it)

UwTeachUndo	BOOL				False	HMI: UNDO unwinder TEACH (PLC resets it)
RwLearnReset	BOOL				False	HMI / Watch: clear learned decels RwStopDecelK, RwStopDecel, RwSlowDecelK (PLC resets it)
UnderSpeed	BOOL				False	Speed too low
I_ControlVoltage	BOOL				False	Auxiliaries ON
I_Side_Door_Closed	BOOL				False	Roll diameter photocell
RwCoreDiameter	REAL				False	Rewinder core outside diameter for TEACH CORE (mm)
UwTeachState	INT				False	Unwinder TEACH CORE: 0 idle 1 teaching 2 done 3 refused
SoftStop	BOOL				False	Soft stop in progress (HMI)
SoftStopEnable	BOOL				False	STOP button = soft stop (False = immediate as before)
SoftStopEndSpd	REAL				False	Soft stop: run command drops below this speed (m/min)
SoftStopDecel	REAL				False	Soft stop decel (mm/s2), 0 = DecelCoarse
RwTeachCore	BOOL				False	HMI: TEACH CORE request (PLC resets it)
RwTeachState	INT				False	TEACH CORE: 0 idle 1 teaching 2 done 3 refused
O_MotorStart	BOOL				False	Motor start
RwSlowDecelK	REAL				False	Learned slow-down decel per mm of rewinder diameter (mm/s2 per mm)
RwStopDecelK	REAL				False	Learned stop decel per mm of rewinder diameter (mm/s2 per mm)
I_OR_Position_Sens	BOOL				False	Rider roll position sensor
WorkSpd_Max	REAL				False	Work speed pot full scale (m/min), 50..VelMax
FinalSpd_Max	REAL				False	Final speed pot full scale (m/min), 5..WorkSpd_Max
I_OR_Hoover_Mode	BOOL				False	Rider roll pressure switch
I_OR_Auto_OS	BOOL				False	Rider roll in automatic
I_OR_Cont_OS	BOOL				False	Rider roll in continuous
I_Start_PB	BOOL				False	Start button
I_Motor_Thermal	BOOL				False	Motor thermal switch
I_FanTemp_Ok	BOOL				False	Motor (fan) thermal OK
I_Stop_PB	BOOL				False	Stop button
I_Hub_Position	BOOL				False	Hub/chuck in extraction position
I_FrontGuard_Closed	BOOL				False	Front guard closed
RwMaxSlowFallback	REAL				False	Rewinder max: slow-down diameter while K not learned (mm)
RwMaxSlowAtDiam	REAL				False	Rewinder max: slow-down starts at diameter (mm)
O_Alarm_Array	BOOL				False	Alarm lamp
O_RwUnload	BOOL				False	Unloaders enable - opening
I_Prot_ByPass	BOOL				False	Guards-bypassed selector switch
O_Prot_ByPass_Lamp	BOOL				False	Guards bypassed lamp
O_Brake	BOOL				False	Brake command

Fan_Timeout	REAL				False	Fans stop timeout (from panel)
O_CoolingFan	BOOL				False	Brake and motor fans
O_OR_Auto_Lamp	BOOL				False	Rider roll auto/continuous lamp
O_OverRider_Up	BOOL				False	Raise rider roll
O_OverRider_Hold	BOOL				False	Lock rider roll
Fix10Vdc	INT				False	Fixed 10 Vdc generator
InvRef	INT				False	Inverter speed reference
Encoder_On	BOOL				False	Encoder enable
Encoder_Reset	BOOL				False	Encoder reset
OverRider_TimeOFF	REAL				False	Rider roll photocell OFF timeout (panel)
OverRider_TimeON	REAL				False	Rider roll photocell ON timeout (panel)
VelMax	REAL				False	Maximum speed
I_Jog_PB	BOOL				False	Slow speed button
O_Motor_Lamp	BOOL				False	Motor ON lamp
AlarmOn	BOOL				False	Alarm active
BuzzerOn	BOOL				False	Panel buzzer command
I_VFD_Alarm	BOOL				False	Inverter in alarm
Msg	INT				False	Message
Index_B	INT				False	Active alarm index
MotStop	BOOL				False	Motor stopped
Clock1s	BOOL				False	1 s clock
ManMode	BOOL				False	Manual mode
Counter_Reset	BOOL				False	Panel counter reset button
I_FrontGuard_Open	BOOL				False	Front guard open
O_RwLoading_Enable	BOOL				False	Unloaders enable - closing
JogSpd_SP	REAL				False	Speed in manual
O_AccDec2	BOOL				False	Second acc/dec ramp
AccelTrim	REAL				False	Acceleration coefficient in run
DecelTrim	REAL				False	Deceleration coefficient in run
DecelCoarse	REAL				False	Acceleration coefficient at start (orig. comment; name suggests decel)
AccelCoarse	REAL				False	Deceleration coefficient at start (orig. comment; name suggests accel)
Speed_Display	REAL				False	Set speed (displayed)
O_AcousticSignal	BOOL				False	Horn / acoustic signal
Pot1_Work_Velocity	INT				False	Potentiometer 1 value - Work speed -
O_FrontGuard_ByPass	BOOL				False	Front guard bypass
Pot2_Slow_Velocity	INT				False	Potentiometer 2 value - Final speed -
LowSpd_Display	REAL				False	Slow-down speed (displayed)
Encoder_Pulse	INT				False	Encoder pulses
RwSlowDecelLrn	REAL				False	Learned real slow-down decel (mm/s2) - 0 = not learned
UwSlowMargin_m	REAL				False	Slow-down metres
UwStop_at_M_Val	REAL				False	Total metres
I_AutoStop_OS	BOOL				False	Stop on metre counter

I_Input_11_Spare_1255	BOOL				False	Slow down at end of roll
I_T_one_TensionOK	BOOL				False	Slow-down photocell
I_WebBreak_Monitor	BOOL				False	Paper break monitoring ON
I_WebBreak_PH	BOOL				False	Paper break photocell
I_Vacuum_OK	BOOL				False	Extractor fan thermal OK
CountReset_Req	BOOL				False	Metre counter reset request
Psw	INT				False	Password
I_RsetAlarm	BOOL				False	Alarm reset
EncoderWheel_Circumf	REAL				False	Encoder wheel diameter
AlarmCodeSV	ARRAY[0..32] OF BOOL				False	Panel alarms
SafeGuard_ByPass	BOOL				False	Guards bypassed
FrontGuard_Closed	BOOL				False	Front guard closed
Core_Measure	BOOL				False	(?) Stored core value
UwDiam_Measure	REAL				False	Measured distance
UwCoreDistance	REAL				False	(?) Core measurement
RwSlowFallbackMm	REAL				False	
AI_Diam_Unwinder	INT				False	Analog value - parent roll measurement
RwMatChangePct	REAL				False	
UwCoreDiameter	REAL				False	(?) Shaft diameter
Measure_ON	BOOL				False	(?) Measurement enabled
UwDiam_Act	REAL				False	(?) Measured roll diameter
UwStopDiam_Limit	REAL				False	(?) Stop diameter
UwSlow_at_D_Val	REAL				False	(?) Slow-down diameter
Alarm_Reset	BOOL				False	(?) Alarm reset from panel
HMI_Screen	ARRAY[0..1] OF DINT				False	HMI panel page
Psw22	BOOL				False	(?) Password 22
RwMatReset	BOOL				False	
Psw44	BOOL				False	(?) Password 44
I_Roll_OK	BOOL				False	(?) Roll OK
AI_Diam_Rewinder	INT				False	Rewinder DT35 raw - I/O Map Unit2 Ch2
RwStopEnable	BOOL				False	Rewinder diameter stop enabled
RwTargetDiam	REAL				False	Rewinder final diameter (mm)
RwUseSlow	BOOL				False	Slow down before target
RwSlowMargin_m	REAL				False	Metres at slow speed before stop
UwRawMin	INT				False	
UwRawMax	INT				False	
RwSlowDecel	REAL				False	Slow-down decel mm/s2 - 0 = auto from DecStart
RwResetHyst	REAL				False	Reset requests below target - this (mm)
RwDist0V	REAL				False	DT35 distance at 0 V (mm)
RwDist10V	REAL				False	DT35 distance at 10 V (mm)
RwRawMin	INT				False	Raw below = sensor fault

RwRawMax	INT				False	Raw above = sensor fault
RwStopOnFault	BOOL				False	Stop on sensor fault
RwAxisDist	REAL				False	Sensor to shaft centre (mm) - teach
RwFilter_K	REAL				False	Distance filter per 100 ms (0..1)
RwMinPulses	DINT				False	Min pulses in window to learn
RwLearn_K	REAL				False	Material factor filter weight
RwMat_K	REAL				False	Learned material factor
RwLearnStop	BOOL				False	Learn stop decel on every stop
RwMinLearnSpd	REAL				False	Min speed to learn a stop (m/min)
RwStopDecel	REAL				False	Learned stop decel (pulses/s2)
RwTreact	REAL				False	Stop reaction time (s)
RwTstopDef	REAL				False	Stop time until decel learned (s)
RwTrim	REAL				False	Stop length trim (>1 = earlier)
RwDist	REAL				False	Filtered distance (mm)
RwDiam	REAL				False	Rewinder diameter (mm)
RwSpeed	REAL				False	Line speed from encoder (m/min)
RwRemain_m	REAL				False	Metres to target diameter
RwStopLen_m	REAL				False	Stopping length now (m)
RwSlowLen_m	REAL				False	Slow-down length now (m)
RwStop_at_D_Req	BOOL				False	Rewinder diameter STOP request
RwSlow_at_D_Req	BOOL				False	Rewinder diameter SLOW request
RwSensFault	BOOL				False	Rewinder DT35 sensor fault
UwStopEnable	BOOL				False	Unwinder metre-counter stop ON (HMI)
SlowAtDisp	REAL				False	HMI: slow-down starts at (m or mm, see SlowAtMode)
UwSlow_at_D_Req	BOOL				False	
UwSlowDiamAct	REAL				False	
RwJogMaxDiam	REAL				False	
JogInhibit	BOOL				False	
SCurveTime	REAL				False	
TOneDecel	BOOL				False	
HornStartBeep_s	REAL				False	
HornBeepLen_s	REAL				False	
HornBeepPeriod	REAL				False	
HornStopBeep_s	REAL				False	
HornTest	BOOL				False	
RwLockDiam	REAL				False	
UwSensTest	BOOL				False	
RwLockMinGrow_mm	REAL				False	
RwMatLocked	BOOL				False	
LongThick_mm	REAL				False	
RwDiamRev	REAL				False	
RwWave_mm	REAL				False	
UwStopLenShow	BOOL				False	
UwSlowDiamShow	BOOL				False	
SoftCancelAccel	REAL				False	
SpdHold	BOOL				False	
SlowAtMode	INT				False	HMI: 0=none 1=unwinder [m] 2=rewinder [mm]
SlowAtShow	BOOL				False	HMI: show slow-down display

UwSlow_at_M_Val	REAL				False	Unwinder: slow-down starts at (m)
UwSlow_at_M_Req	BOOL				False	Unwinder metre slow-down request (latched)
RwSlow_at_D_Val	REAL				False	Rewinder: slow-down starts at diameter (mm)
RwStop_at_D_Val	REAL				False	Rewinder: stop command at diameter (mm)
RwMaxDiam	REAL				False	Rewinder max diameter protection (mm)
RwMaxStop	BOOL				False	Rewinder max diameter STOP (latched)
RwMaxSlow	BOOL				False	Rewinder max diameter SLOW (latched)
UwEmptyHyst	REAL				False	
RwThickMin_mm	REAL				False	Learning: min plausible paper thickness (mm)
RwThickMax_mm	REAL				False	Learning: max plausible paper thickness (mm)
RwDecelMin_mm	REAL				False	Learning: min plausible stop decel (mm/s2)
RwDecelMax_mm	REAL				False	Learning: max plausible stop decel (mm/s2)
RwLearnFailMax	INT				False	Learning: failed windows before alarm (12 = 2 min)
RwStopFailMax	INT				False	Learning: failed stops before alarm
GSM_Learned_OK	BOOL				False	Rewinder learning fault (alarm 16)
RwThick_mm	REAL				False	Learned paper thickness (mm)
Psw33	BOOL				False	
RwAutoMaxSpd	REAL				False	
RwWinLen_m	REAL				False	
RwWinGrow_mm	REAL				False	
GsmAsk	BOOL				False	
GsmAskYes	BOOL				False	
DecelWeb	REAL				False	
AccelWeb	REAL				False	
RwRatioD	REAL				False	
FFEnable	BOOL				False	
UwStopDiamShow	BOOL				False	
UwStop_at_D_Val	REAL				False	
Clock10ms	BOOL				False	
Sec_to_Min	REAL				False	
Psw48	BOOL				False	
JogRun	BOOL				False	
GuardStopOK	BOOL				False	V11: stopped for real -> guard bypass allowed (Output 13)
SlowDown	BOOL				False	Slow down
WarningOn	bool				False	
AlarmSilenced	BOOL				False	
PaperGSM	REAL				False	
WebWidth_mm	REAL				False	
TensFactor	REAL				False	
TensShowKg	BOOL				False	
TensShowLb	BOOL				False	
TensShowN	BOOL				False	

	TensIdeal_Disp	REAL				False	
	TensUnit	INT				False	
	TensUnitPB	BOOL				False	
	TensCheckEnable	BOOL				False	
	TensLostWarn	BOOL				False	
	TensWaitSpd	REAL				False	
	TensWait	BOOL				False	
	TensionOK	BOOL				False	
	UwSensFault	BOOL				False	
	Thickness_Fault	BOOL				False	
	AutoStopDone	BOOL				False	
	AutoStopWhy	INT				False	
	AutoEnRefused	BOOL				False	
	AutoEnWhy	INT				False	
	TensWaitHz	REAL				False	
	StartBlocked	BOOL				False	
	GuardOpen	BOOL				False	
	SideDoorOpen	BOOL				False	

1-8-1-1-2.General

0 Remote 10.146.233.11 ... 1s clock.



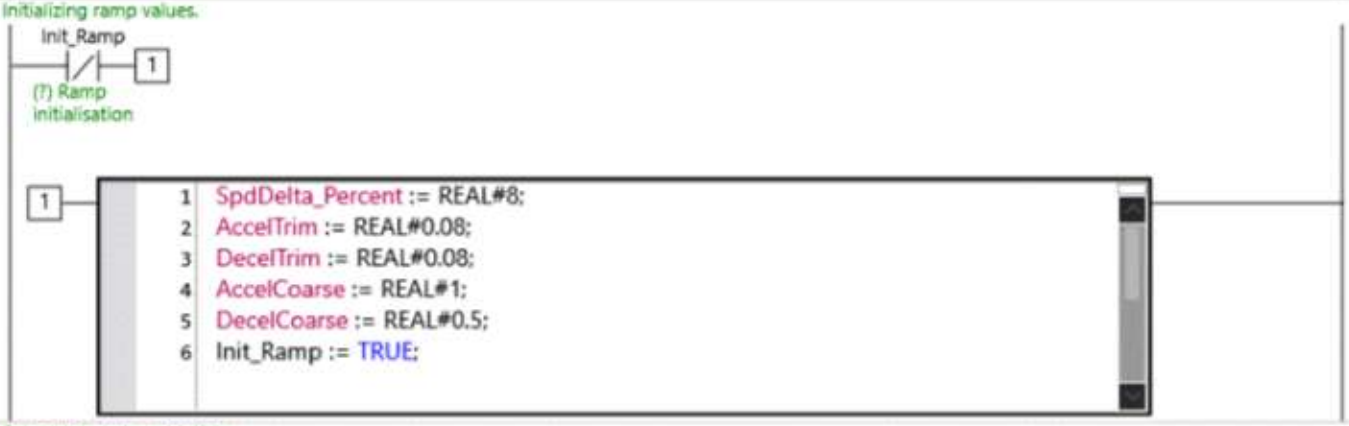
1 100ms clock.



2 10ms clock.



3 Initializing ramp values.

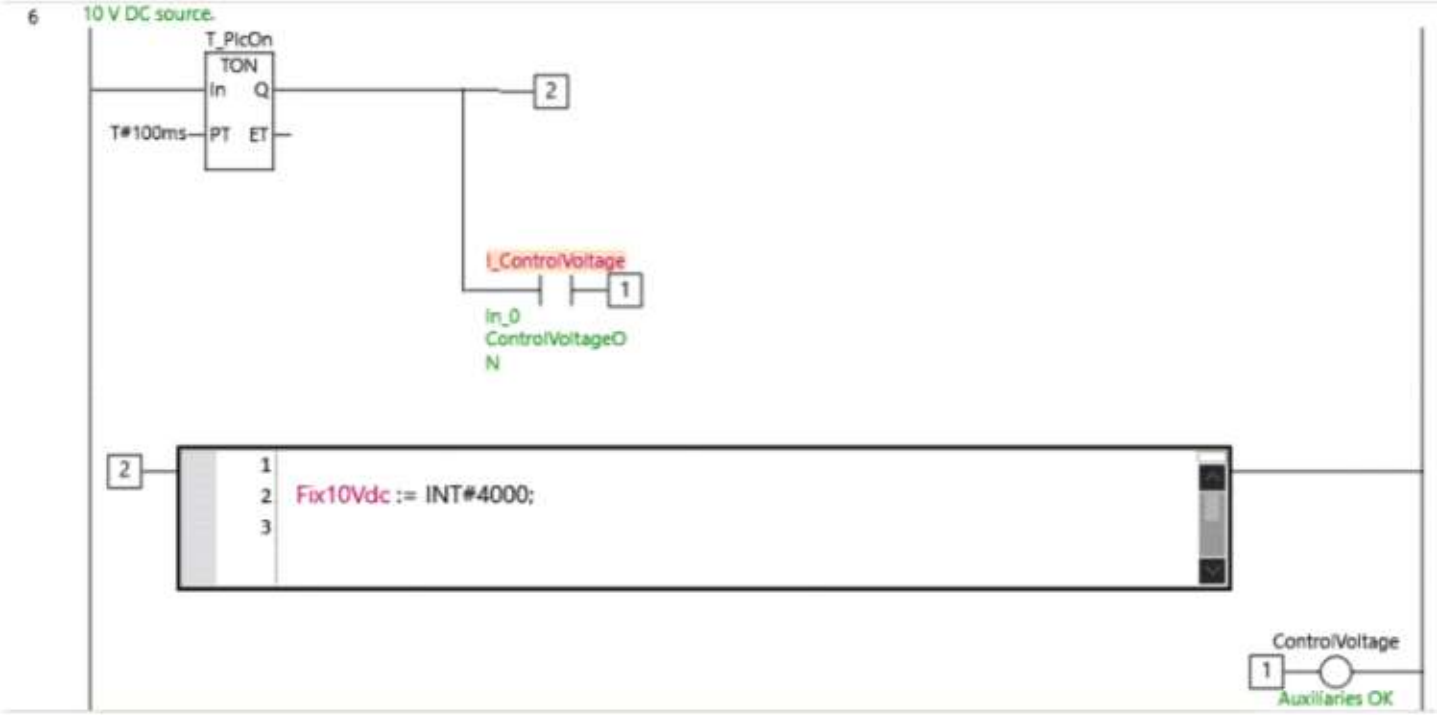


4 Set analog scale resolution.



5 Variable for converting m/min to mm/sec and vice versa.




6
10 V DC source.
T_PlcOn
TON
In Q
T#100ms PT ET
2
I_ControlVoltage
1
In_0
ControlVoltageO
N
2
1
2 Fix10Vdc := INT#4000;
3
ControlVoltage
1
Auxiliaries OK

7 START blocked by JOG

```
// ==== V11: REPLACE the code of General rung 7 ("START blocked by JOG", printed V9).
// START push-button is ignored while the JOG (slow speed) button is held and for
// 2 s after it is released. A START held during the block must be released and
// pressed again (no start by itself when the 2 s are over).
// Without this: START pressed during a jog sets Slow_Inhibit -> releasing the jog
// no longer stops (General rung 11) and VelSet jumps to the work speed.

T_JogStartBlk(In := I_Jog_PB, PT := T#2s);      // TOF: ON while held + 2 s
StartHeldBlk := (StartHeldBlk OR (I_Start_PB AND T_JogStartBlk.Q)) AND I_Start_PB;
StartPB_OK := I_Start_PB AND NOT T_JogStartBlk.Q AND NOT StartHeldBlk;

// V10: JogRun = a JOG really running: JOG button held, NOT a normal START run
// (Slow_Inhibit = normal run, set by START) and NOT a soft stop. Used instead of
// the raw I_Jog_PB for: guard bypass (Output 13), 2nd drive ramp (Output 12),
// pot reading (Motor_Manager 8), feed-forward (Motor_Manager 14/17), AutoStop learning.
// Without it, holding JOG during a 500 m/min run bypassed the front gate / lateral
// doors safety (23KA4 // 5SA1) and switched the drive to its 2nd ramp.

JogRun := I_Jog_PB AND NOT Slow_Inhibit AND NOT SoftStop AND NOT SideDoorOpen;

// Guard bypass (Output_Manager rung 13, 23KA4 // 5SA1): ON when the run command is
// off (machine stopped by the operator).

GuardStopOK := NOT StartMotor_Cmd;

// START refused (JOG still allowed - it has its own guard bypass):
//  - no paper at the web break photocell (monitoring ON)
//  - front gate or side door open, unless the key bypass 5SA1 (I_Prot_ByPass) is ON

WebBreakNow := I_WebBreak_Monitor AND I_WebBreak_PH;
GuardOpen   := (NOT I_FrontGuard_Closed OR NOT I_Side_Door_Closed) AND NOT I_Prot_ByPass;
StartBlocked := WebBreakNow OR GuardOpen;

// Side door open (no key bypass): no JOG, and stop any JOG / run in progress
SideDoorOpen := NOT I_Side_Door_Closed AND NOT I_Prot_ByPass;
```

8

```
// ==== General: NEW ST rung (no contact), directly BELOW rung 7 (JogRun / GuardStopOK)
// ==== Comment: "T-one Alarm 0: tension established"
// T-one: F.9 = 0 (alarm when tension ABOVE threshold), F.10 = threshold (kg), F.11 = full scale
// Relay pins 15-16 (Alarm 0): +24 V -> pin 15, pin 16 -> Inline IB IL 24 DI 8 IN05, pin 1.3
// I_T_one_TensionOK = node#[1.0] _IN05 (Pin 1.3) (was I_SlowSens_PH, used in no rung)
// Relay closed = tension > F.10 (not latched). Broken wire / T-one off = "no tension" (fail-safe).

T_TensOn(In := I_T_one_TensionOK, PT := T#500ms);       // stable 0.5 s
T_TensOff(In := NOT I_T_one_TensionOK, PT := T#2s);     // ignore short dips (splice, wave, chatter)

IF T_TensOn.Q OR NOT TensCheckEnable THEN               // HMI switch OFF = old behaviour
    TensionOK := TRUE;
ELSIF T_TensOff.Q THEN
    TensionOK := FALSE;
END_IF;

// Latch: tension reached once since START

IF NOT StartMotor_Cmd THEN
    TensionSeen := FALSE;
ELSIF TensionOK THEN
    TensionSeen := TRUE;
END_IF;

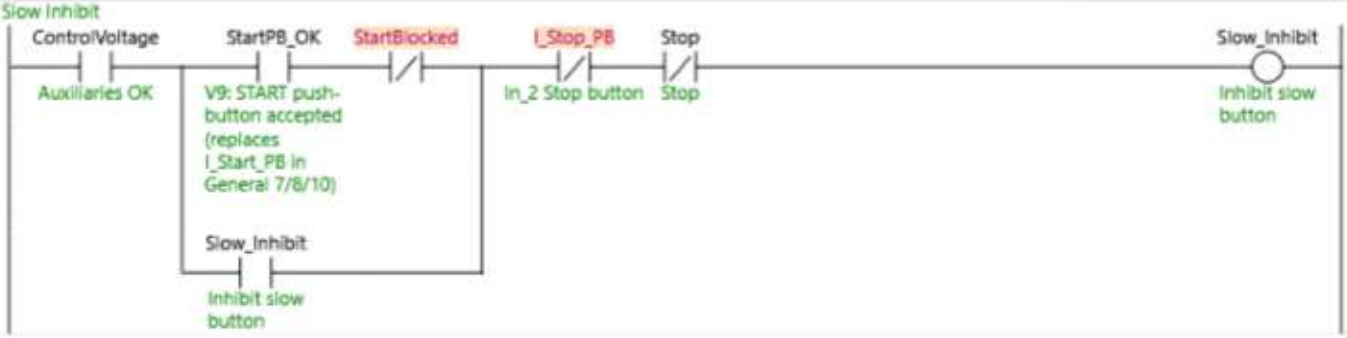
// After START: hold at TensWaitSpd until the tension is there (JOG not affected)

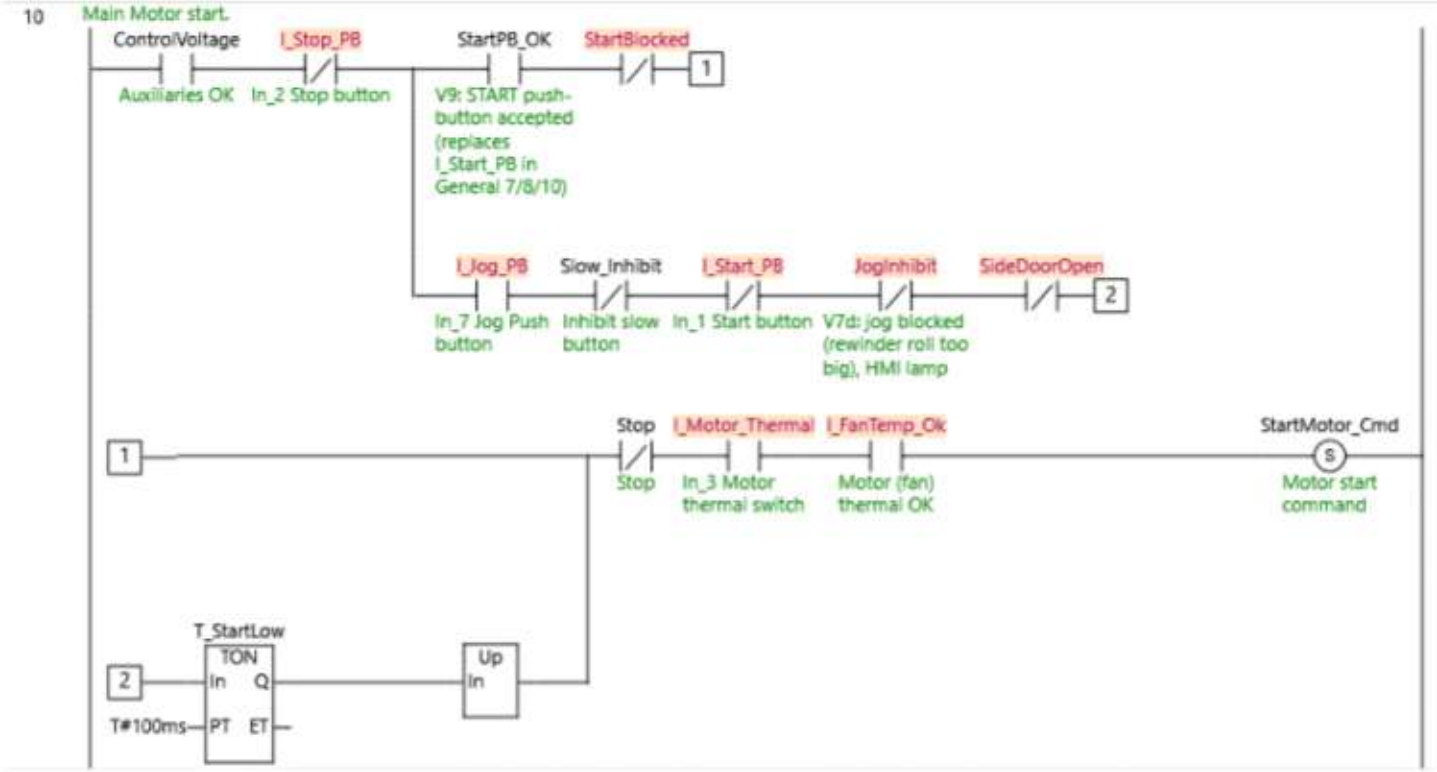
TensWait := StartMotor_Cmd AND NOT TensionSeen AND NOT JogRun;

// Tension lost later while running -> warning only (no slow-down)

TensLostWarn := StartMotor_Cmd AND TensionSeen AND NOT TensionOK;
```

9 Slow Inhibit




10
Main Motor start.
ControlVoltage
Auxiliaries OK
I_Stop_PB
In_2 Stop button
StartPB_OK
V9: START push-button accepted (replaces I_Start_PB in General 7/8/10)
StartBlocked
1
I_Jog_PB
In_7 Jog Push button
Slow_Inhibit
Inhibit slow button
I_Start_PB
In_1 Start button
JogInhibit
V7d: jog blocked (rewinder roll too big), HMI lamp
SideDoorOpen
2
1
Stop
Stop
I_Motor_Thermal
In_3 Motor thermal switch
I_FanTemp_Ok
Motor (fan) thermal OK
StartMotor_Cmd
S
Motor start command
T_StartLow
TON
2
In
Q
T#100ms
PT
ET
Up
In

11 Soft Stop Process

```
// ==== V12: REPLACE the code of General rung 10 ("Soft Stop Process", printed V9).
// v6m: STOP button = soft stop. StopNow replaces I_Stop_PB in rung 9.
// V12: an end-of-roll stop (FinalStopRamp, AutoStop section 11b) also starts the
// soft stop, at FinalStopDecel (Motor_Manager rung 17): START cannot cancel it.

StopNow := FALSE;
T_StopLong(In := I_Stop_PB, PT := T#1s);

IF NOT StartMotor_Cmd THEN
    SoftStop := FALSE;
ELSIF I_Stop_PB AND NOT StopPB_Mem THEN            // STOP pressed (edge)

    IF SoftStop OR NOT SoftStopEnable OR I_Jog_PB
      OR (Velocity_M_min <= SoftStopEndSpd) THEN
        StopNow := TRUE;                           // 2nd press / jog / slow / disabled
    ELSE
        SoftStop := TRUE;                          // start soft stop
    END_IF;
END_IF;

StopPB_Mem := I_Stop_PB;

IF T_StopLong.Q THEN
    StopNow := TRUE;                               // held 1 s = immediate
END_IF;

// V12: end-of-roll stop ramp (FinalStopRamp set by AutoStop section 11b)

IF FinalStopRamp AND StartMotor_Cmd AND NOT SoftStop THEN
    SoftStop := TRUE;
END_IF;

// Side door opened while running (not JOG): soft stop. START cannot cancel it

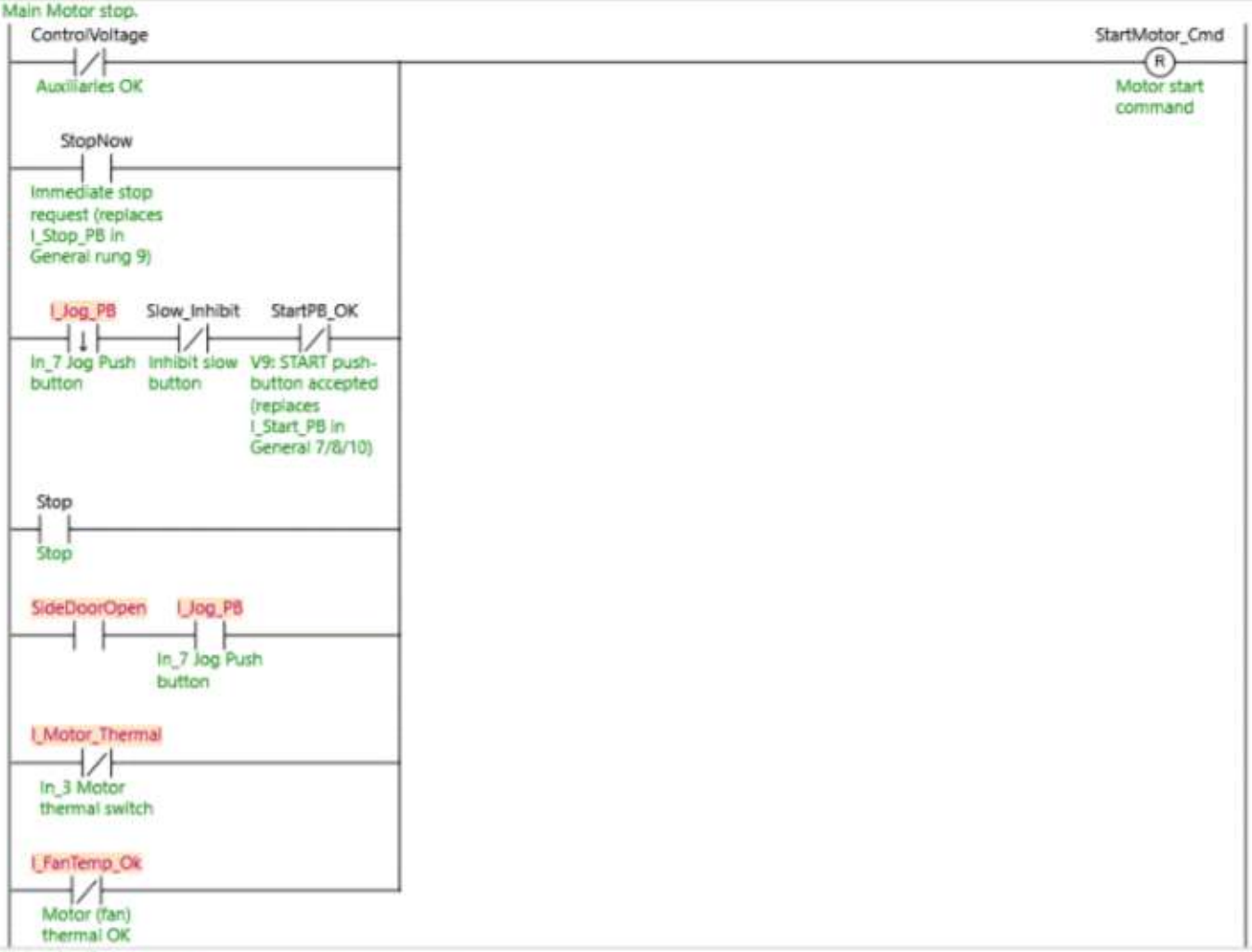
IF SoftStop AND I_Start_PB AND NOT FinalStopRamp AND NOT SideDoorOpen
THEN
    SoftStop := TRUE;
END_IF;

IF SoftStop AND I_Start_PB AND NOT FinalStopRamp THEN
    SoftStop := FALSE;                             // START cancels the soft stop
END_IF;

T_SoftStop(In := SoftStop, PT := T#120s);

IF SoftStop AND ((Velocity_M_min <= SoftStopEndSpd) OR T_SoftStop.Q) THEN
    StopNow := TRUE;                               // almost stopped (or timeout)
END_IF;
```

12 Main Motor stop.



13 Hub extraction clearance.



14 Brake control.

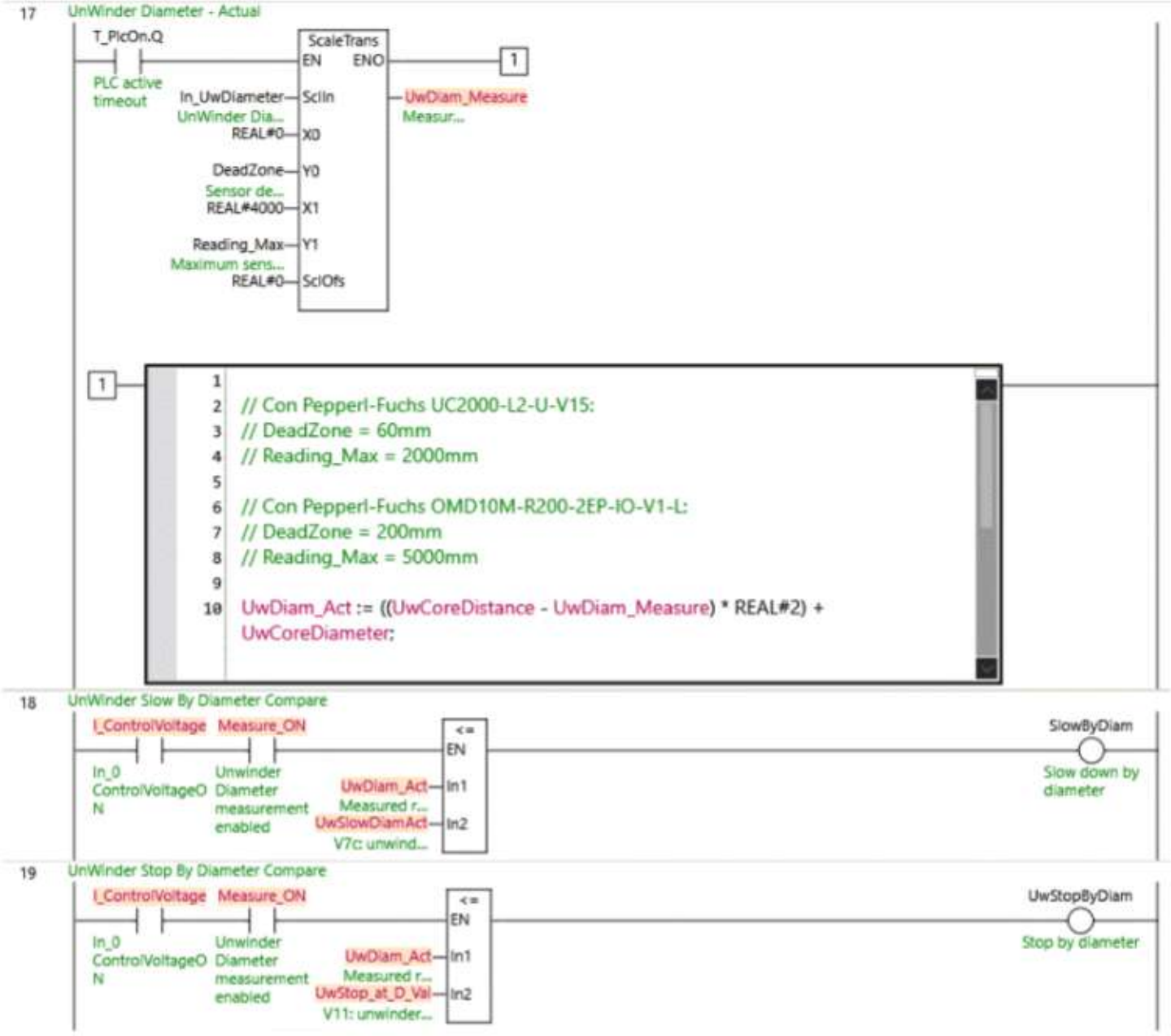


15 Cooling Fan control.

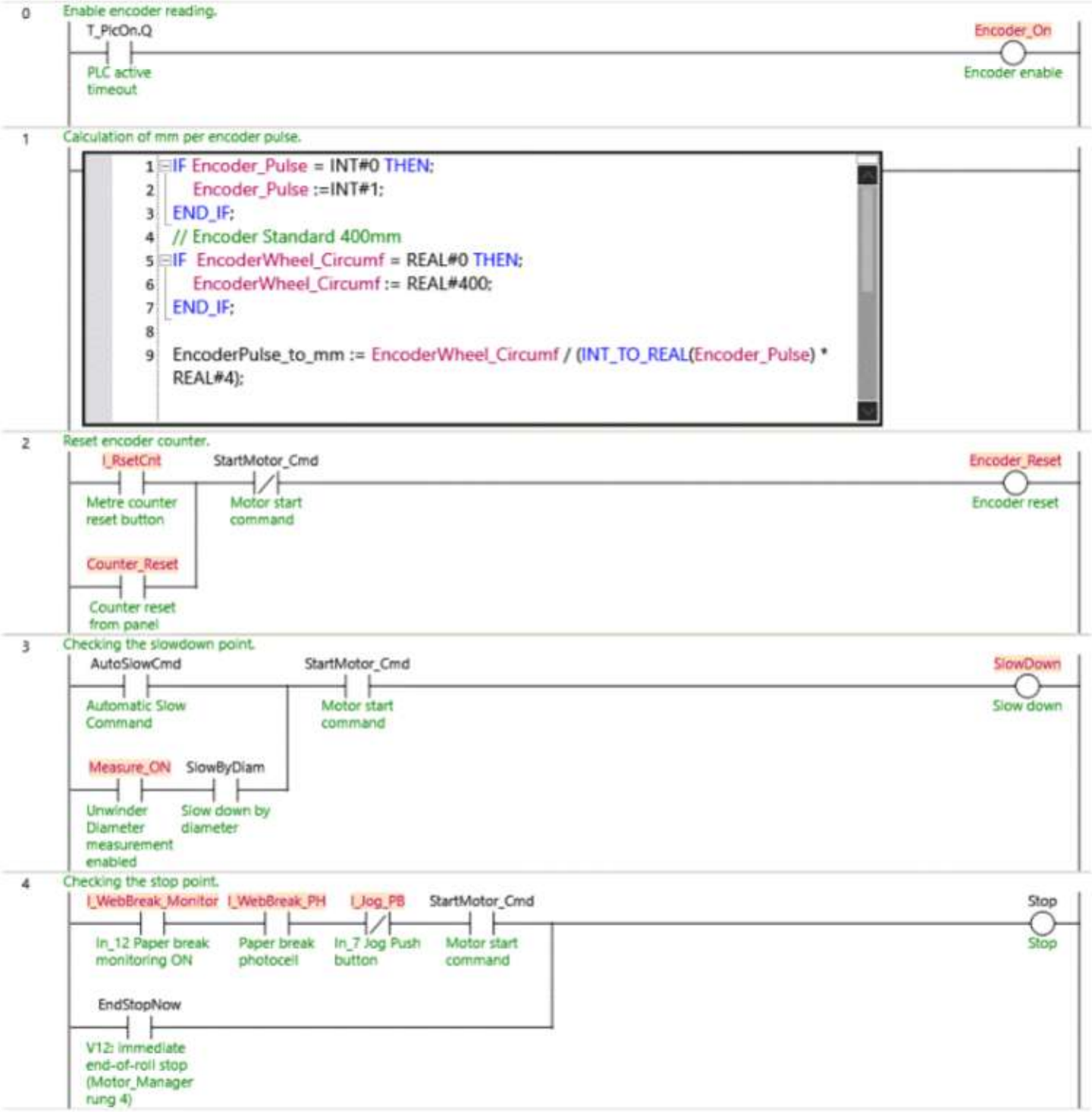


16 UnWinder Diameter - Input

```
1  In_UwDiameter := INT_TO_REAL(AI_Diam_Unwinder)/REAL#1;
```


17
UnWinder Diameter - Actual
T_PlcOn.Q
PLC active timeout
ScaleTrans
EN
ENO
1
In_UwDiameter
UnWinder Dia...
REAL#0
SclIn
X0
DeadZone
Sensor de...
REAL#4000
Y0
X1
Reading_Max
Maximum sens...
REAL#0
Y1
SclOfs
UwDiam_Measure
Measur...
1
1
// Con Pepperl-Fuchs UC2000-L2-U-V15:
// DeadZone = 60mm
// Reading_Max = 2000mm
// Con Pepperl-Fuchs OMD10M-R200-2EP-IO-V1-L:
// DeadZone = 200mm
// Reading_Max = 5000mm
UwDiam_Act := ((UwCoreDistance - UwDiam_Measure) * REAL#2) +
UwCoreDiameter;
18
UnWinder Slow By Diameter Compare
I_ControlVoltage
In_0 ControlVoltageON
Measure_ON
Unwinder Diameter measurement enabled
<=
EN
UwDiam_Act
Measured r...
In1
UwSlowDiamAct
V7c: unwind...
In2
SlowByDiam
Slow down by diameter
19
UnWinder Stop By Diameter Compare
I_ControlVoltage
In_0 ControlVoltageON
Measure_ON
Unwinder Diameter measurement enabled
<=
EN
UwDiam_Act
Measured r...
In1
UwStop_at_D_Val
V11: unwinder...
In2
UwStopByDiam
Stop by diameter

1-8-1-1-3.Motor_Manager



5 Check the maximum and minimum values for working speed and slow speed.

```
IF VelMax < REAL#600 THEN;
      VelMax := REAL#600;
   ELSIF VelMax > REAL#1000 THEN;
      VelMax := REAL#1000;
END_IF;
Vel_Max_mms := VelMax * REAL#1000 / REAL#60;

IF JogSpd_SP < REAL#10 THEN;
      JogSpd_SP := REAL#10;
ELSIF JogSpd_SP > REAL#70 THEN;
   JogSpd_SP := REAL#70;
END_IF;

// v6y: rewinder AUTO max speed. 0 = no limit

IF RwAutoMaxSpd > VelMax THEN
   RwAutoMaxSpd := VelMax;
ELSIF (RwAutoMaxSpd > REAL#0) AND (RwAutoMaxSpd < REAL#50) THEN
   RwAutoMaxSpd := REAL#50;
END_IF;

IF NOT StartMotor_Cmd THEN          // taken over at standstill only
   RwAutoLimAct := RwStopEnable AND (RwAutoMaxSpd > REAL#0);
   RwAutoMaxSpdAct := RwAutoMaxSpd;
END_IF;

// ---- V7: add at the END of Motor_Manager rung 5 ("Check the maximum and
minimum values...")
// ---- after the JogSpd_SP clamp, before Output_Ratio. Same style as the VelMax /
JogSpd clamps.
// V7: limits on settings that stop the machine or break the auto stop if 0 / absurd

IF SpdDelta_Percent < REAL#2 THEN              // fine band [m/min]: 0 = speed
hunting
   SpdDelta_Percent := REAL#2;
ELSIF SpdDelta_Percent > REAL#15 THEN
   SpdDelta_Percent := REAL#15;
END_IF;

IF AccelTrim < REAL#0.02 THEN                  // trim ramps: 0 = no speed correction
   AccelTrim := REAL#0.02;
ELSIF AccelTrim > REAL#0.5 THEN
   AccelTrim := REAL#0.5;
END_IF;

IF DecelTrim < REAL#0.02 THEN
   DecelTrim := REAL#0.02;
ELSIF DecelTrim > REAL#0.5 THEN
   DecelTrim := REAL#0.5;
END_IF;

IF AccelCoarse < REAL#0.2 THEN                 // coarse ramps: 0 = never reaches
speed
   AccelCoarse := REAL#0.2;
ELSIF AccelCoarse > REAL#4 THEN
   AccelCoarse := REAL#4;
END_IF;
```

```
IF DecelCoarse < REAL#0.2 THEN
    DecelCoarse := REAL#0.2;
ELSIF DecelCoarse > REAL#4 THEN
    DecelCoarse := REAL#4;
END_IF;

IF DecelWeb > REAL#0 THEN                    // V7 web ramps: 0 = old coarse ramp (allowed)
    IF DecelWeb < REAL#50 THEN
        DecelWeb := REAL#50;
    ELSIF DecelWeb > REAL#1000 THEN
        DecelWeb := REAL#1000;
    END_IF;
END_IF;

IF AccelWeb > REAL#0 THEN
    IF AccelWeb < REAL#50 THEN
        AccelWeb := REAL#50;
    ELSIF AccelWeb > REAL#1000 THEN
        AccelWeb := REAL#1000;
    END_IF;
END_IF;

IF SoftStopDecel > REAL#1500 THEN            // 0 = DecelCoarse (allowed)
    SoftStopDecel := REAL#1500;
END_IF;

IF RwMaxDiam > REAL#1450 THEN                // rewinder MECHANICAL limit [mm]
    RwMaxDiam := REAL#1450;
ELSIF RwMaxDiam < REAL#300 THEN
    RwMaxDiam := REAL#300;
END_IF;

IF RwMaxSlowFallback >= RwMaxDiam THEN       // fallback slow-down below the max
    RwMaxSlowFallback := RwMaxDiam - REAL#30;
END_IF;

// ---- V9 (V7d): add at the END of Motor_Manager rung 5, after the V7 clamps
// ---- (last lines now: RwMaxSlowFallback clamp), before Output_Ratio / Input_Ratio.
// V7d: jog / slow start button blocked on a big rewinder roll.
// RwJogMaxDiam (mm, HMI, Retain): 0 = no limit, else min 150.
// DT35 fault -> jog allowed (needed to thread / clear the machine).

IF (RwJogMaxDiam > REAL#0) AND (RwJogMaxDiam < REAL#150) THEN
    RwJogMaxDiam := REAL#150;
END_IF;

JogInhibit := (RwJogMaxDiam > REAL#0) AND (RwDiam > RwJogMaxDiam) AND NOT RwSensFault;

Output_Ratio := (VelMax * Mt_To_mm) / UINT_TO_REAL(Analog_Res); //REAL#4000;
Input_Ratio := UINT_TO_REAL(Analog_Res) / VelMax ;
```

6 Reading the current value of the potentiometers. v6n: requested pot ranges, limited, applied only with the motor stopped !

```
IF WorkSpd_Max_Set < 50.0 THEN
    WorkSpd_Max_Set := 50.0;
ELSIF WorkSpd_Max_Set > VelMax THEN
    WorkSpd_Max_Set := VelMax;
END_IF;

IF FinalSpd_Max_Set < 5.0 THEN
    FinalSpd_Max_Set := 5.0;
ELSIF FinalSpd_Max_Set > WorkSpd_Max_Set THEN
    FinalSpd_Max_Set := WorkSpd_Max_Set;
END_IF;

IF NOT StartMotor_Cmd THEN
    WorkSpd_Max := WorkSpd_Max_Set;          // motor stopped: apply
    FinalSpd_Max := FinalSpd_Max_Set;
END_IF;

SpdMax_Pending := (WorkSpd_Max <> WorkSpd_Max_Set)
            OR (FinalSpd_Max <> FinalSpd_Max_Set)
            OR (RwAutoLimAct <> (RwStopEnable AND (RwAutoMaxSpd > REAL#0)))
            OR (RwAutoLimAct AND (RwAutoMaxSpdAct <> RwAutoMaxSpd));

// v6i: pots scaled to their own full scale (m/min)

IF Pot1_Work_Velocity > INT#5 THEN
    Pot1_Mem := REAL_TO_INT(INT_TO_REAL(Pot1_Work_Velocity) * WorkSpd_Max /
VelMax);
ELSE
    Pot1_Mem := INT#0;
END_IF;

IF Pot2_Slow_Velocity > INT#5 THEN
    Pot2_Mem := REAL_TO_INT(INT_TO_REAL(Pot2_Slow_Velocity) * FinalSpd_Max /
VelMax);
ELSE
    Pot2_Mem := INT#0;
END_IF;
```

7 Updating the set speed.

```
IF ManMode = FALSE THEN;

    Speed_Display := UINT_TO_REAL(Pot1_Mem) / Input_Ratio;
    LowSpd_Display := UINT_TO_REAL(Pot2_Mem) / Input_Ratio;

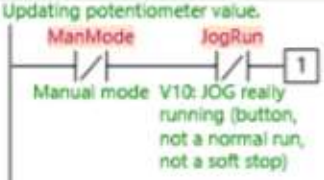
  // v6y: rewinder AUTO stop -> work speed limited

  IF RwAutoLimAct AND (Speed_Display > RwAutoMaxSpdAct) THEN
     Speed_Display := RwAutoMaxSpdAct;
  END_IF;

ELSE;
  Speed_Display := Pot1_Work_Velocity / (Analog_Res / UINT#40); // v6i: raw pot. %
  LowSpd_Display := Pot2_Slow_Velocity / (Analog_Res / UINT#40);

END_IF;
```

8 Updating potentiometer value.




```
IF SoftStop THEN
   VelSet := 0.0;                    // v6m: soft stop
ELSIF SlowDown = FALSE  THEN
   VelSet := UINT_TO_REAL(Pot1_Mem) / Input_Ratio;

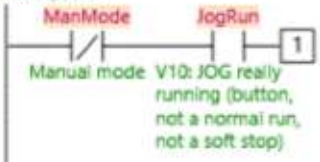
   // v6y: rewinder AUTO stop -> work speed limited

   IF RwAutoLimAct AND (VelSet > RwAutoMaxSpdAct) THEN
      VelSet := RwAutoMaxSpdAct;
   END_IF;

ELSE
   VelSet := UINT_TO_REAL(Pot2_Mem) / Input_Ratio;
END_IF;

VelSet_mms := VelSet * Mt_To_mm;
```

9 Slow speed.




```
VelSet := JogSpd_SP;
VelSet_mms := VelSet * Mt_To_mm;
```

10 Check the difference between the set speed and the detected speed.

```
//SpdDelta := VelSet_mms * SpdDelta_Percent / REAL#100;

SpdDelta := SpdDelta_Percent;
SpdDeltaMax := VelSet + SpdDelta;
SpdDeltaMin := VelSet - SpdDelta;

// Too fast

IF Velocity_M_min > SpdDeltaMax THEN
    OverSpeed := TRUE;
ELSE
    OverSpeed := FALSE;
END_IF;

// Too slow

IF Velocity_M_min < SpdDeltaMin THEN
    UnderSpeed := TRUE;
ELSE
    UnderSpeed := FALSE;
END_IF;
```

11 Speed calculation.

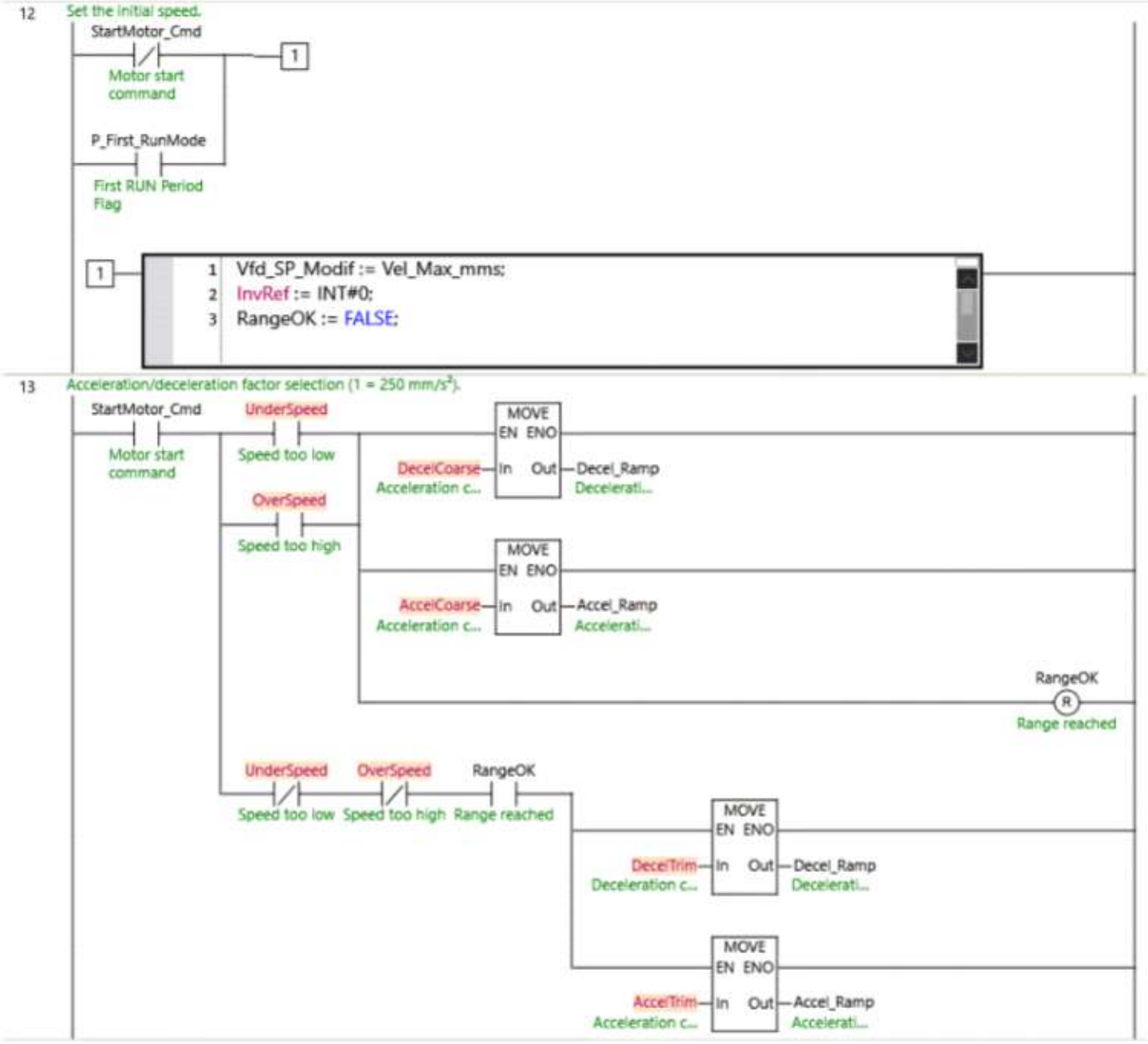
```
// ==== V9: REPLACE Motor_Manager rungs 11 ("Read the current value of the encoder", MOVE on
// ==== Clock100ms rising edge) and 12 ("Speed calculation", Clock100ms falling edge) by ONE
// ==== ST rung with this code (no contact in front). Comment: "V9: speed over exactly 100 ms".
// Old: window = time between the rising and falling edge of Clock100ms, assumed 50 ms,
// but with the 4 ms PrimaryTask it is 12 or 13 scans (48 / 52 ms) -> the speed jumps
// -4 % / +4 % (418 set: display 397 / 431).
// New: exactly 25 scans = 100 ms (same as AutoStop SpdAct_Pps), updated 10 times/s.

SpdScanCnt := SpdScanCnt + 1;

IF SpdScanCnt >= 25 THEN
    SpdScanCnt := 0;
    DeltaPulse := Encoder_Pos - MemPos;
    MemPos := Encoder_Pos;

    IF ABS(DeltaPulse) > 100000 THEN            // counter reset (Encoder_Reset)
        DeltaPulse := 0;
    END_IF;

    Velocity := DINT_TO_REAL(DeltaPulse) * EncoderPulse_to_mm * REAL#10;   // mm/s
    Velocity_M_min := Velocity / Mt_To_mm;
END_IF;
```


12
Set the initial speed.
StartMotor_Cmd
Motor start command
1
P_First_RunMode
First RUN Period Flag
1
1 Vfd_SP_Modif := Vel_Max_mms;
2 InvRef := INT#0;
3 RangeOK := FALSE;
13
Acceleration/deceleration factor selection (1 = 250 mm/s²).
StartMotor_Cmd
Motor start command
UnderSpeed
Speed too low
OverSpeed
Speed too high
MOVE
EN ENO
DecelCoarse
Acceleration c...
In Out
Decel_Ramp
Decelerati...
MOVE
EN ENO
AccelCoarse
Acceleration c...
In Out
Accel_Ramp
Accelerati...
RangeOK
R
Range reached
UnderSpeed
Speed too low
OverSpeed
Speed too high
RangeOK
Range reached
MOVE
EN ENO
DecelTrim
Deceleration c...
In Out
Decel_Ramp
Decelerati...
MOVE
EN ENO
AccelTrim
Acceleration c...
In Out
Accel_Ramp
Accelerati...

14 Ratio drive counts per m/min of web at the current rewinder diameter.

```
// ==== V10: REPLACE the code of Motor_Manager rung 14 (printed V9) ("Ratio drive counts per m/min of web
// ==== at the current rewinder diameter." = V7 rung A) with this file. (V7b jog line inside.)
// Ratio drive counts per m/min of web at the current rewinder diameter.
// The motor drives the rewinder: web speed per Hz grows with the diameter, so
// counts per m/min = RwRatioD / diameter. RwRatioD (Retain) is adapted by rung B.

IF (RwRatioD > 0.0) AND (DecelDiam >= 100.0) THEN
    WebRatio := RwRatioD / DecelDiam;
ELSE
    WebRatio := 0.0;
END_IF;

// Fallback only (FFEnable = FALSE): old integrating loop (rung 16) with the coarse
// ramps in real web mm/s2. With FFEnable = TRUE, rung B drives the reference.
// V7b: never in jog -> jog keeps the rung 13 ramps exactly as v6y. V10: JogRun (jog really running).

IF NOT FFEnable AND NOT JogRun AND StartMotor_Cmd AND (UnderSpeed OR OverSpeed)
   AND (WebRatio > 0.0) THEN

    IF DecelWeb > 0.0 THEN
        Decel_Ramp := DecelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
        IF Decel_Ramp > 8.0 THEN
            Decel_Ramp := 8.0;
        END_IF;
    END_IF;

    IF AccelWeb > 0.0 THEN
        Accel_Ramp := AccelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;

        IF Accel_Ramp > 8.0 THEN
            Accel_Ramp := 8.0;
        END_IF;
    END_IF;
END_IF;
```

15 Soft Stop - SoftStopDecel (mm/s2), same for every roll.

```
// v6r: soft stop at a REAL web decel SoftStopDecel (mm/s2), same for every roll.
// Ratio drive counts / web m/min frozen at the soft stop start (encoder = exact).

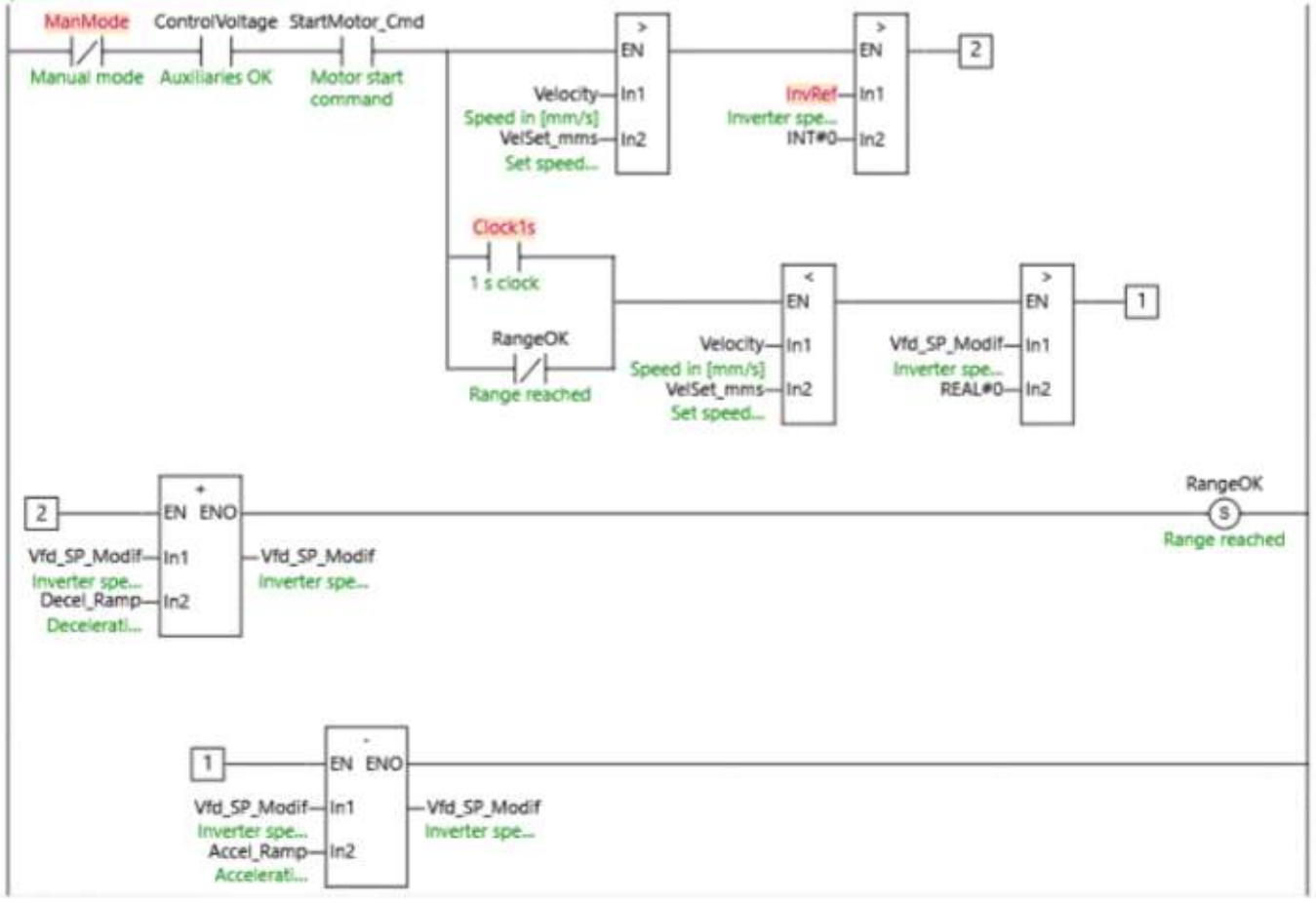
IF SoftStop AND NOT SoftStopMem THEN
    IF Velocity_M_min > 1.0 THEN
        SoftStopRatio := INT_TO_REAL(InvRef) / Velocity_M_min;
    ELSE
        SoftStopRatio := 0.0;
    END_IF;
END_IF;

SoftStopMem := SoftStop;

IF SoftStop AND (SoftStopDecel > 0.0) THEN
    IF SoftStopRatio > 0.0 THEN
        Decel_Ramp := SoftStopDecel * 0.06 * SoftStopRatio * Vel_Max_mms /
1000000.0;
    ELSE
        Decel_Ramp := SoftStopDecel / 250.0;          // fallback (old v6m behaviour)
    END_IF;

    IF Decel_Ramp > 8.0 THEN
        Decel_Ramp := 8.0;                        // max 9 Hz/s asked from the PLC
    END_IF;
END_IF;
```

16 Speed correction factor calculation.



17 Slow adaptation of RwRatioD on the relative speed error.

```
// ==== V12: REPLACE the code of Motor_Manager rung 17 (printed V9) ("Slow adaptation of RwRatioD on the
// ==== relative speed error." = V7 rung B) with this file. (V7a + V7b + V7f + V9 + V10 + V12 inside.)
// The PLC sends the drive STRAIGHT to the frequency of the set speed
// (VelSet x WebRatio), along a ramp in REAL web mm/s2 that does not depend on the
// measured speed -> no hunting, whatever the roll inertia / web delay.
// The measured speed only adapts RwRatioD slowly (~10 s), once at target.
// It overwrites Vfd_SP_Modif after rung 16, so rung 16 has no effect in AUTO.
// FFEnable = FALSE -> old loop (rung 16) exactly as before.
// V7b: not in jog (V10: JogRun = jog really running): at 10 m/min the feed-forward gives
// only ~0.4 Hz, too little to move the web -> the old loop (rung 16) runs the jog.

FFActive := FFEnable AND NOT ManMode AND NOT JogRun AND StartMotor_Cmd
        AND ControlVoltage AND (WebRatio > 0.0) AND (Vel_Max_mms > 0.0);

IF NOT StartMotor_Cmd THEN
   RampRef := 0.0;
   FFRate := 0.0;
   FFSteadyTicks := 0;
   FFSlowAcc := FALSE;
END_IF;

// V9: START during a soft stop: the decel flattens and the speed is HELD
// (SpdHold, HMI lamp) until START is pushed AGAIN -> gentle ramp up at
// SoftCancelAccel (mm/s2) to the set speed. The hold only blocks speeding up:
// a lower target (pot down, auto slow-down, RwAutoMaxSpd) still slows down.
// STOP during the hold = new soft stop (General rung 9).

IF (SoftCancelAccel > 0.0) AND (SoftCancelAccel < 20.0) THEN
   SoftCancelAccel := 20.0;
ELSIF SoftCancelAccel > 1000.0 THEN
   SoftCancelAccel := 1000.0;
END_IF;

FFStartEdge := I_Start_PB AND NOT FFStartMem;
FFStartMem := I_Start_PB;

IF FFSoftMem AND NOT SoftStop AND StartMotor_Cmd AND FFActive THEN   // soft stop cancelled
   SpdHold := TRUE;
   FFSlowAcc := FALSE;
ELSIF SpdHold AND FFStartEdge THEN                    // 2nd START: resume
   SpdHold := FALSE;
   FFSlowAcc := TRUE;
END_IF;

IF SoftStop OR NOT StartMotor_Cmd OR NOT FFActive THEN    // FFEnable OFF: old loop
   SpdHold := FALSE;
   FFSlowAcc := FALSE;
END_IF;

FFSoftMem := SoftStop;

// V7f: S-curve time (s). 0 = off, max 2 s
```

```
53
54  IF SCurveTime < 0.0 THEN
55      SCurveTime := 0.0;
56  ELSIF SCurveTime > 2.0 THEN
57      SCurveTime := 2.0;
58  END_IF;
59
60  IF FFActive THEN
61      IF NOT FFWasActive THEN
62          RampRef := Vel_Max_mms - Vfd_SP_Modif;          // bumpless start / switch-
    over
63          FFRate := 0.0;
64      END_IF;
65
66      // Target reference (mm/s of the 0..10 V scale) for the set speed
67      FFTarget := VelSet * WebRatio * Output_Ratio;
68
69      IF FFTarget > Vel_Max_mms THEN
70          FFTarget := Vel_Max_mms;
71      ELSIF FFTarget < 0.0 THEN
72          FFTarget := 0.0;
73      END_IF;
74
75      // Tension: hold at TensWaitSpd until the T-one reports tension
    (I_T_one_TensionOK)
76
77  //    IF TensWait AND (VelSet > TensWaitSpd) THEN
78  //        FFTarget := TensWaitSpd * WebRatio * Output_Ratio;
79  //    END_IF;
80
81      // Tension: hold at a minimum DRIVE FREQUENCY until the T-one reports tension
82      // (independent of the rewinder diameter). TensWaitHz = 0 -> old m/min hold.
83
84      IF TensWait THEN
85
86          IF TensWaitHz > REAL#0.0 THEN
87              TensRef := TensWaitHz * REAL#4000.0 / REAL#60.0 * Output_Ratio;
88          ELSE
89              TensRef := TensWaitSpd * WebRatio * Output_Ratio;
90          END_IF;
91
92          IF FFTarget > TensRef THEN
93              FFTarget := TensRef;
94          END_IF;
95      END_IF;
96
97
98      // V9: hold after a cancelled soft stop (the S-curve flattens it smoothly):
99      // never above the present reference, lower targets still allowed
100
101     IF SpdHold THEN
102         IF FFTarget > RampRef THEN
103             FFTarget := RampRef;
104         END_IF;
105
106     ELSIF FFSlowAcc AND (ABS(FFTarget - RampRef) <= FFTarget * 0.005 + 2.0) THEN
107         FFSlowAcc := FALSE;                          // back at the set speed
108     END_IF;
109
```

```
110     // Ramp steps per 4 ms scan: web mm/s2 -> reference mm/s per scan
111
112     IF FFSlowAcc AND (SoftCancelAccel > 0.0) THEN
113         StepAcc := SoftCancelAccel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;  //
    V9
114     ELSIF AccelWeb > 0.0 THEN
115         StepAcc := AccelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
116     ELSE
117         StepAcc := AccelCoarse;
118     END_IF;
119
120     IF SoftStop AND FinalStopRamp AND (FinalStopDecel > 0.0) THEN
121         StepDec := FinalStopDecel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;  //
    V12 end-of-roll
122     ELSIF SoftStop AND (SoftStopDecel > 0.0) THEN
123         StepDec := SoftStopDecel * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
124     ELSIF DecelWeb > 0.0 THEN
125         StepDec := DecelWeb * 0.06 * WebRatio * Vel_Max_mms / 1000000.0;
126     ELSE
127         StepDec := DecelCoarse;
128     END_IF;
129
130     IF StepAcc > 8.0 THEN
131         StepAcc := 8.0;
132     END_IF;
133
134     IF StepDec > 8.0 THEN
135         StepDec := 8.0;
136     END_IF;
137
138     // Ramp generator
139     // V7f: S-curve. FFRate (reference per scan) changes at most FFJerk per scan,
140     // so the full ramp is reached after SCurveTime (s) and is rounded off the
141     // same way before the target -> the T-one / brake air can follow.
142     // SCurveTime = 0 -> plain linear ramp as V7a.
143
144     IF SCurveTime > 0.0 THEN
145         IF (FFRate > 0.0) OR ((FFRate = 0.0) AND (FFTarget > RampRef)) THEN
146             FFJerk := StepAcc * 0.004 / SCurveTime;
147         ELSE
148             FFJerk := StepDec * 0.004 / SCurveTime;
149         END_IF;
150
151         // rate allowed so that it can still fall to 0 at the target
152         // (no MIN(), argument in a REAL variable first: Sysmac data type rules,
153         //  same pattern as SqrtArg / SQRT(SqrtArg) in AutoStop)
154
155         IF FFTarget >= RampRef THEN
156             FFSqrtArg := REAL#2.0 * FFJerk * (FFTarget - RampRef);
157             FFRateSet := SQRT(FFSqrtArg);
158
159             IF FFRateSet > StepAcc THEN
160                 FFRateSet := StepAcc;
161             END_IF;
162         ELSE
163             FFSqrtArg := REAL#2.0 * FFJerk * (RampRef - FFTarget);
164             FFRateSet := SQRT(FFSqrtArg);
165
166             IF FFRateSet > StepDec THEN
```

```
            FFRateSet := StepDec;
          END_IF;

          FFRateSet := -FFRateSet;
        END_IF;

        IF FFRateSet > FFRate + FFJerk THEN
          FFRate := FFRate + FFJerk;
        ELSIF FFRateSet < FFRate - FFJerk THEN
          FFRate := FFRate - FFJerk;
        ELSE
          FFRate := FFRateSet;
        END_IF;

        RampRef := RampRef + FFRate;

        IF (ABS(FFTarget - RampRef) <= 3.0) AND (ABS(FFRate) <= 2.0 * FFJerk) THEN
          RampRef := FFTarget;                    // at target (pot noise)
          FFRate := 0.0;
        END_IF;
      ELSE
        FFRate := 0.0;

        IF RampRef < FFTarget - StepAcc THEN
          RampRef := RampRef + StepAcc;
        ELSIF RampRef > FFTarget + StepDec THEN
          RampRef := RampRef - StepDec;
        ELSE
          RampRef := FFTarget;                    // at target
        END_IF;
      END_IF;

      // V7a: "at target" = within 0.5 % of the target (the pot and RwDiam noise
      // move FFTarget by a few counts every scan -> the counter never reached 500)

      IF (ABS(FFTarget - RampRef) <= FFTarget * 0.005 + 2.0)
        AND NOT SpdHold AND NOT FFSlowAcc AND NOT TensWait THEN          // V9:
    not during the hold

        IF FFSteadyTicks < 30000 THEN
          FFSteadyTicks := FFSteadyTicks + 1;
        END_IF;
      ELSE
        FFSteadyTicks := 0;
      END_IF;

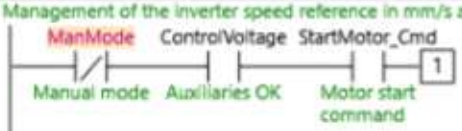
      // Slow adaptation of RwRatioD on the relative speed error (2 s after the
      // target is reached, not at the 60 Hz limit, not during soft stop / jog)

      IF (FFSteadyTicks >= 500) AND NOT SoftStop AND NOT JogRun
        AND (VelSet > 15.0) AND (Velocity_M_min > 10.0) AND NOT RwSensFault
        AND (FFTarget < Vel_Max_mms * 0.98) THEN
        FFErr := (VelSet - Velocity_M_min) / VelSet;

        IF FFErr > 0.05 THEN
          FFErr := 0.05;
        ELSIF FFErr < -0.05 THEN
          FFErr := -0.05;
```

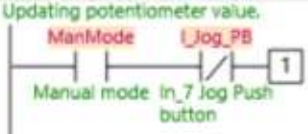
```
224       END_IF;
225
226       IF ABS(FFErr) > 0.003 THEN                  // dead band 0.3 % (encoder noise)
227         RwRatioD := RwRatioD * (1.0 + 0.0004 * FFErr);  // ~10 s time constant
228       END_IF;
229
230       IF RwRatioD < 1000.0 THEN
231         RwRatioD := 1000.0;
232       ELSIF RwRatioD > 8000.0 THEN
233         RwRatioD := 8000.0;
234       END_IF;
235     END_IF;
236
237     Vfd_SP_Modif := Vel_Max_mms - RampRef;              // rung 17 -> InvRef
238 END_IF;
239
240 FFWasActive := FFActive;
241
242 // V7f: T-one decel signal (for a T-one "stop / boost" input, spare PLC output):
243 // ON while the feed-forward ramps down (more than 1 % above the target, so the
244 // slow drift of the target with the diameter does not pulse it) or soft stop.
245
246 TOneDecel := (FFActive AND (RampRef - FFTarget > FFTarget * 0.01 + 5.0))
247           OR (SoftStop AND StartMotor_Cmd);
```

18 Management of the inverter speed reference in mm/s and conversion to analog points (0–4000 points = 0–10 VDC = 0–10,000 mm/s).

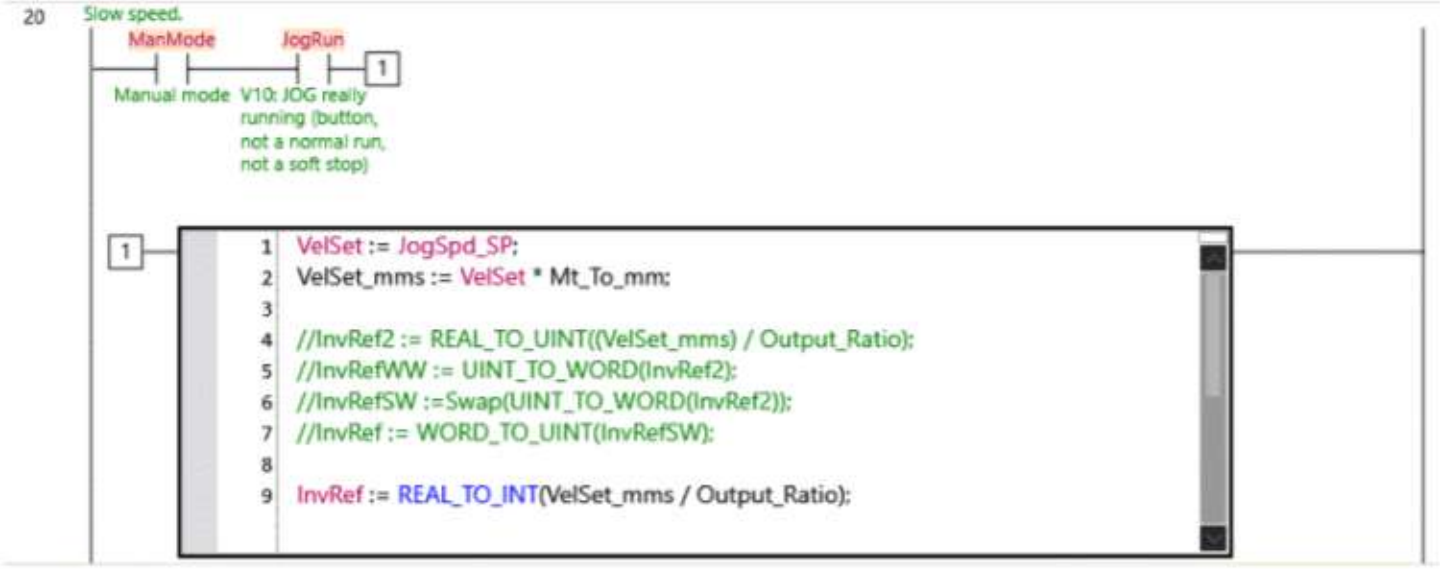



```
1  //InvRef2 := REAL_TO_UINT((Vel_Max_mms - Vfd_SP_Modif ) / Output_Ratio);
2  //InvRefWW := UINT_TO_WORD(InvRef2);
3  //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
4  //InvRef := WORD_TO_UINT(InvRefSW);
5
6  InvRef := REAL_TO_INT((Vel_Max_mms - Vfd_SP_Modif ) / Output_Ratio);
```

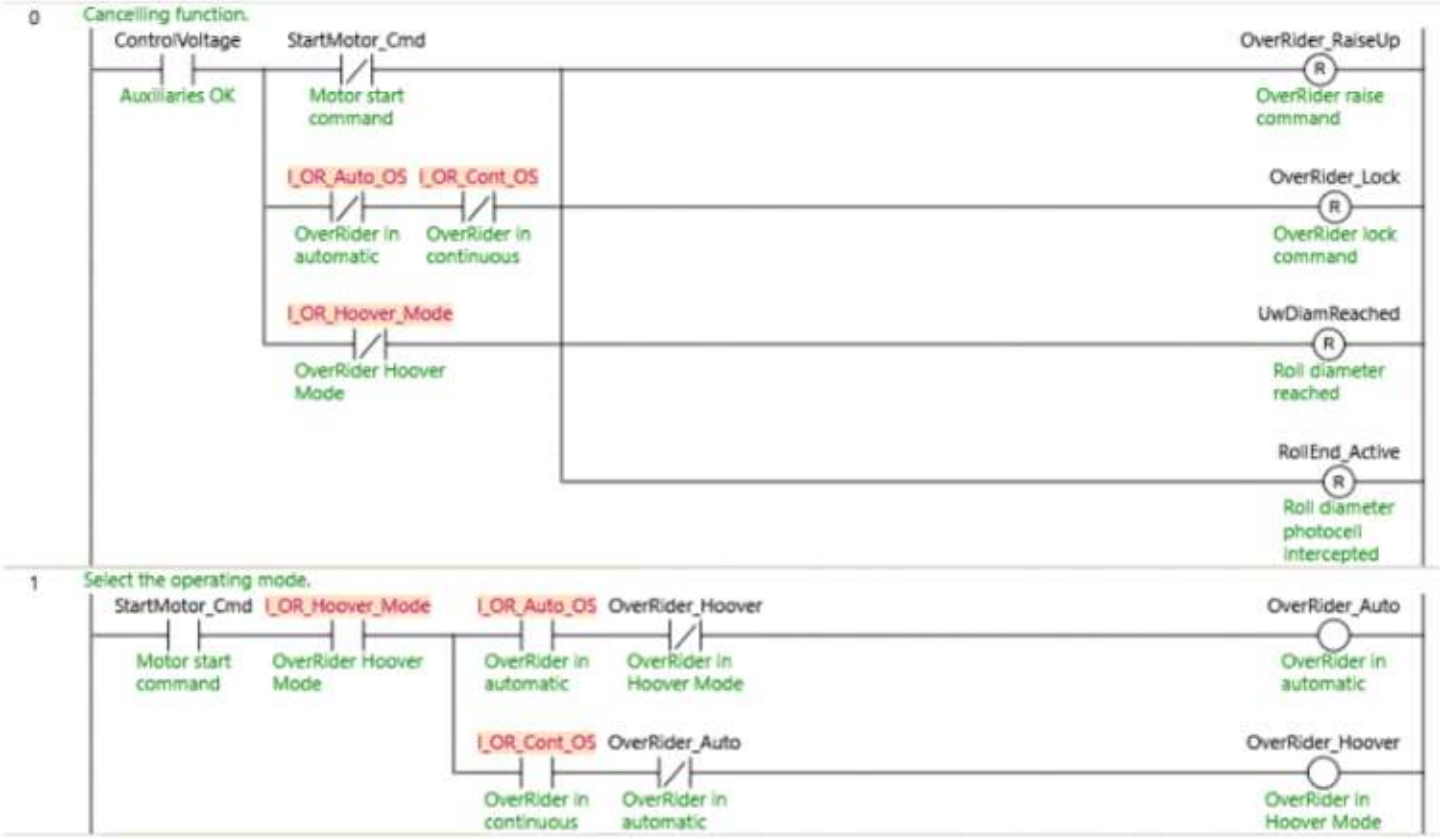
19 Updating potentiometer value.

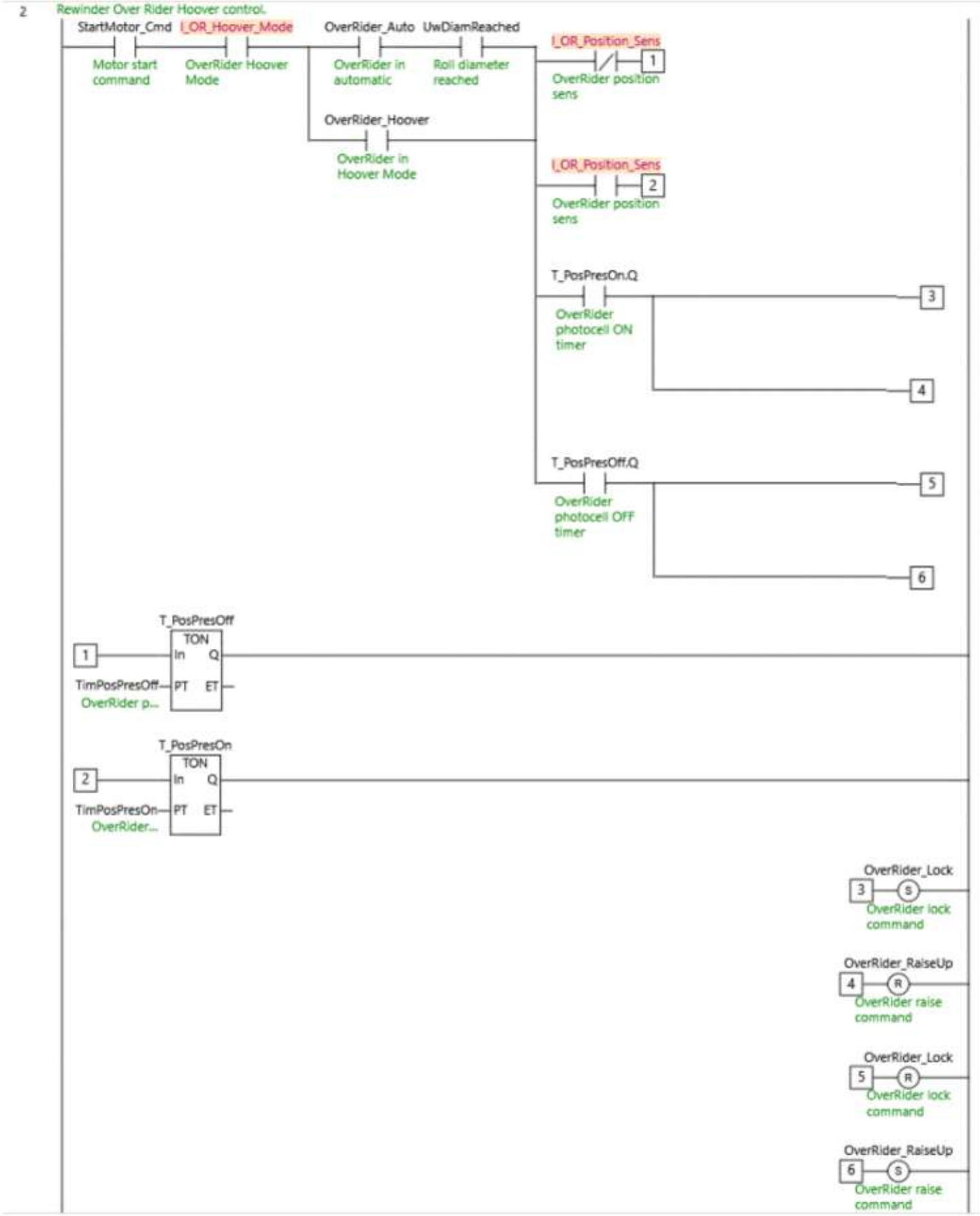



```
1  IF SoftStop THEN
2    InvRef := INT#0;                    // v6m: drive ramps down on its own
3  ELSIF SlowDown = FALSE THEN
4    InvRef := Pot1_Work_Velocity;
5  ELSE
6    InvRef := Pot2_Slow_Velocity;
7  END_IF;
```


20
Slow speed.
ManMode
JogRun
1
Manual mode
V10: JOG really running (button, not a normal run, not a soft stop)
1
1 VelSet := JogSpd_SP;
2 VelSet_mms := VelSet * Mt_To_mm;
3
4 //InvRef2 := REAL_TO_UINT((VelSet_mms) / Output_Ratio);
5 //InvRefWW := UINT_TO_WORD(InvRef2);
6 //InvRefSW :=Swap(UINT_TO_WORD(InvRef2));
7 //InvRef := WORD_TO_UINT(InvRefSW);
8
9 InvRef := REAL_TO_INT(VelSet_mms / Output_Ratio);

1-8-1-1-4.OverRider_Manager

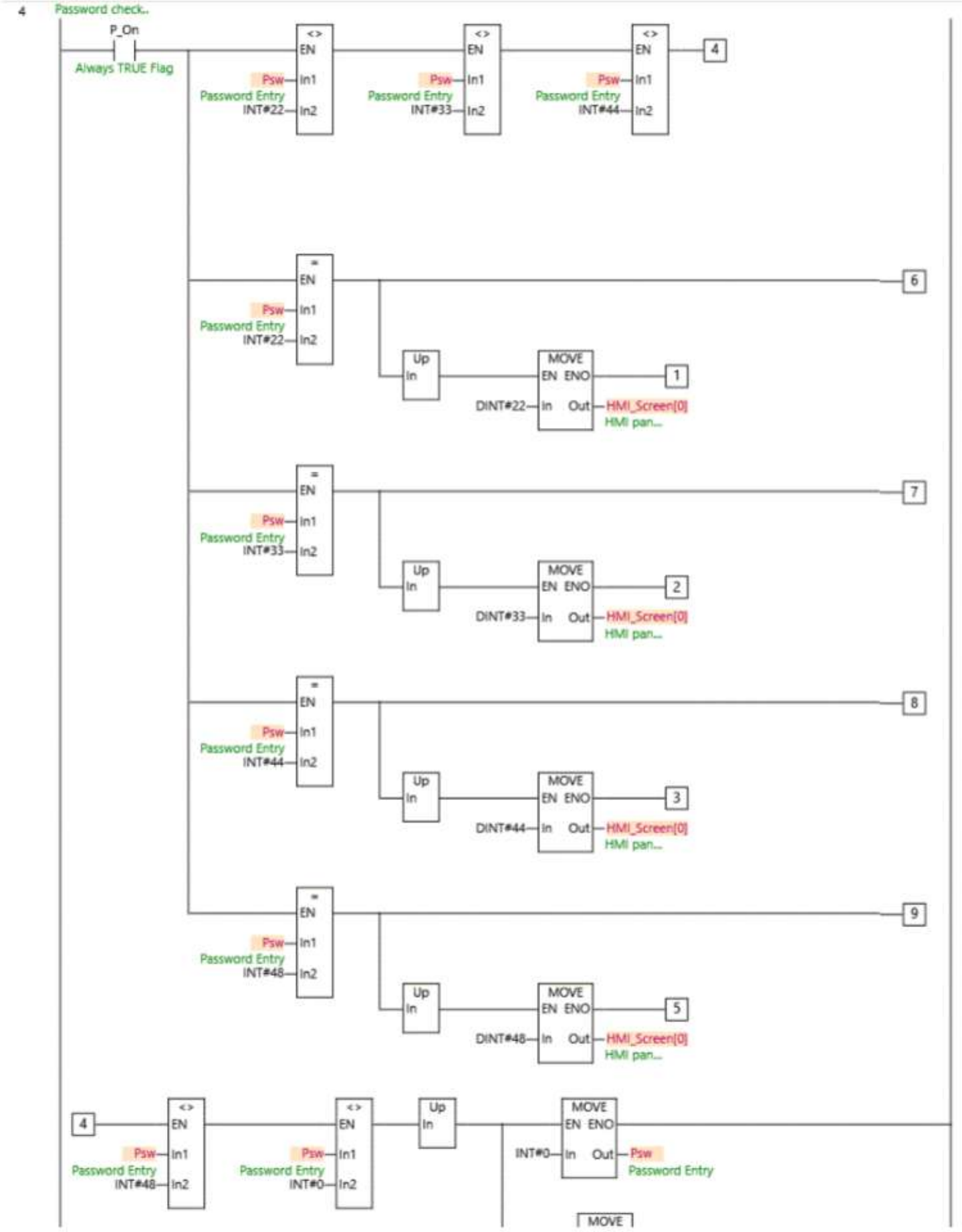


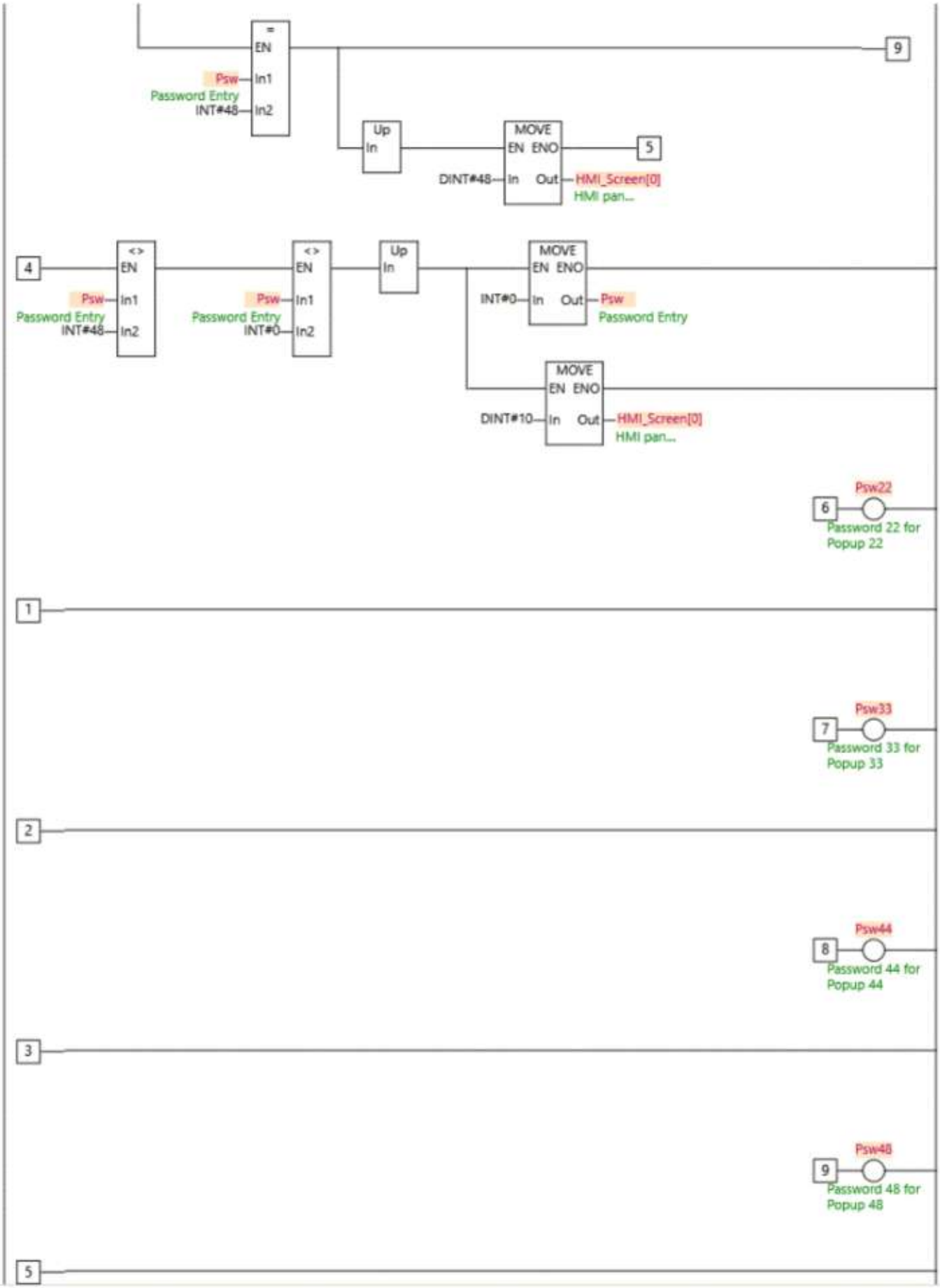

2
Rewinder Over Rider Hoover control.
StartMotor_Cmd
Motor start command
I_OR_Hoover_Mode
OverRider Hoover Mode
OverRider_Auto
OverRider in automatic
UwDiamReached
Roll diameter reached
I_OR_Position_Sens
1
OverRider position sens
OverRider_Hoover
OverRider in Hoover Mode
I_OR_Position_Sens
2
OverRider position sens
T_PosPresOn.Q
OverRider photocell ON timer
3
4
T_PosPresOff.Q
OverRider photocell OFF timer
5
6
T_PosPresOff
TON
In
Q
PT
ET
TimPosPresOff
OverRider p...
T_PosPresOn
TON
In
Q
PT
ET
TimPosPresOn
OverRider...
OverRider_Lock
S
OverRider lock command
OverRider_RaiseUp
R
OverRider raise command
OverRider_Lock
R
OverRider lock command
OverRider_RaiseUp
S
OverRider raise command

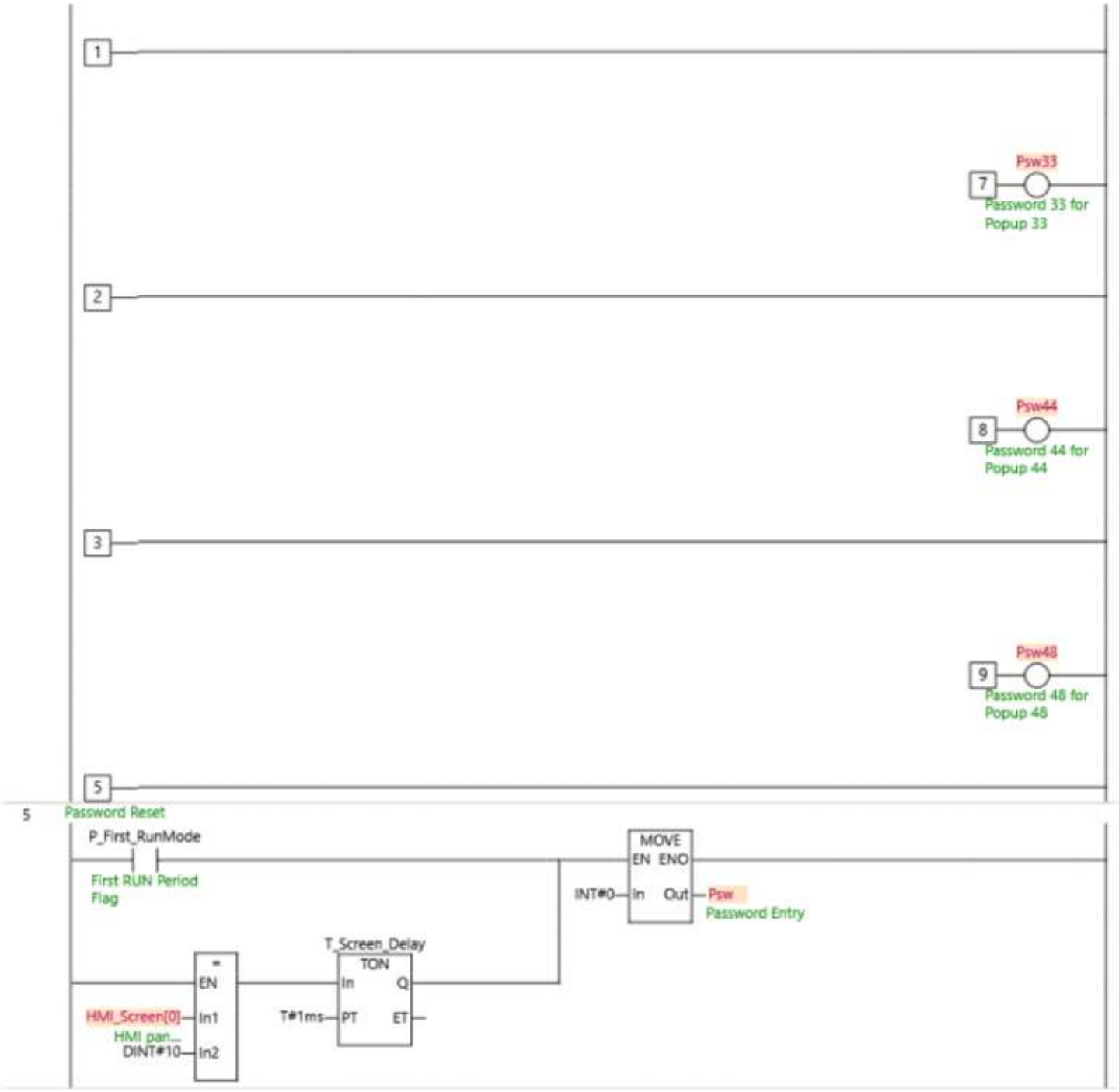

3
Rider roll raise command.
ControlVoltage
Auxiliaries OK
OverRider_RaiseUp
S
OverRider raise command
4
I_ControlVoltage
In_0
ControlVoltageO
N
OverRider_RaiseUp
R
OverRider raise command

1-8-1-1-5.Console

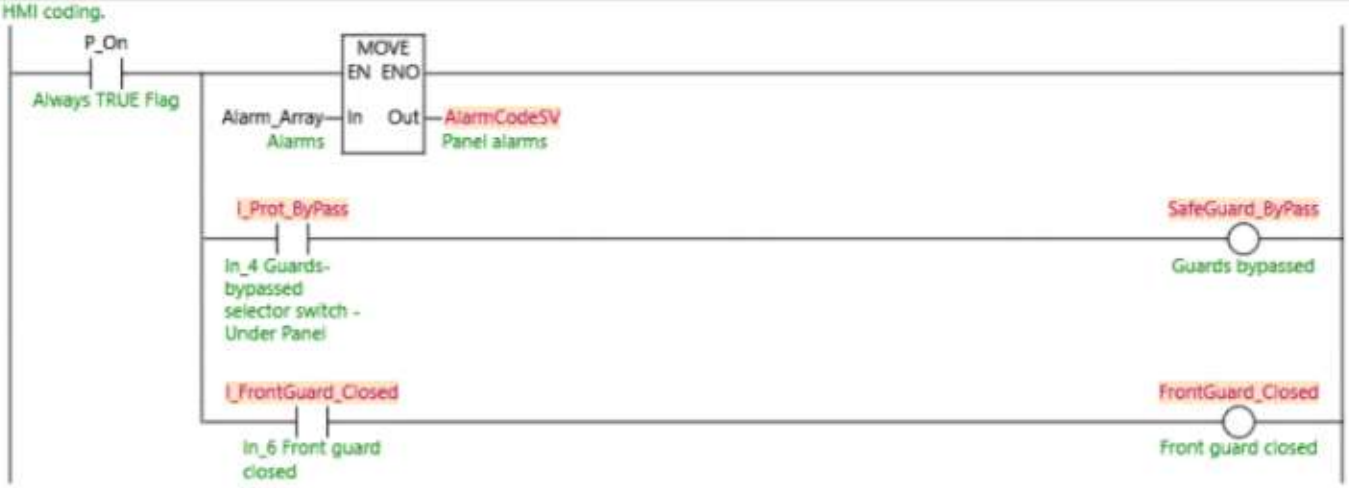



4
Password check..
P_On
Always TRUE Flag
<>
EN
In1
In2
Psw
Password Entry
INT#22
INT#33
INT#44
4
=
6
7
8
9
Up
In
MOVE
EN ENO
In Out
DINT#22
DINT#33
DINT#44
DINT#48
HMI_Screen[0]
HMI pan...
1
2
3
5
INT#48
INT#0
MOVE


=
EN
Psw
Password Entry
INT#48
In1
In2
9
Up
In
MOVE
EN ENO
5
DINT#48
In
Out
HMI_Screen[0]
HMI pan...
4
<>
EN
Psw
Password Entry
INT#48
In1
In2
<>
EN
Psw
Password Entry
INT#0
In1
In2
Up
In
MOVE
EN ENO
INT#0
In
Out
Psw
Password Entry
MOVE
EN ENO
DINT#10
In
Out
HMI_Screen[0]
HMI pan...
Psw22
6
Password 22 for
Popup 22
1
Psw33
7
Password 33 for
Popup 33
2
Psw44
8
Password 44 for
Popup 44
3
Psw48
9
Password 48 for
Popup 48
5


1
Psw33
7
Password 33 for Popup 33
2
Psw44
8
Password 44 for Popup 44
3
Psw48
9
Password 48 for Popup 48
5
5 Password Reset
P_First_RunMode
First RUN Period Flag
MOVE
EN ENO
INT#0 In Out Psw
Password Entry
T_Screen_Delay
TON
In Q
T#1ms PT ET
=
EN
HMI_Screen[0] In1
HMI pan...
DINT#10 In2

6 HMI coding.


P_On
Always TRUE Flag
MOVE
EN ENO
Alarm_Array
Alarms
In Out
AlarmCodeSV
Panel alarms
I_Prot_ByPass
In_4 Guards-bypassed selector switch - Under Panel
SafeGuard_ByPass
Guards bypassed
I_FrontGuard_Closed
In_6 Front guard closed
FrontGuard_Closed
Front guard closed

1-8-1-1-6.Alarms

0 Check Present alarms.

```
// Alarm / warning handling + panel buzzer (BuzzerOn).
// Alarms: buzzer pulses (Clock1s). ALARM RESET (panel or HMI) silences it 1 min;
//         a NEW alarm during the silence sounds again at once.
// Warnings: ONE 1 s beep each time a new warning appears.
// PrimaryTask 4 ms: 250 scans = 1 s, 15000 scans = 60 s.

AlarmOn := FALSE;
WarningOn := FALSE;
AlarmCnt := 0;
WarnCnt := 0;

FOR Index_A := 0 TO 32 DO

   IF Alarm_Array[Index_A] THEN

      CASE Index_A OF
         5, 13, 15, 16, 17, 18, 19, 20, 21:        // warnings
            WarningOn := TRUE;
            WarnCnt := WarnCnt + 1;

            IF NOT AlarmMem[Index_A] THEN
               WarnBeepTicks := 250;              // NEW warning -> 1 s beep
            END_IF;
      ELSE                                        // alarms
         AlarmOn := TRUE;
         AlarmCnt := AlarmCnt + 1;
      END_CASE;

   END_IF;
   AlarmMem[Index_A] := Alarm_Array[Index_A];
END_FOR;

// ALARM RESET button (panel I_RsetAlarm or HMI Alarm_Reset): silence 1 min

RstEdge := (I_RsetAlarm OR Alarm_Reset) AND NOT RstMem;
RstMem := I_RsetAlarm OR Alarm_Reset;

IF RstEdge AND AlarmOn THEN
   SilenceTicks := 15000;
END_IF;

IF (AlarmCnt > AlarmCntMem) OR NOT AlarmOn THEN
   SilenceTicks := 0;                             // new alarm / no alarm: not silenced
END_IF;

IF SilenceTicks > 0 THEN
   SilenceTicks := SilenceTicks - 1;
END_IF;

IF WarnBeepTicks > 0 THEN
   WarnBeepTicks := WarnBeepTicks - 1;
END_IF;

AlarmCntMem := AlarmCnt;
WarnCntMem := WarnCnt;

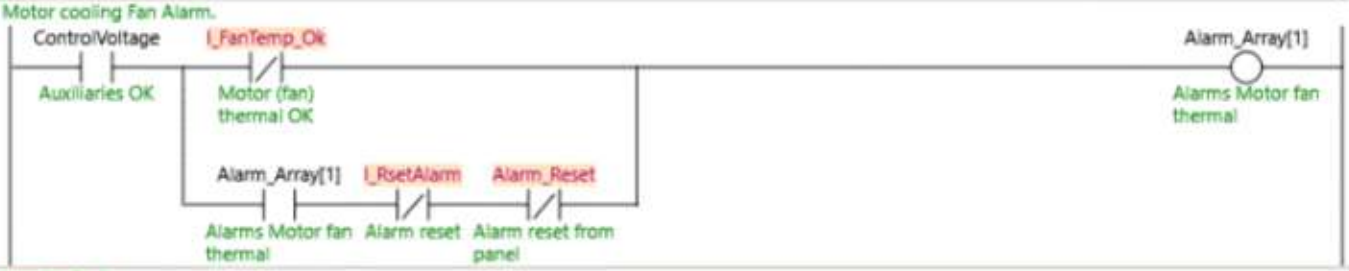
```

```
57  AlarmSilenced := AlarmOn AND (SilenceTicks > 0);
58
59  BuzzerOn := ControlVoltage
60          AND ((AlarmOn AND NOT AlarmSilenced AND Clock1s) OR (WarnBeepTicks
    > 0));
```

1

```
1   // ---- Alarm 14 (ALARM): unwinder sonar (NX-AD2603 Ch1, P+F UC2000) fault ----
2   // Raw value out of range for 1 s (no echo / broken wire / sensor off).
3   // Set when the measurement is used (unwinder AutoStop ON or motor running).
4   // latched, RESET only once the sensor is OK again (same as alarm 10, DT35).
5
6   T_UwSensFlt(In := (AI_Diam_Unwinder < UwRawMin) OR (AI_Diam_Unwinder >
    UwRawMax), PT := T#1s);
7   UwSensTest := T_UwSensFlt.Q;
8
9   IF ControlVoltage AND Measure_ON AND UwSensTest AND (UwStopEnable OR
    StartMotor_Cmd) THEN
10      UwSensFault := TRUE;
11  ELSIF NOT UwSensTest AND (I_RsetAlarm OR Alarm_Reset) THEN
12      UwSensFault := FALSE;
13  END_IF;
14
15  // ---- Alarm 17 (WARNING): Paper thickness not learned ----
16  // RwMat_K = 0 -> diameter / stop-length targets use the fixed fallback.
17  // Only above 300 mm rewinder diameter (learning needs ~20 mm growth after
    NEW PAPER),
18  // so it does not beep at every new roll. Not latched.
19
20  Thickness_Fault := ControlVoltage AND StartMotor_Cmd
21              AND (RwMat_K <= 0.0) AND (RwDiam > 300.0);
```

2 Motor cooling Fan Alarm.



3 Inverter fault.



4 Machine in emergency mode.



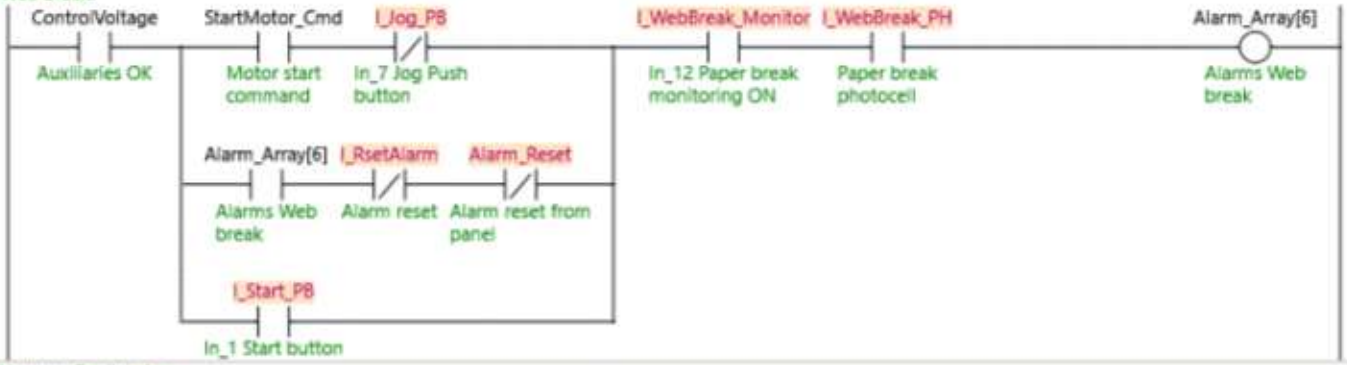
5 Main Motor thermal cutout.



6 Front Guard - Warning



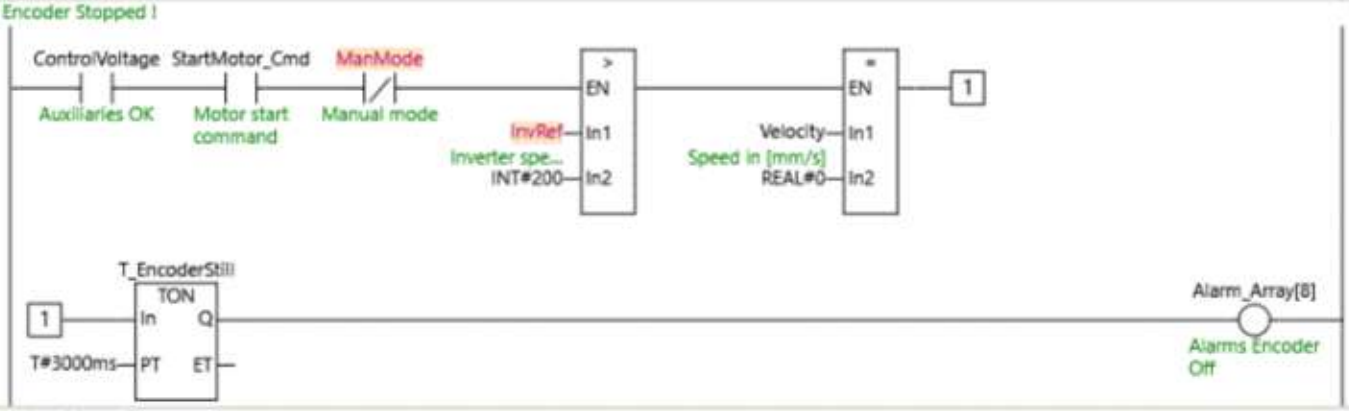
7 Web Break.



8 Vacuum Cooling fan.



9 Encoder Stopped !



10 Hub not in position.



11 Rewinder Diameter Sensor Fault.



12 Rewinder Diameter Sensor Fault - Reset



13 Rewinder max diameter Machine protection.



14 Rewinder max diameter Machine protection - Reset



15 Emergency Stop by Holding Stop PB 2s



16 Web Tension Monitor - Warning



17 Unwinder Diameter Sensor Fault.



18 Unwinder Roll is Empty - Warning



19 Unwinder Roll is Empty - Reset



20 Rewinder GSM learning fault



21 Rewinder learning fault - Reset



22 Paper Thickness Learn Process - Not Done ! - Warning



23 Side Door Opened - Warning



24 Side Door Opened - Warning - Reset



25 Front Gate Not Closed ! - Warning



26 Front Gate Not Closed ! - Warning - Reset



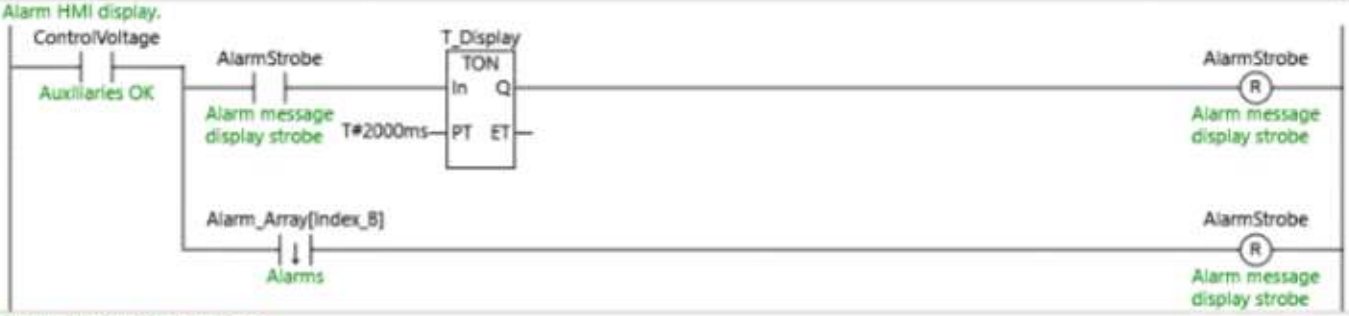
27 AutoStop Done ! - Warning



28 AutoStop Set Enable Monitor - Warning



29 Alarm HMI display.



30 Strobe displays alarm message.



31 Alarm presence check cycle.

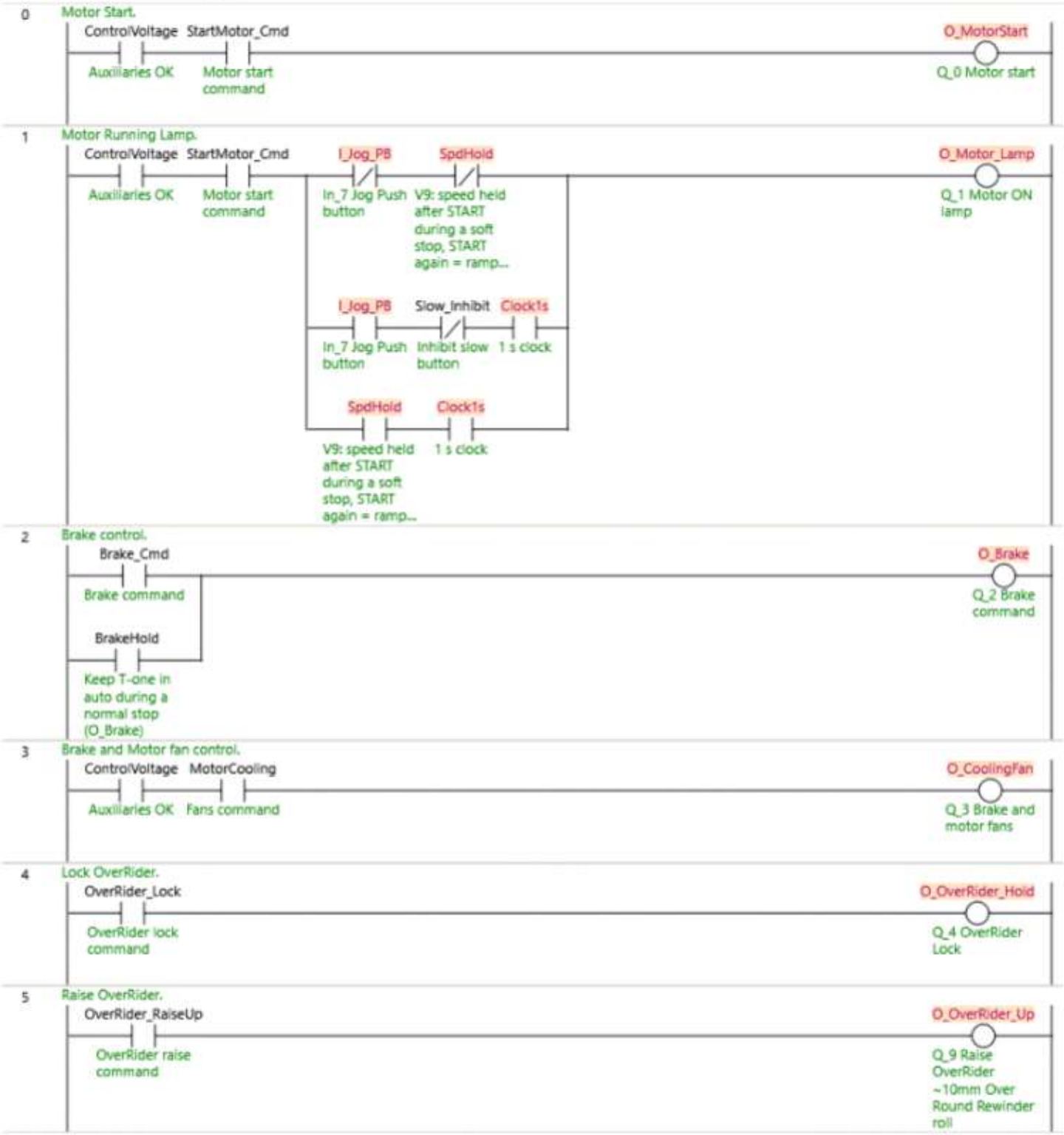


32 Initializing alarm message.




33
AlarmOn
WarningOn
Alarm active
HMI Warning Display
MOVE
EN ENO
INT#33
In
Out
Msg
Message...

1-8-1-1-7.Output_Manager



6 Rider roll auto/continuous lamp.



7 Alarm lamp.



8 Guards bypassed lamp.



9 Unloaders enable - opening.



10 Unloaders enable - closing.



11 Horn / acoustic signal.

```
// ==== V9: REPLACE Output_Manager rung 11 ("Horn / acoustic signal.", ladder) by ONE ST rung
// ==== with this code (no contact in front). Comment: "V9: horn, beeps set on the HMI".
// Old: horn blinking with Clock1s (0.5 s ON / 0.5 s OFF) during the whole slow-down
//      and the crawl at final speed -> very annoying.
// New, slow-down / final speed (StartMotor_Cmd AND SlowDown AND NOT Stop):
//      HornStartBeep_s beep when it starts (1 s).
//      then a HornBeepLen_s beep (0.2 s) every HornBeepPeriod (5 s).
//      then ONE HornStopBeep_s beep (3 s) when the machine has stopped (< 2 m/min).
//      Each setting 0 = that beep OFF. Front guard moving: unchanged (Clock1s blinking).
// PrimaryTask 4 ms: 250 scans = 1 s.

// HMI limits

IF HornStartBeep_s < REAL#0 THEN
    HornStartBeep_s := REAL#0;
ELSIF HornStartBeep_s > REAL#5 THEN
    HornStartBeep_s := REAL#5;
END_IF;

IF HornBeepLen_s < REAL#0 THEN
    HornBeepLen_s := REAL#0;
ELSIF HornBeepLen_s > REAL#2 THEN
    HornBeepLen_s := REAL#2;
END_IF;

IF HornBeepPeriod < REAL#0 THEN
    HornBeepPeriod := REAL#0;
ELSIF (HornBeepPeriod > REAL#0) AND (HornBeepPeriod < HornBeepLen_s + REAL#0.5) THEN
    HornBeepPeriod := HornBeepLen_s + REAL#0.5;    // always a pause between beeps
ELSIF HornBeepPeriod > REAL#30 THEN
    HornBeepPeriod := REAL#30;
END_IF;

IF HornStopBeep_s < REAL#0 THEN
    HornStopBeep_s := REAL#0;
ELSIF HornStopBeep_s > REAL#10 THEN
    HornStopBeep_s := REAL#10;
END_IF;

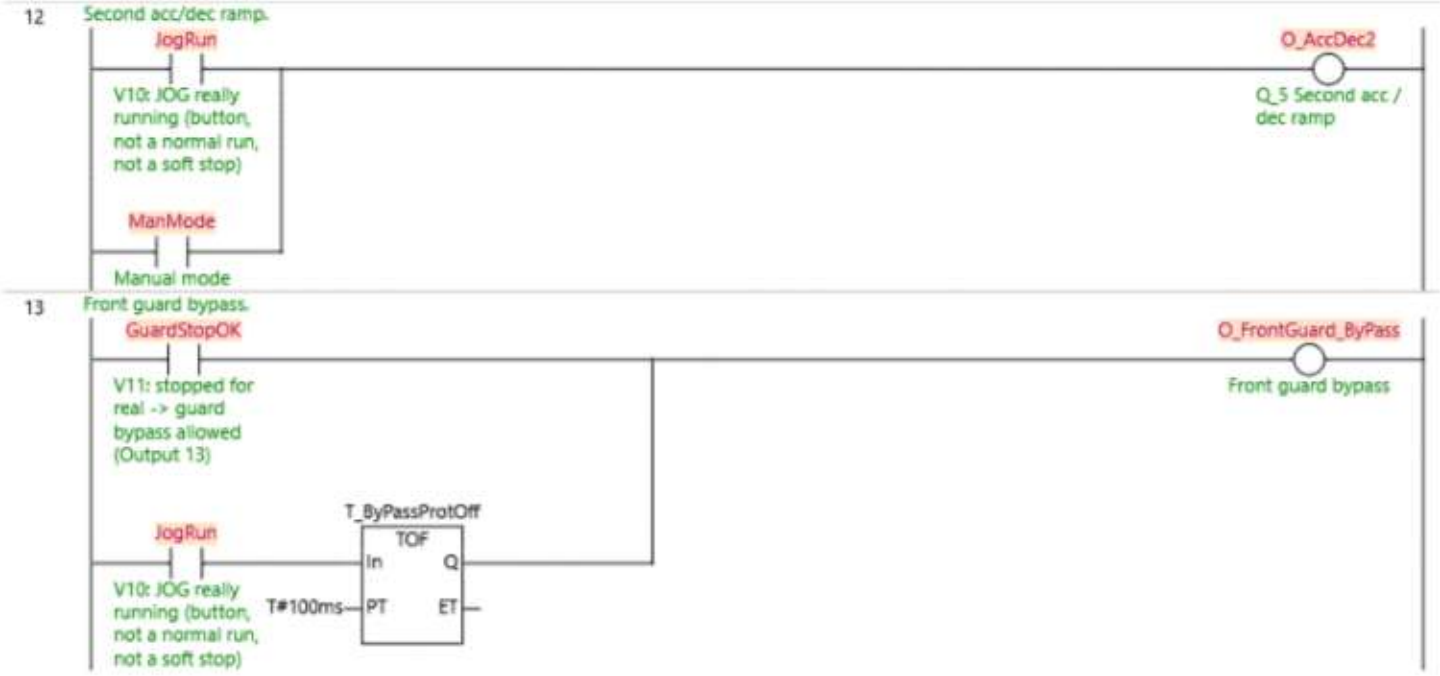
HornStartScans := REAL_TO_INT(HornStartBeep_s * REAL#250);
HornLenScans   := REAL_TO_INT(HornBeepLen_s * REAL#250);
HornPer        := REAL_TO_INT(HornBeepPeriod * REAL#250);

// Slow-down / final speed beeps

HornRun := StartMotor_Cmd AND SlowDown AND NOT Stop AND NOT JogRun;  // no horn while jogging

IF HornRun THEN
    IF NOT HornRunMem THEN
        HornTicks := 0;                    // slow-down starts
```

```
51      END_IF;
52
53      IF HornTicks < 30000 THEN
54          HornTicks := HornTicks + 1;
55      END_IF;
56
57      IF HornTicks <= HornStartScans THEN
58          HornSlowBeep := TRUE;                    // start beep
59          HornCnt := 0;
60      ELSIF (HornPer > 0) AND (HornLenScans > 0) THEN
61          HornCnt := HornCnt + 1;
62          IF HornCnt >= HornPer THEN
63              HornCnt := 0;
64          END_IF;
65
66          HornSlowBeep := (HornCnt >= HornPer - HornLenScans);   // beep at the end
    of each period
67      ELSE
68          HornSlowBeep := FALSE;                   // periodic beeps OFF
69      END_IF;
70  ELSE
71      HornTicks := 0;
72      HornCnt := 0;
73      HornSlowBeep := FALSE;
74  END_IF;
75
76  // End of a slow-down by a stop (auto stop, STOP button): one beep at standstill
77  // (Stop is set in Motor_Manager before StartMotor_Cmd drops on the next scan)
78
79  IF HornRunMem AND NOT HornRun AND (Stop OR NOT StartMotor_Cmd) THEN
80      HornStopPend := TRUE;
81  END_IF;
82
83  IF StartMotor_Cmd AND NOT HornStartMem THEN
84      HornStopPend := FALSE;                       // restarted: no stop beep
85      HornStopTicks := 0;
86  END_IF;
87
88  HornStartMem := StartMotor_Cmd;
89
90  IF HornStopPend AND (Velocity_M_min < REAL#2) THEN
91      HornStopPend := FALSE;
92      HornStopTicks := REAL_TO_INT(HornStopBeep_s * REAL#250);
93  END_IF;
94
95  IF HornStopTicks > 0 THEN
96      HornStopTicks := HornStopTicks - 1;
97  END_IF;
98
99  HornRunMem := HornRun;
100
101 // HMI test button: horn while pressed (machine stopped only)
102
103 O_AcousticSignal := (NOT I_FrontGuard_Closed AND NOT I_FrontGuard_Open AND
    Clock1s)
104                 OR HornSlowBeep OR (HornStopTicks > 0)
105                 OR (HornTest AND NOT StartMotor_Cmd);
```


12
Second acc/dec ramp.
JogRun
V10: JOG really running (button, not a normal run, not a soft stop)
ManMode
Manual mode
O_AccDec2
Q_5 Second acc / dec ramp
13
Front guard bypass.
GuardStopOK
V11: stopped for real -> guard bypass allowed (Output 13)
JogRun
V10: JOG really running (button, not a normal run, not a soft stop)
T_ByPassProtOff
TOF
In
Q
T#100ms
PT
ET
O_FrontGuard_ByPass
Front guard bypass

1-8-1-1-8.AutoStop_Control

0 Auto Stop Process 1-5

```
(*
================================================================
====================
  AutoStop_Control  V10  -  PART 1 of 2 (rung 0): sections 1 - 5c
  Project : 04820KA001 Mono Canada II  (NX1P2, PrimaryTask 4 ms)
  Rewinder (Rw) : SICK DT35 0-10 V -> AI_Diam_Rewinder (AD2603 Ch2)
  Unwinder (Uw) : meter counter mode + sonar diameter (UwDiam_Act)

  NAMING  Uw_ = unwinder   Rw_ = rewinder
        _mm = millimeters   _m = meters
        _Pls = encoder pulses (counts)   _Pps = pulses per second
        _Pps2 = pulses per second squared

  V10: JogRun (jog really running) instead of the raw jog button (5, 5b):
       long baseline also restarted if the encoder goes 2 m back (4b).
  V9 : RwDiamRev = DT35 averaged over exactly one revolution (2b), used by
       the thickness learning (removes the out-of-round wave), RwWave_mm.
     GSM popup: once per real new roll (empty 3 s, then > stop + 100 mm
       for 3 s); NEW PAPER closes it (section 10).
     paper thickness: long-baseline learning (D^2 growth / length) from the
       start point, LOCKED at RwLockDiam (300 mm): no more change, no
       paper-change detection, until NEW PAPER / GSM popup YES (4b).
  V8 : = V7 + V7c unwinder DIAMETER slow-down follows the speed (6c).
       V7e ManMode slow-down model as before V7 (section 6),
       V7f S-curve extra length in the slow-down (section 6)
  V7 : FEED-FORWARD speed loop (Motor_Manager rung 8): reference = set speed
     x RwRatioD / diameter along a ramp in real web mm/s2 (DecelWeb /
     AccelWeb), RwRatioD adapted slowly -> no hunting on big rolls.
     Section 6: slow-down length = one ramp at DecelWeb (FFEnable), else
     web coarse ramp + trim tail; learning only as fallback.
  v6y: rewinder AUTO stop (RwStopEnable) limits the work speed to
     RwAutoMaxSpd (500 m/min), taken over at standstill only; section 6
     uses the limited speed.
  v6x: new parent roll on the unwinder -> GsmAsk (HMI popup "did you change
     the paper?"); GsmAskYes forgets the learned thickness.
  v6w: paper thickness window is LENGTH-based instead of 10 s, modulated by
     the diameter (RwWinGrow_mm 2.5 mm of growth, 4 m .. RwWinLen_m 30 m):
     same accuracy at every speed and roll size; stopped window = dropped.
     ALARM RESET also clears LearnFailCnt / StopFailCnt (alarm 16).
  v6v: BrakeHold released on the SIGNED web speed (Velocity_M_min < 5),
     so a web pulled back after the stop releases the T-one at once.
  v6u: stop learning: a stop shorter than the reaction distance or faster
     than RwDecelMax_mm is ignored (not a fault). Only a too slow stop
     (< RwDecelMin_mm) counts toward alarm 16.
  v6t: rewinder slow-down fallback while the paper thickness is not
     learned: RwSlowReq at RwTargetDiam - RwSlowFallbackMm (30 mm).
  v6s: paper change: a learning window more than RwMatChangePct (25 %) away
     from RwMat_K is ignored once, taken directly the 2nd time in a row
     (automatic). RwMatReset (HMI "NEW PAPER") forgets RwMat_K.
  v6p: section 5c: RwLearnReset clears RwStopDecelK, RwStopDecel and
     RwSlowDecelK (the v6f seed that refilled RwStopDecelK is removed).
  v6l: previous value saved before every TEACH CORE (RwAxisDist_Prev,
       UwCoreDist_Prev, Retain) + UNDO buttons (RwTeachUndo,
       UwTeachUndo): swap current <-> previous. State 4 = RESTORED.
  v6k: unwinder TEACH CORE (Core_Measure) works like the rewinder one:
       same checks, 2 s stable with sonar AVERAGE, status UwTeachState.
```

```
55          Replaces General rung 15 (one immediate sonar reading).
56      v6j: TEACH CORE (section 14): HMI button RwTeachCore, empty core on the
57          rewinder, machine stopped -> after 2 s stable:
58          RwAxisDist := RwDist + RwCoreDiameter / 2. Status in RwTeachState.
59      v6i: auto: pots scaled to WorkSpd_Max / FinalSpd_Max (Motor_Manager rung 6,
60          Pot1_Mem / Pot2_Mem). ManMode keeps the raw pots (percent),
61          so section 6 uses the raw pots in ManMode.
62      v6h: BrakeHold keeps O_Brake (T-one "reel brake automatic") ON during a
63          normal stop until the web is almost stopped (< 5 m/min, max 15 s).
64          Before: T-one switched to "stop" with the run command, while the
65          rewinder was decelerating hard -> unwinder overran, web on the
66          floor (field 10/20/2026). Used in Output_Manager rung 2.
67      v6g: slow-down AND stop decel proportional to the rewinder diameter.
68          The ATV930 ramp is in Hz/s (motor = rewind shaft), so the web
69          decel [mm/s2] = shaft decel x radius. Field data 10/20/2026:
70          24 mm/s2 at 120 mm -> 90 mm/s2 at 600 mm.
71          Learned: RwSlowDecelK, RwStopDecelK [mm/s2 per mm of diameter].
72          Live:    RwSlowDecelLrn [mm/s2], RwStopDecel [pulses/s2] = K x DecelDiam.
73      v6f: learn the REAL slow-down deceleration (section 5b) -> RwSlowDecelLrn
74          (mm/s2). The ATV930 stretches its ramp with a heavy roll (no
75          braking resistor), so the PLC ramp (DecelCoarse) was too
76          optimistic: the stop came before final speed was reached.
77          Priority in section 6: RwSlowDecel (manual) > RwSlowDecelLrn
78          (learned) > DecelCoarse/DecelTrim model.
79      v6e: GSM_Learned_OK (replaces global RwLearnFault): TRUE only when
80          RwMat_K is learned, RwStopDecel is learned (if RwLearnStop) and
81          there is no learning fault. Alarm 16 uses GSM_LearnFault.
82      v6d: exact 100 ms tick (25 x 4 ms scan counter, the self-resetting
83          TON ran ~104-108 ms -> speed ~4-8 % too high)
84         standstill = <= 2 counts / 100 ms (encoder jitter)
85         stop learning skipped after jog / manual / slow-button runs and
86          when the auxiliaries drop (E-stop): RunWasJog, StopLearnValid
87         work / final speed from the potentiometers in BOTH modes
88          (Speed_Display / LowSpd_Display are not m/min in ManMode)
89         slow-down length includes the DecelTrim tail inside the
90          SpdDelta_Percent band (Motor_Manager rungs 14/15)
91      v6c: learning supervision -> RwLearnFault (alarm 16), learned paper
92          thickness RwThick_mm, plausibility limits for K and stop decel
93      v6b: RwMaxSlowAtDiam display + RwMaxSlowFallback (slow-down warning
94          while RwMat_K is not learned yet)
95      v6 : consistent internal names, units in names.
96         Globals renamed: RwRemain_m, RwStopLen_m, RwSlowLen_m,
97                     RwSlowMargin_m, UwSlowMargin_m
98      v5 : max diameter protection (always active), sensor plausibility,
99         12SA5 master (section 9), alarms in section Alarms (ladder)
100     v4b: AUTO interlock + single HMI display (SlowAtDisp / SlowAtMode)
101
        ================================================================
        ==================== *)
102
103 (* ---- 1. 100 ms tick, mm per encoder pulse ------------------------------ *)
104 (* v6d: 25 scans x 4 ms PrimaryTask = exactly 100 ms.                    *)
105 (*      Change 25 if the PrimaryTask period is changed.                  *)
106
107 TickCnt := TickCnt + 1;
108
109 IF TickCnt >= 25 THEN
110     TickCnt := 0;
111     Tick := TRUE;
```

```
112 ELSE
113     Tick := FALSE;
114 END_IF;
115
116 IF (Encoder_Pulse > 0) AND (EncoderWheel_Circumf > 0.0) THEN
117     MmPerPulse := EncoderWheel_Circumf / (INT_TO_REAL(Encoder_Pulse) * 4.0);
118 ELSE
119     MmPerPulse := 0.0;
120 END_IF;
121
122 (* ---- 2. Rewinder DT35 -> filtered distance -> diameter ---------- *)
123
124 RwDistRaw_mm := RwDist0V + (INT_TO_REAL(AI_Diam_Rewinder) / 4000.0)
125             * (RwDist10V - RwDist0V);
126
127 IF Tick THEN
128     IF NOT RwFiltInit THEN
129         RwDist := RwDistRaw_mm;
130         RwFiltInit := TRUE;
131     ELSE
132         RwDist := RwDist + RwFilter_K * (RwDistRaw_mm - RwDist);
133     END_IF;
134 END_IF;
135
136 RwSensFault := (AI_Diam_Rewinder < RwRawMin) OR (AI_Diam_Rewinder >
    RwRawMax)
137             OR (RwDist > RwAxisDist);
138
139 RwDiam := 2.0 * (RwAxisDist - RwDist);
140
141 IF RwDiam < 0.0 THEN
142     RwDiam := 0.0;
143 END_IF;
144
145 (* ---- 2b. V9: diameter averaged over exactly ONE revolution ---------- *)
146 (* The out-of-round roll gives a wave at the rotation frequency (seen in   *)
147 (* the 10/29 traces: 2.4 Hz at 444 m/min / 973 mm). Averaging the raw     *)
148 (* DT35 distance over exactly one revolution cancels it (and all its      *)
149 (* harmonics), whatever the phase -> no shaft position sensor needed:     *)
150 (* one revolution = PI x D of web on the encoder. Each scan is weighted   *)
151 (* by its encoder pulses = by roll ANGLE, so ramps and standstill do not  *)
152 (* bias it (stopped: nothing added). Used only by the thickness learning. *)
153 (* RwWave_mm = peak-peak diameter wave in the last revolution (HMI).      *)
154
155 RevDP := ABS(Encoder_Pos - RevPrevPos);
156 RevPrevPos := Encoder_Pos;
157
158 IF RwSensFault OR (MmPerPulse <= 0.0) OR (RwDiam < 50.0) OR (RevDP > 100000)
    THEN
159     RevW := 0.0;
160     RevSum := 0.0;
161     RwDiamRevOk := RwDiamRevOk AND NOT RwSensFault;
162 ELSIF RevDP > 0 THEN
163
164     IF RevW <= 0.0 THEN
165         RevMin := RwDistRaw_mm;
166         RevMax := RwDistRaw_mm;
167     END_IF;
168
```

```
169     RevSum := RevSum + RwDistRaw_mm * DINT_TO_REAL(RevDP);
170     RevW := RevW + DINT_TO_REAL(RevDP);
171
172     IF RwDistRaw_mm < RevMin THEN
173         RevMin := RwDistRaw_mm;
174     END_IF;
175
176     IF RwDistRaw_mm > RevMax THEN
177         RevMax := RwDistRaw_mm;
178     END_IF;
179
180     IF RevW >= 3.14159 * RwDiam / MmPerPulse THEN
181         RwDiamRev := 2.0 * (RwAxisDist - RevSum / RevW);
182         RwWave_mm := 2.0 * (RevMax - RevMin);
183         RwDiamRevOk := TRUE;
184         RevW := 0.0;
185         RevSum := 0.0;
186     END_IF;
187 END_IF;
188
189 (* v6g: diameter used by the decel model (last valid on sensor fault)      *)
190
191 IF NOT RwSensFault THEN
192     DecelDiam := RwDiam;
193 ELSE
194     DecelDiam := RwLastValidDiam;
195 END_IF;
196
197 IF DecelDiam < 100.0 THEN
198     DecelDiam := 100.0;                    (* core *)
199 END_IF;
200
201 (* ---- 3. Actual speed from encoder, 100 ms window ---------------------- *)
202
203 IF Tick THEN
204
205     IF NOT SpdInit THEN
206         EncPrev := Encoder_Pos;
207         SpdInit := TRUE;
208     END_IF;
209
210     SpdAct_Pps := DINT_TO_REAL(ABS(Encoder_Pos - EncPrev)) * 10.0;
211     EncPrev := Encoder_Pos;
212 END_IF;
213
214 RwSpeed := SpdAct_Pps * MmPerPulse * 0.06;        (* m/min *)
215
216 (* ---- 4a. v6s: NEW PAPER button: forget the learned thickness ---------- *)
217 (* RwMat_K = 0 -> the first valid 10 s window sets it directly.           *)
218 (* HMI momentary button: must be HELD 3 s, then the reset is done once and *)
219 (* the PLC clears the bit (HMI lamp goes off = done).                     *)
220
221 T_MatReset(In := RwMatReset, PT := T#3s);
222
223 IF T_MatReset.Q THEN
224     RwMat_K := 0.0;
225     LearnFailCnt := 0;
226     MatChangeCnt := 0;
227     RwMatLocked := FALSE;                  (* V9 *)
```

```
228     LongActive := FALSE;
229     GsmAsk := FALSE;                    (* V9: NEW PAPER answers the popup *)
230     GsmArmed := FALSE;
231     RwMatReset := FALSE;
232   END_IF;
233
234   (* ---- 4b. V9: long-baseline thickness learning + LOCK ---------- *)
235   (* From the start point (first run after NEW PAPER / GSM YES / power-on   *)
236   (* while not locked): K = (D^2 - D0^2) / (2 x pulses). The waving of the  *)
237   (* roll (+/-0.4 mm) only matters at the two ends, so the error falls with *)
238   (* the growth (50 mm growth: ~1 %). After 20 mm growth it replaces the    *)
239   (* 30 m windows. At RwLockDiam (and >= RwLockMinGrow_mm growth) the value
      *)
240   (* is LOCKED (Retain): windows and paper-change detection stop.          *)
241   (* RwLockDiam = 0 -> never locked (windows + long baseline as learning).  *)
242
243   IF Tick AND NOT RwMatLocked THEN
244
245     IF LongActive AND (RwDiamRev < LongStartD - 30.0) AND RwDiamRevOk THEN
246       LongActive := FALSE;              (* new core: restart *)
247     END_IF;
248
249     IF Encoder_Reset OR (Encoder_Pos < LongStartPos - 20000) THEN
250       LongActive := FALSE;              (* metre counter reset / 2 m back *)
251     END_IF;
252
253     IF NOT LongActive AND O_MotorStart AND RwDiamRevOk
254       AND (RwDiamRev > 0.0) THEN
255       LongStartPos := Encoder_Pos;
256       LongStartD := RwDiamRev;
257       LongActive := TRUE;
258     END_IF;
259
260     LongValid := FALSE;
261
262     IF LongActive AND RwDiamRevOk AND (MmPerPulse > 0.0) THEN
263       LongGrow_mm := RwDiamRev - LongStartD;
264       LongPls := DINT_TO_REAL(ABS(Encoder_Pos - LongStartPos));
265
266       IF (LongGrow_mm >= 20.0) AND (LongPls > 0.0) THEN
267         LongK := (RwDiamRev * RwDiamRev - LongStartD * LongStartD) / (2.0 *
    LongPls);
268         LongThick_mm := LongK * 3.14159 / (2.0 * MmPerPulse);
269
270         IF (LongThick_mm >= RwThickMin_mm) AND (LongThick_mm <=
    RwThickMax_mm) THEN
271           LongValid := TRUE;
272           RwMat_K := LongK;
273           MatChangeCnt := 0;
274           LearnFailCnt := 0;
275
276           IF (RwLockDiam > 0.0) AND (RwDiamRev >= RwLockDiam)
277             AND (LongGrow_mm >= RwLockMinGrow_mm) THEN
278             RwMatLocked := TRUE;        (* solid from now on *)
279             LongActive := FALSE;
280           END_IF;
281         END_IF;
282       END_IF;
283     END_IF;
```

```
284 END_IF;
285
286 IF RwMatLocked THEN
287     LongValid := FALSE;
288     LearnFailCnt := 0;                    (* no thickness alarm *)
289 END_IF;
290
291 (* ---- 4. Learn material factor  K = dDiam/dPulse * Diam ---------- *)
292
293 IF Tick THEN
294     LearnWinTicks := LearnWinTicks + 1;
295
296     IF NOT LearnInit THEN
297         LearnWinStartPos := Encoder_Pos;  LearnWinStartDiam := RwDiamRev;  (* V9 *)
298         LearnWinTicks := 0;
299         LearnWinRunning := O_MotorStart;
300         LearnInit := TRUE;
301     ELSE
302         LearnWinRunning := LearnWinRunning AND O_MotorStart;
303         (* v6w: LENGTH-based window, modulated by the diameter: closes    *)
304         (* when the roll should have grown RwWinGrow_mm (2.5 mm), so the  *)
305         (* accuracy vs DT35 noise is the same at every speed and size.    *)
306         (* Pulses = growth x diameter / K (K = learned RwMat_K).          *)
307         (* Limits: min 4 m, max RwWinLen_m (30 m). K not learned: max.    *)
308         (* Motor stopped during the window -> window dropped (no fault).  *)
309
310         IF MmPerPulse > 0.0 THEN
311             IF (RwMat_K > 0.0) AND (RwWinGrow_mm > 0.0) THEN
312                 LearnWinLen_Pls := RwWinGrow_mm * MAX(RwDiam, REAL#100.0) /
    RwMat_K;
313             ELSE
314                 LearnWinLen_Pls := RwWinLen_m * 1000.0 / MmPerPulse;
315             END_IF;
316
317             IF LearnWinLen_Pls < 4000.0 / MmPerPulse THEN
318                 LearnWinLen_Pls := 4000.0 / MmPerPulse;          (* 4 m *)
319             END_IF;
320
321             IF (RwWinLen_m > 0.0)
322               AND (LearnWinLen_Pls > RwWinLen_m * 1000.0 / MmPerPulse) THEN
323                 LearnWinLen_Pls := RwWinLen_m * 1000.0 / MmPerPulse;
324             END_IF;
325         ELSE
326             LearnWinLen_Pls := 300000.0;
327         END_IF;
328
329         IF (DINT_TO_REAL(ABS(Encoder_Pos - LearnWinStartPos)) >= LearnWinLen_Pls)
330           OR NOT LearnWinRunning THEN
331             LearnWin_Pls := ABS(Encoder_Pos - LearnWinStartPos);
332             LearnWinDiamDelta := RwDiamRev - LearnWinStartDiam;  (* V9 *)
333             (* A window SHOULD learn: ran all the time, enough length,  *)
334             (* DT35 OK. Then it must give a plausible paper thickness. *)
335
336             IF LearnWinRunning AND (LearnWin_Pls >= RwMinPulses)
337               AND NOT RwSensFault
338               AND NOT RwMatLocked AND NOT LongValid     (* V9 *)
339               AND RwDiamRevOk AND (LearnWinStartDiam > 0.0) THEN
340                 LearnWinOk := FALSE;
341
```

```
            IF (LearnWinDiamDelta > 0.0) AND (MmPerPulse > 0.0) THEN
               MatK_Window := (LearnWinDiamDelta / DINT_TO_REAL(LearnWin_Pls))
                        * ((RwDiamRev + LearnWinStartDiam) / 2.0);
               (* K = 2 * thickness * mm-per-pulse / PI          *)
               ThickWin_mm := MatK_Window * 3.14159 / (2.0 * MmPerPulse);

               IF (ThickWin_mm >= RwThickMin_mm)
                 AND (ThickWin_mm <= RwThickMax_mm) THEN
                  LearnWinOk := TRUE;

                  IF RwMat_K <= 0.0 THEN
                     RwMat_K := MatK_Window;
                     MatChangeCnt := 0;
                  ELSIF (RwMatChangePct > 0.0)
                       AND (ABS(MatK_Window - RwMat_K)
                            > RwMat_K * RwMatChangePct / 100.0) THEN
                     (* v6s: far from the learned value: 1 window = ignored *)
                     (* (outlier). 2 in a row = paper changed -> take it.  *)
                     MatChangeCnt := MatChangeCnt + 1;

                     IF MatChangeCnt >= 2 THEN
                        RwMat_K := MatK_Window;
                        MatChangeCnt := 0;
                     END_IF;
                  ELSE
                     MatChangeCnt := 0;
                     RwMat_K := RwMat_K + RwLearn_K * (MatK_Window - RwMat_K);
                  END_IF;
               END_IF;
            END_IF;

            IF LearnWinOk THEN
               LearnFailCnt := 0;
            ELSIF LearnFailCnt < 1000 THEN
               LearnFailCnt := LearnFailCnt + 1;  (* no growth / implausible *)
            END_IF;
         END_IF;
         LearnWinStartPos := Encoder_Pos;  LearnWinStartDiam := RwDiamRev;  (* V9
*)
         LearnWinTicks := 0;
         LearnWinRunning := O_MotorStart;
      END_IF;
   END_IF;
END_IF;

(* ---- 5. Learn real stop deceleration on every stop ---------------------- *)
(* v6d: a run with jog / manual / slow button uses the 2nd ramp (O_AccDec2) *)
(*      -> its stop must not be learned. Latched for the whole run because  *)
(*      the jog button is already released on the stop edge.                *)

IF O_MotorStart AND NOT MotorRunMem THEN          (* start edge *)
   RunWasJog := FALSE;
END_IF;

IF O_MotorStart AND (ManMode OR JogRun) THEN
   RunWasJog := TRUE;
END_IF;

IF MotorRunMem AND NOT O_MotorStart THEN          (* stop command edge *)
```

```
    StopMeasActive := TRUE;
    StopStartPos := Encoder_Pos;
    StopStartSpd_Pps := SpdAct_Pps;
    StopStartDiam := DecelDiam;
    StandstillTicks := 0;
    (* STOP button and automatic stop take the same path (drive DI1 and    *)
    (* O_Brake -> T-one drop together, electrical plan sheets 1/9/15),     *)
    (* so both are learned.                                                *)
    StopLearnValid := ControlVoltage AND NOT RunWasJog AND NOT ManMode;
END_IF;

MotorRunMem := O_MotorStart;

IF O_MotorStart THEN
    StopMeasActive := FALSE;                (* restarted: abort *)
ELSIF StopMeasActive AND Tick THEN

    IF NOT ControlVoltage THEN
        StopLearnValid := FALSE;            (* E-stop during the stop *)
    END_IF;

    IF SpdAct_Pps <= 20.0 THEN              (* <= 2 counts / 100 ms *)
        StandstillTicks := StandstillTicks + 1;
    ELSE
        StandstillTicks := 0;
    END_IF;

    IF StandstillTicks >= 3 THEN            (* standstill 300 ms *)
        StopMeasActive := FALSE;
        StopMeas_Pls := DINT_TO_REAL(ABS(Encoder_Pos - StopStartPos));

        IF RwLearnStop AND StopLearnValid
          AND (StopStartSpd_Pps * MmPerPulse * 0.06 >= RwMinLearnSpd) THEN
          (* This stop SHOULD learn: result must be plausible       *)
          (* v6u: a stop shorter than the reaction distance, or     *)
          (* faster than RwDecelMax_mm, is NOT a fault: the web     *)
          (* stopped within a few mm, nothing to learn (trace       *)
          (* 10/28/2026: 23 m/min -> stopped in 102 pulses = 10 mm). *)
          (* Only a too SLOW stop (< RwDecelMin_mm) counts as fail.  *)
          StopLearnOk := FALSE;

          IF StopMeas_Pls > StopStartSpd_Pps * RwTreact THEN
            StopDecelMeas := (StopStartSpd_Pps * StopStartSpd_Pps)
                           / (2.0 * (StopMeas_Pls - StopStartSpd_Pps * RwTreact));
            StopDecelMeas_mm := StopDecelMeas * MmPerPulse;  (* mm/s2 *)

            IF (StopDecelMeas_mm >= RwDecelMin_mm)
              AND (StopDecelMeas_mm <= RwDecelMax_mm) THEN
              StopLearnOk := TRUE;
              (* v6g: learn per mm of diameter *)
              DecelK_Meas := StopDecelMeas_mm / ((StopStartDiam + DecelDiam) /
2.0);

              IF RwStopDecelK <= 0.0 THEN
                RwStopDecelK := DecelK_Meas;
              ELSE
                RwStopDecelK := RwStopDecelK + 0.3 * (DecelK_Meas -
RwStopDecelK);
              END_IF;
```

```
                END_IF;

                StopFailCnt := 0;
            ELSIF StopDecelMeas_mm < RwDecelMin_mm THEN

                IF StopFailCnt < 1000 THEN
                    StopFailCnt := StopFailCnt + 1;   (* stop too long *)
                END_IF;
            END_IF;                          (* too fast: ignored *)
        END_IF;                              (* too short: ignored *)
    END_IF;
  END_IF;
END_IF;

(* ---- 5b. v6f: learn real slow-down deceleration ---------------------- *)
(* From the SlowDown edge (at speed) until final speed is reached, or     *)
(* until the stop comes first. a = (v0^2 - v1^2) / (2 * distance).        *)
(* Not learned in jog / manual / slow button (2nd ramp).                  *)

IF SlowDown AND NOT SlowMem AND O_MotorStart
  AND NOT ManMode AND NOT JogRun
  AND (SpdAct_Pps > SpdFinal_Pps + SpdBand_Pps) THEN
  SlowMeasActive := TRUE;
  SlowStartPos := Encoder_Pos;
  SlowStartSpd_Pps := SpdAct_Pps;
  SlowStartDiam := DecelDiam;
END_IF;
SlowMem := SlowDown;

IF SlowMeasActive THEN

  IF NOT SlowDown OR ManMode OR JogRun THEN
    SlowMeasActive := FALSE;                 (* aborted *)
  ELSIF Tick AND ((SpdAct_Pps <= SpdFinal_Pps + 0.25 * SpdBand_Pps)
                  OR NOT O_MotorStart) THEN      (* final reached or stopped *)
    SlowMeasActive := FALSE;
    SlowMeas_Pls := DINT_TO_REAL(ABS(Encoder_Pos - SlowStartPos));

    IF (SlowMeas_Pls > 0.0) AND (MmPerPulse > 0.0)
      AND ((SlowStartSpd_Pps - SpdAct_Pps) * MmPerPulse * 0.06 >= 10.0) THEN
      SlowDecelMeas_mm := (SlowStartSpd_Pps * SlowStartSpd_Pps
                           - SpdAct_Pps * SpdAct_Pps)
                          / (2.0 * SlowMeas_Pls) * MmPerPulse;  (* mm/s2 *)

      IF (SlowDecelMeas_mm >= RwDecelMin_mm)
        AND (SlowDecelMeas_mm <= RwDecelMax_mm) THEN
        (* v6g: learn per mm of diameter *)
        DecelK_Meas := SlowDecelMeas_mm / ((SlowStartDiam + DecelDiam) /
2.0);

        IF RwSlowDecelK <= 0.0 THEN
          RwSlowDecelK := DecelK_Meas;
        ELSE
          RwSlowDecelK := RwSlowDecelK + 0.5 * (DecelK_Meas -
RwSlowDecelK);
        END_IF;
      END_IF;
    END_IF;
  END_IF;
```

```
513 END_IF;
514
515 (* ---- 5c. v6g: live decels = learned K x current diameter ---------- *)
516 (* v6p: the one-time v6f seed is removed (it re-filled RwStopDecelK from    *)
517 (* RwStopDecel, so the learning could never be reset).                      *)
518 (* RwLearnReset (HMI / Watch tab, PLC resets it): clear the learned decels. *)
519 (* Both are re-learned at the next auto slow-down / stop >= RwMinLearnSpd.  *)
520
521 IF RwLearnReset THEN
522     RwStopDecelK := 0.0;
523     RwStopDecel := 0.0;
524     RwSlowDecelK := 0.0;
525     StopFailCnt := 0;
526     RwLearnReset := FALSE;
527 END_IF;
528
529 IF (RwStopDecelK > 0.0) AND (MmPerPulse > 0.0) THEN
530     RwStopDecel := RwStopDecelK * DecelDiam / MmPerPulse;     (* pulses/s2 *)
531 END_IF;
532
533 IF RwSlowDecelK > 0.0 THEN
534     RwSlowDecelLrn := RwSlowDecelK * DecelDiam;               (* mm/s2 *)
535 ELSE
536     RwSlowDecelLrn := 0.0;
537 END_IF;
538
```

1 Auto Stop Process 6-14

```
(*
 ================================================================
 ====================
   AutoStop_Control  V12  -  PART 2 of 2 (rung 1): sections 6 - 14c
   V12: END-OF-ROLL STOP RAMP: auto / diameter stops ramp down at
FinalStopDecel
        (soft stop mechanism) instead of the drive DEC ramp (3 s = 20 Hz/s,
        up to ~8 m/s2 of web on a big roll): stop lengths use it (6, 11b).
   V11: 3-state unwinder display + stop command at UwStop_at_D_Val (section 8).
   (Sysmac: an inline ST rung holds max 1000 rows -> split in 2 rungs.
    Rung 0 = part 1 runs first, then this rung, every scan.)

 ================================================================
 ==================== *)

(* ---- 6. Lengths in pulses: stop, slow-down, remaining ---------- *)
(* Stop length uses the ACTUAL speed (precision)                    *)

IF RwStopDecel > 0.0 THEN
   StopLen_Pls := SpdAct_Pps * RwTreact
                + (SpdAct_Pps * SpdAct_Pps) / (2.0 * RwStopDecel);
ELSE
   StopLen_Pls := SpdAct_Pps * (RwTreact + RwTstopDef / 2.0);
END_IF;

(* V12: end-of-roll stop ramp at FinalStopDecel (web mm/s2) + S-curve    *)

IF (FinalStopDecel > 0.0) AND FFEnable AND NOT ManMode AND (MmPerPulse >
0.0) THEN
   StopLen_Pls := SpdAct_Pps * RwTreact
                + (SpdAct_Pps * SpdAct_Pps) / (2.0 * FinalStopDecel / MmPerPulse)
                + SpdAct_Pps * 0.5 * SCurveTime;
END_IF;

StopLen_Pls := StopLen_Pls * RwTrim;

(* Slow-down uses SpdRef = MAX(actual, work-speed setpoint):            *)
(* stable value, shown to the operator even before start                *)

SlowLenBase_Pls := 0.0;  RwSlowLen_Pls := 0.0;  UwSlowLen_Pls := 0.0;

IF MmPerPulse > 0.0 THEN
   (* v6f: manual > learned > PLC ramp model                           *)
   (* V7: manual > feed-forward (one ramp at DecelWeb, real web mm/s2)  *)
   (*     > old loop with web ramps (coarse DecelWeb + trim tail)      *)
   (*     > learned > old PLC ramp model                               *)

   IF RwSlowDecel > 0.0 THEN
      SlowDecel_Pps2 := RwSlowDecel / MmPerPulse;
      SlowOneRamp := TRUE;
   ELSIF FFEnable AND (DecelWeb > 0.0) AND (WebRatio > 0.0) AND NOT
ManMode THEN
      SlowDecel_Pps2 := DecelWeb / MmPerPulse;
      SlowOneRamp := TRUE;
   ELSIF (DecelWeb > 0.0) AND (WebRatio > 0.0) AND NOT ManMode THEN  (* V7e
*)
      SlowDecel_Pps2 := DecelWeb / MmPerPulse;
      SlowOneRamp := FALSE;
```

```
51    ELSIF RwSlowDecelLrn > 0.0 THEN
52       SlowDecel_Pps2 := RwSlowDecelLrn / MmPerPulse;  (* real, incl. trim tail *)
53       SlowOneRamp := TRUE;
54    ELSE
55       SlowDecel_Pps2 := DecelCoarse * 250.0 / MmPerPulse; (* 1 = 250 mm/s2 *)
56       SlowOneRamp := FALSE;
57    END_IF;
58
59    (* v6d: DecelTrim ramp used inside the +/- SpdDelta_Percent band *)
60
61    IF (DecelTrim > 0.0) AND (WebRatio > 0.0) AND (Vel_Max_mms > 0.0) THEN
62       (* V7: trim ramp (reference units) converted to the real web decel *)
63       TrimDecel_Pps2 := DecelTrim * 1000000.0 / (0.06 * WebRatio * Vel_Max_mms)
64                         / MmPerPulse;
65    ELSIF DecelTrim > 0.0 THEN
66       TrimDecel_Pps2 := DecelTrim * 250.0 / MmPerPulse;
67    ELSE
68       TrimDecel_Pps2 := SlowDecel_Pps2;
69    END_IF;
70
71    SpdBand_Pps := SpdDelta_Percent * 1000.0 / 60.0 / MmPerPulse;
72
73    (* v6d: speeds from the potentiometers, valid in auto AND ManMode     *)
74    (*      (m/min = Pot / Input_Ratio, same as Motor_Manager rung 8)     *)
75
76    IF (Input_Ratio > 0.0) AND ManMode THEN
77       (* v6i: ManMode drives the inverter with the raw pots (rung 17) *)
78       SpdFinal_Pps := INT_TO_REAL(Pot2_Slow_Velocity) / Input_Ratio * 1000.0 /
   60.0 / MmPerPulse;
79       SpdWork_Pps  := INT_TO_REAL(Pot1_Work_Velocity) / Input_Ratio * 1000.0 /
   60.0 / MmPerPulse;
80    ELSIF Input_Ratio > 0.0 THEN
81       SpdFinal_Pps := INT_TO_REAL(Pot2_Mem) / Input_Ratio * 1000.0 / 60.0 /
   MmPerPulse;
82       SpdWork_Pps  := INT_TO_REAL(Pot1_Mem) / Input_Ratio * 1000.0 / 60.0 /
   MmPerPulse;
83    ELSE
84       SpdFinal_Pps := LowSpd_Display * 1000.0 / 60.0 / MmPerPulse;
85       SpdWork_Pps  := Speed_Display  * 1000.0 / 60.0 / MmPerPulse;
86    END_IF;
87
88    (* v6y: rewinder AUTO stop -> work speed limited to RwAutoMaxSpdAct *)
89    (* (limit + value taken over only at standstill, Motor_Manager rung 5) *)
90
91    IF RwAutoLimAct AND NOT ManMode
92       AND (SpdWork_Pps > RwAutoMaxSpdAct * 1000.0 / 60.0 / MmPerPulse) THEN
93       SpdWork_Pps := RwAutoMaxSpdAct * 1000.0 / 60.0 / MmPerPulse;
94    END_IF;
95
96    SpdRef_Pps := MAX(SpdAct_Pps, SpdWork_Pps);
97
98    IF RwStopDecel > 0.0 THEN
99       StopLenFinal_Pls := SpdFinal_Pps * RwTreact
100                         + (SpdFinal_Pps * SpdFinal_Pps) / (2.0 * RwStopDecel);
101   ELSE
102      StopLenFinal_Pls := SpdFinal_Pps * (RwTreact + RwTstopDef / 2.0);
103   END_IF;
104
105   (* V12: final stop from the final speed on the end-of-roll ramp *)
```

```
106
107   IF (FinalStopDecel > 0.0) AND FFEnable AND NOT ManMode THEN
108      StopLenFinal_Pls := SpdFinal_Pps * RwTreact
109                        + (SpdFinal_Pps * SpdFinal_Pps) / (2.0 * FinalStopDecel /
      MmPerPulse)
110                        + SpdFinal_Pps * 0.5 * SCurveTime;
111   END_IF;
112
113   SlowLenBase_Pls := StopLenFinal_Pls * RwTrim;
114
115   IF (SpdRef_Pps > SpdFinal_Pps) AND (SlowDecel_Pps2 > 0.0) THEN
116
117      IF SlowOneRamp OR (TrimDecel_Pps2 <= 0.0) THEN
118         (* manual or learned decel: one constant ramp *)
119         SlowLenBase_Pls := SlowLenBase_Pls
120            + (SpdRef_Pps * SpdRef_Pps - SpdFinal_Pps * SpdFinal_Pps)
121             / (2.0 * SlowDecel_Pps2);
122      ELSE
123         (* auto: coarse ramp down to the band, trim ramp inside it *)
124         SpdKnee_Pps := SpdFinal_Pps + SpdBand_Pps;
125
126         IF SpdRef_Pps > SpdKnee_Pps THEN
127            SlowLenBase_Pls := SlowLenBase_Pls
128               + (SpdRef_Pps * SpdRef_Pps - SpdKnee_Pps * SpdKnee_Pps)
129                / (2.0 * SlowDecel_Pps2)
130               + (SpdKnee_Pps * SpdKnee_Pps - SpdFinal_Pps * SpdFinal_Pps)
131                / (2.0 * TrimDecel_Pps2);
132         ELSE
133            SlowLenBase_Pls := SlowLenBase_Pls
134               + (SpdRef_Pps * SpdRef_Pps - SpdFinal_Pps * SpdFinal_Pps)
135                / (2.0 * TrimDecel_Pps2);
136         END_IF;
137      END_IF;
138   END_IF;
139
140   (* V7f: S-curve rounding: the slow-down ends SCurveTime later ->      *)
141   (*      extra length = (SpdRef - SpdFinal) / 2 x SCurveTime          *)
142
143   IF FFEnable AND NOT ManMode AND (RwSlowDecel <= 0.0) AND (DecelWeb >
      0.0)
144      AND (WebRatio > 0.0) AND (SCurveTime > 0.0)
145      AND (SpdRef_Pps > SpdFinal_Pps) THEN
146      SlowLenBase_Pls := SlowLenBase_Pls
147                        + (SpdRef_Pps - SpdFinal_Pps) * 0.5 * SCurveTime;
148   END_IF;
149
150   RwSlowLen_Pls := SlowLenBase_Pls + RwSlowMargin_m * 1000.0 / MmPerPulse;
151
152   IF UwSlowMargin_m > 0.0 THEN
153      UwMarginUsed_m := UwSlowMargin_m;
154   ELSE
155      UwMarginUsed_m := RwSlowMargin_m;
156   END_IF;
157
158   UwSlowLen_Pls := SlowLenBase_Pls + UwMarginUsed_m * 1000.0 / MmPerPulse;
159 END_IF;
160
161 IF RwDiam >= RwTargetDiam THEN
```

```
ELSIF RwMat_K > 0.0 THEN
    RwRemain_Pls := (RwTargetDiam * RwTargetDiam - RwDiam * RwDiam)
                  / (2.0 * RwMat_K);
ELSE
    RwRemain_Pls := 1.0E9;                    (* K not learned: plain target *)
END_IF;

RwRemain_m  := RwRemain_Pls  * MmPerPulse / 1000.0;
RwStopLen_m := StopLen_Pls   * MmPerPulse / 1000.0;
RwSlowLen_m := RwSlowLen_Pls * MmPerPulse / 1000.0;

(* ---- 6b. Live operator displays ------------------------------------------ *)
(* Rewinder: diameter where slow-down / stop command will happen          *)

IF RwMat_K > 0.0 THEN
    SqrtArg := RwTargetDiam * RwTargetDiam - 2.0 * RwMat_K * RwSlowLen_Pls;
    IF SqrtArg > 0.0 THEN
        RwSlow_at_D_Val := SQRT(SqrtArg);
    ELSE
        RwSlow_at_D_Val := 0.0;
    END_IF;

    SqrtArg := RwTargetDiam * RwTargetDiam - 2.0 * RwMat_K * StopLen_Pls;

    IF SqrtArg > 0.0 THEN
        RwStop_at_D_Val := SQRT(SqrtArg);
    ELSE
        RwStop_at_D_Val := 0.0;
    END_IF;
ELSE
    (* v6t: thickness not learned: fixed slow-down point before the target *)

    IF RwUseSlow AND (RwSlowFallbackMm > 0.0) THEN
        RwSlow_at_D_Val := RwTargetDiam - RwSlowFallbackMm;
    ELSE
        RwSlow_at_D_Val := RwTargetDiam;
    END_IF;
    RwStop_at_D_Val := RwTargetDiam;
END_IF;

(* Unwinder: meter count where the slow-down will start                   *)

IF MmPerPulse > 0.0 THEN
    UwSlow_at_M_Val := UwStop_at_M_Val - UwSlowLen_Pls * MmPerPulse / 1000.0;
ELSE
    UwSlow_at_M_Val := UwStop_at_M_Val;
END_IF;

IF UwSlow_at_M_Val < 0.0 THEN
    UwSlow_at_M_Val := 0.0;
END_IF;

(* Unwinder slow-down request, latched                                    *)

IF (Current_Metre < UwSlow_at_M_Val - 50.0) OR (UwStop_at_M_Val <= 0.0) THEN
    UwSlow_at_M_Req := FALSE;                 (* counter reset / longer roll *)
END_IF;
```

```
221 IF (UwStop_at_M_Val > 0.0) AND (Current_Metre >= UwSlow_at_M_Val) THEN
222     UwSlow_at_M_Req := TRUE;
223 END_IF;
224
225 (* ---- 6c. V7c: unwinder DIAMETER slow-down follows the speed ---------- *)
226 (* Works without AutoStop (only Measure_ON, like General rung 16).       *)
227 (* Paper left above the stop diameter, in pulses:                         *)
228 (*     (UwDiam_Act^2 - UwStopDiam_Limit^2) / (2 x RwMat_K)                *)
229 (* (same paper as the rewinder -> same learned K). Slow down when it is   *)
230 (* shorter than UwSlowLen_Pls = decel from the speed (DecelWeb) + final    *)
231 (* stop + UwSlowMargin_m. Solved for the diameter:                        *)
232 (*     UwSlowDiamCalc = SQRT(UwStopDiam_Limit^2 + 2 x RwMat_K x
    UwSlowLen_Pls)    *)
233 (* K not learned / UwSlow_at_D_Req OFF -> fixed UwSlow_at_D_Val as before. *)
234 (* Latched: once slowed down, the point is frozen (+10 mm hysteresis), so  *)
235 (* the lower speed does not move it back and speed up again. Released by   *)
236 (* a new roll (diameter 30 mm above the frozen point).                     *)
237
238 IF UwSlow_at_D_Req AND (RwMat_K > 0.0) AND (UwStopDiam_Limit > 0.0)
239    AND (UwSlowLen_Pls > 0.0) THEN
240     SqrtArg := UwStopDiam_Limit * UwStopDiam_Limit + 2.0 * RwMat_K *
    UwSlowLen_Pls;
241     UwSlowDiamCalc := SQRT(SqrtArg);            (* same pattern as 6b *)
242 ELSE
243     UwSlowDiamCalc := UwSlow_at_D_Val;
244 END_IF;
245
246 IF UwSlowDiamCalc < UwStopDiam_Limit + 10.0 THEN
247     UwSlowDiamCalc := UwStopDiam_Limit + 10.0;
248 ELSIF UwSlowDiamCalc > 1500.0 THEN
249     UwSlowDiamCalc := 1500.0;
250 END_IF;
251
252 IF UwSlowDiamLatch AND (UwDiam_Act > UwSlowDiamAct + 30.0) THEN
253     UwSlowDiamLatch := FALSE;                   (* new roll *)
254 END_IF;
255
256 IF NOT UwSlowDiamLatch THEN
257     UwSlowDiamAct := UwSlowDiamCalc;            (* live point, HMI *)
258     IF Measure_ON AND (UwDiam_Act > 0.0) AND (UwDiam_Act <=
    UwSlowDiamCalc) THEN
259         UwSlowDiamLatch := TRUE;
260         UwSlowDiamAct := UwSlowDiamCalc + 10.0;    (* frozen + hysteresis *)
261     END_IF;
262 END_IF;
263
264 (* ---- 7. Rewinder requests: latched, reset when a new core is loaded ---- *)
265
266 IF NOT RwStopEnable
267    OR ((RwDiam < RwTargetDiam - RwResetHyst) AND NOT RwSensFault) THEN
268     RwStop_at_D_Req := FALSE;
269     RwSlow_at_D_Req := FALSE;
270 END_IF;
271
272 IF RwStopEnable THEN
273     IF (RwRemain_Pls <= StopLen_Pls) OR (RwSensFault AND RwStopOnFault) THEN
274         RwStop_at_D_Req := TRUE;
275     END_IF;
276     IF RwUseSlow AND (RwRemain_Pls <= RwSlowLen_Pls) THEN
```

```
277         RwSlow_at_D_Req := TRUE;
278     END_IF;
279
280     (* v6t: thickness not learned (RwMat_K = 0): slow down at a fixed      *)
281     (* distance before the target instead of stopping from full speed.    *)
282
283     IF RwUseSlow AND (RwMat_K <= 0.0) AND (RwSlowFallbackMm > 0.0)
284       AND NOT RwSensFault
285       AND (RwDiam >= RwTargetDiam - RwSlowFallbackMm) THEN
286         RwSlow_at_D_Req := TRUE;
287     END_IF;
288 END_IF;
289
290 (* ---- 8. One AUTO at a time + single "SLOW DOWN AT" display ---------- *)
291
292 // ---- Refuse AUTO STOP when its target is already reached (HMI warning 20) ----
293 // AutoEnWhy (HMI bits): 1 = unwinder metres already reached, 2 = rewinder roll
    >= target
294
295 IF I_AutoStop_OS AND RwStopEnable AND NOT RwArmMem AND NOT RwSensFault
296     AND (RwTargetDiam > REAL#0.0) AND (RwDiam >= RwTargetDiam -
    RwResetHyst) THEN
297     RwStopEnable := FALSE;                    // refused
298     AutoEnRefused := TRUE;
299     AutoEnWhy := 2;
300 END_IF;
301
302 IF I_AutoStop_OS AND UwStopEnable AND NOT UwArmMem
303     AND (UwStop_at_M_Val > REAL#0.0) AND (Current_Metre >= UwStop_at_M_Val)
    THEN
304     UwStopEnable := FALSE;                    // refused
305     AutoEnRefused := TRUE;
306     AutoEnWhy := 1;
307 END_IF;
308
309 IF UwStopEnable AND NOT UwEnMem THEN          (* unwinder AUTO switched
    ON *)
310     RwStopEnable := FALSE;
311 END_IF;
312
313 IF RwStopEnable AND NOT RwEnMem THEN          (* rewinder AUTO switched
    ON *)
314     UwStopEnable := FALSE;
315 END_IF;
316
317 T_AutoEnRef(In := AutoEnRefused, PT := T#5s);     // warning shown 5 s
318
319 IF T_AutoEnRef.Q OR I_RsetAlarm OR Alarm_Reset THEN
320     AutoEnRefused := FALSE;
321 END_IF;
322
323 RwArmMem := I_AutoStop_OS AND RwStopEnable;       // armed = selector 12SA5
    AND enable
324 UwArmMem := I_AutoStop_OS AND UwStopEnable;
325 UwEnMem := UwStopEnable;
326 RwEnMem := RwStopEnable;
327
328 IF RwStopEnable THEN
329     SlowAtDisp := RwSlow_at_D_Val;            (* mm *)
```

```
330     SlowAtMode := 2;
331   ELSIF UwStopEnable THEN
332     SlowAtDisp := UwSlow_at_M_Val;              (* m  *)
333     SlowAtMode := 1;
334   ELSE
335     SlowAtDisp := 0.0;
336     SlowAtMode := 0;                          (* hide display *)
337   END_IF;
338
339   SlowAtShow := (SlowAtMode <> 0);
340
341 (* V11: unwinder column, same place, 3 states:                    *)
342 (*   AUTO STOP UNWINDER (12SA5 + unwinder AUTO) -> STOP LENGTH [m]   *)
343 (*   normal run, before the slow-down          -> SLOW AT Ø [mm]     *)
344 (*   slow-down reached (decel / final speed)   -> STOP AT Ø [mm]     *)
345 (* V11: stop by diameter EARLIER, so the web is stopped AT UwStopDiam_Limit:  *)
346 (* the stop command comes at UwStop_at_D_Val = UwStopDiam_Limit + the
    diameter  *)
347 (* of the paper unwound during the stop from the ACTUAL speed      *)
348 (* (StopLen_Pls, section 6, learned stop decel):                  *)
349 (*   UwStop_at_D_Val = SQRT(UwStopDiam_Limit^2 + 2 x RwMat_K x StopLen_Pls)
    *)
350 (* General rung 18 (UwStopByDiam) compares with UwStop_at_D_Val.  *)
351 (* K not learned -> UwStopDiam_Limit (as before). Max UwStopDiam_Limit + 100
    mm.    *)
352
353   UwStopLenShow := I_AutoStop_OS AND (UwStopEnable OR (AutoEnRefused AND
    (AutoEnWhy = 1)));
354   UwSlowDiamShow := NOT UwStopLenShow AND Measure_ON AND NOT
    UwSlowDiamLatch;
355   UwStopDiamShow := NOT UwStopLenShow AND Measure_ON AND
    UwSlowDiamLatch;
356
357 IF (RwMat_K > 0.0) AND (StopLen_Pls > 0.0) THEN
358     SqrtArg := UwStopDiam_Limit * UwStopDiam_Limit + 2.0 * RwMat_K *
    StopLen_Pls;
359     UwStop_at_D_Val := SQRT(SqrtArg);
360   ELSE
361     UwStop_at_D_Val := UwStopDiam_Limit;
362   END_IF;
363
364 IF UwStop_at_D_Val > UwStopDiam_Limit + 100.0 THEN
365     UwStop_at_D_Val := UwStopDiam_Limit + 100.0;
366   ELSIF UwStop_at_D_Val < UwStopDiam_Limit THEN
367     UwStop_at_D_Val := UwStopDiam_Limit;
368   END_IF;
369
370 (* ---- 9. Automatic stop master = selector 12SA5 (I_AutoStop_OS) -------- *)
371
372   AutoSlowCmd := I_AutoStop_OS AND
373             ((UwStopEnable AND UwSlow_at_M_Req) OR (RwStopEnable AND
    RwSlow_at_D_Req));
374
375   AutoStopCmd := I_AutoStop_OS AND
376             ((UwStopEnable AND (UwStop_at_M_Val > 0.0)
377               AND (Current_Metre >= UwStop_at_M_Val))
378              OR (RwStopEnable AND RwStop_at_D_Req));
379
380 (* ---- 10. Unwinder roll empty (for alarm 15) ---------------------------- *)
```

```
381
382 IF UwDiam_Act > (UwStopDiam_Limit + UwEmptyHyst) THEN
383     UnWinder_Empty := FALSE;                    (* new roll loaded *)
384 END_IF;
385
386 IF (UwDiam_Act > 0.0) AND (UwDiam_Act <= UwStopDiam_Limit) THEN
387     UnWinder_Empty := TRUE;                     (* roll empty *)
388 END_IF;
389
390 (* v6x: new parent roll loaded (empty -> not empty) -> HMI popup        *)
391
392 (* "DID YOU CHANGE THE PAPER (GSM)?". YES = forget the learned thickness *)
393 (* (same as NEW PAPER). NO = the HMI resets GsmAsk.                      *)
394 (* V9: ask only ONCE per REAL new roll: the roll must have been empty for *)
395 (* 3 s (arms the question), then the new roll must read more than        *)
396 (* UwStopDiam_Limit + 100 mm for 3 s. The sonar bouncing around
    UwStopDiam_Limit    *)
397 (* (empty core, hands, arms) no longer re-opens the popup.              *)
398
399 IF Tick THEN
400     IF UnWinder_Empty THEN
401
402         IF UwEmptyTicks < 1000 THEN
403             UwEmptyTicks := UwEmptyTicks + 1;
404         END_IF;
405
406         IF UwEmptyTicks >= 30 THEN
407             GsmArmed := TRUE;
408         END_IF;
409
410         UwFullTicks := 0;
411     ELSE
412         UwEmptyTicks := 0;
413
414         IF GsmArmed AND Measure_ON AND (UwDiam_Act > UwStopDiam_Limit +
    100.0) THEN
415             UwFullTicks := UwFullTicks + 1;
416
417             IF UwFullTicks >= 30 THEN
418                 GsmAsk := TRUE;
419                 GsmArmed := FALSE;
420                 UwFullTicks := 0;
421             END_IF;
422         ELSE
423             UwFullTicks := 0;
424         END_IF;
425     END_IF;
426 END_IF;
427
428 UwEmptyMem := UnWinder_Empty;
429
430 IF GsmAskYes THEN
431     RwMat_K := 0.0;
432     RwMatLocked := FALSE;                       (* V9 *)
433     LongActive := FALSE;
434     LearnFailCnt := 0;
435     MatChangeCnt := 0;
436     GsmAskYes := FALSE;
437     GsmAsk := FALSE;
```

```
438 END_IF;
439
440 (* V10: HMI popup "UNWINDER DIAMETER SENSOR SONAR IS OFF ! DO YOU WANT
    TO  *)
441 (* TURN IT ON ?" - only at standstill: at power-up and after every stop    *)
442 (* while Measure_ON = FALSE. YES (SonarAskYes, PLC resets it) = sonar ON:  *)
443 (* NO = the HMI resets SonarAsk (asked again after the next stop).        *)
444
445 IF Measure_ON THEN
446     SonarAsk := FALSE;                    (* sonar ON: no popup *)
447 ELSIF NOT StartMotor_Cmd AND (SonarRunMem OR NOT SonarInit) THEN
448     SonarAsk := TRUE;                     (* stopped / power-up *)
449 END_IF;
450
451 SonarInit := TRUE;
452 SonarRunMem := StartMotor_Cmd;
453
454 IF StartMotor_Cmd THEN
455     SonarAsk := FALSE;                    (* never while running *)
456 END_IF;
457
458 IF SonarAskYes THEN
459     Measure_ON := TRUE;
460     SonarAsk := FALSE;
461     SonarAskYes := FALSE;
462 END_IF;
463
464 (* ---- 11. Rewinder max diameter protection - ALWAYS active ---------- *)
465 (* Independent of selector 12SA5 and of the AUTO buttons.                *)
466 (* Mechanical limit 680 mm radius = 1360 mm                             *)
467
468 IF NOT RwSensFault THEN
469     RwLastValidDiam := RwDiam;
470 END_IF;
471
472 IF RwDiam >= RwMaxDiam THEN
473     RwRemainMax_Pls := 0.0;
474 ELSIF RwMat_K > 0.0 THEN
475     RwRemainMax_Pls := (RwMaxDiam * RwMaxDiam - RwDiam * RwDiam)
476                        / (2.0 * RwMat_K);
477 ELSE
478     RwRemainMax_Pls := 1.0E9;             (* K not learned: plain limit *)
479 END_IF;
480
481 (* Display: diameter where the max-protection slow-down will start       *)
482
483 IF RwMat_K > 0.0 THEN
484     SqrtArg := RwMaxDiam * RwMaxDiam - 2.0 * RwMat_K * RwSlowLen_Pls;
485     IF SqrtArg > 0.0 THEN
486         RwMaxSlowAtDiam := SQRT(SqrtArg);
487     ELSE
488         RwMaxSlowAtDiam := 0.0;
489     END_IF;
490 ELSIF RwMaxSlowFallback > 0.0 THEN
491     RwMaxSlowAtDiam := RwMaxSlowFallback;  (* fallback in use *)
492 ELSE
493     RwMaxSlowAtDiam := RwMaxDiam;
494 END_IF;
495
```

```
(* Reset only after the roll is removed (diameter well below the limit)    *)

IF (RwDiam < RwMaxDiam - RwResetHyst) AND NOT RwSensFault THEN
    RwMaxStop := FALSE;
    RwMaxSlow := FALSE;
END_IF;

IF NOT RwSensFault THEN

    IF RwRemainMax_Pls <= StopLen_Pls THEN
        RwMaxStop := TRUE;                    (* speed-compensated stop *)
    END_IF;

    IF RwRemainMax_Pls <= RwSlowLen_Pls THEN
        RwMaxSlow := TRUE;                    (* warning slow-down + horn *)
    END_IF;

    IF (RwMat_K <= 0.0) AND (RwMaxSlowFallback > 0.0)
      AND (RwDiam >= RwMaxSlowFallback) THEN
        RwMaxSlow := TRUE;                    (* fallback while K not learned *)
    END_IF;

ELSIF (AI_Diam_Rewinder < RwRawMin) AND (RwLastValidDiam > RwMaxDiam -
100.0) THEN
    RwMaxStop := TRUE;                        (* roll grew into the DT35 near zone *)
END_IF;

(* Max diameter protection added OUTSIDE the 12SA5 master (section 9)      *)

AutoSlowCmd := AutoSlowCmd OR RwMaxSlow;
AutoStopCmd := AutoStopCmd OR RwMaxStop;

(* ---- 11b. V12: END-OF-ROLL STOP RAMP ---------------------------------- *)
(* Normal end-of-roll stops (AutoStopCmd: unwinder metres, rewinder target,*)
(* rewinder max diameter; UwStopByDiam: unwinder diameter) no longer drop  *)
(* the run command at once (drive DEC ramp 3 s = 20 Hz/s -> up to 8 m/s2  *)
(* of web on a big roll, hard on the rewinder gearbox). With the feed-     *)
(* forward in AUTO they start the SOFT STOP at FinalStopDecel (General     *)
(* rung 10); the run command drops below SoftStopEndSpd. START cannot      *)
(* cancel it. The stop lengths (section 6) use the same ramp, so the web   *)
(* still stops at the target / UwStopDiam_Limit.                           *)
(* Immediate stop (EndStopNow -> Motor_Manager rung 4 Stop):               *)
(*   ramp not usable (FinalStopDecel 0, FFEnable OFF, ManMode, JOG, the    *)
(*   motor stopped = start blocked as before), or safety net: unwinder     *)
(*   10 mm below UwStopDiam_Limit / rewinder at RwMaxDiam during the ramp.*)

IF FinalStopDecel < 0.0 THEN
    FinalStopDecel := 0.0;
ELSIF (FinalStopDecel > 0.0) AND (FinalStopDecel < 50.0) THEN
    FinalStopDecel := 50.0;
ELSIF FinalStopDecel > 2000.0 THEN
    FinalStopDecel := 2000.0;
END_IF;

EndStopReq := AutoStopCmd OR (Measure_ON AND UwStopByDiam);
EndRampOK := (FinalStopDecel > 0.0) AND FFActive AND StartMotor_Cmd;

IF EndStopReq AND EndRampOK THEN
    FinalStopRamp := TRUE;                    (* -> soft stop, General 10 *)
```

```
554 END_IF;
555
556 IF NOT StartMotor_Cmd THEN
557     FinalStopRamp := FALSE;
558 END_IF;
559
560 EndRampHard := FinalStopRamp
561             AND ((Measure_ON AND (UwDiam_Act > 0.0)
562                 AND (UwDiam_Act < UwStopDiam_Limit - 10.0))
563                 OR ((RwDiam >= RwMaxDiam) AND NOT RwSensFault));
564
565 EndStopNow := (EndStopReq AND NOT FinalStopRamp) OR EndRampHard;
566
567 // ---- 11c. Auto stop DONE (HMI warning 19), latched until START / RESET ----
568 // AutoStopWhy: 1 unwinder metres, 2 rewinder diameter, 4 rewinder MAX, 8
    unwinder diameter
569
570 IF StartMotor_Cmd AND NOT AS_RunMem THEN          // new START: clear
571     AutoStopDone := FALSE;
572     AutoStopSeen := FALSE;
573     AutoStopWhy := 0;
574 END_IF;
575
576 IF StartMotor_Cmd AND EndStopReq AND NOT AutoStopSeen AND NOT JogRun
    THEN
577     AutoStopSeen := TRUE;                    // auto stop started while running
578
579     IF RwMaxStop THEN
580         AutoStopWhy := 4;                    // rewinder MAX (protection, highest)
581     ELSIF I_AutoStop_OS AND RwStopEnable AND RwStop_at_D_Req THEN
582         AutoStopWhy := 2;                    // rewinder diameter reached
583     ELSIF Measure_ON AND UwStopByDiam THEN
584         AutoStopWhy := 8;                    // unwinder diameter (sonar)
585     ELSE
586         AutoStopWhy := 1;                    // unwinder metres
587     END_IF;
588 END_IF;
589
590 IF AS_RunMem AND NOT StartMotor_Cmd AND AutoStopSeen THEN
591     AutoStopDone := TRUE;                    // stopped by the auto stop
592 END_IF;
593
594 IF I_RsetAlarm OR Alarm_Reset THEN
595     AutoStopDone := FALSE;
596 END_IF;
597
598 T_AutoStopDisp(In := AutoStopDone, PT := T#8s);   // show the warning 8s
599
600 IF T_AutoStopDisp.Q THEN
601     AutoStopDone := FALSE;
602 END_IF;
603
604 AS_RunMem := StartMotor_Cmd;
605
606 (* ---- 12. Learning supervision -> alarm 16 (section Alarms) ---------- *)
607 (* Learned paper thickness, for display / plausibility check            *)
608
609 IF MmPerPulse > 0.0 THEN
610     RwThick_mm := RwMat_K * 3.14159 / (2.0 * MmPerPulse);
```

```
ELSE
    RwThick_mm := 0.0;
END_IF;

(* Fault: too many consecutive windows / stops that should have          *)
(* learned but did not. Clears by itself on the next good result.        *)
(* v6w: ALARM RESET (panel button or HMI) also clears the fail counters, *)
(* so alarm 16 resets in one press. It comes back only if the learning   *)
(* fails again RwLearnFailMax / RwStopFailMax times in a row.            *)

IF I_RsetAlarm OR Alarm_Reset THEN
    LearnFailCnt := 0;
    StopFailCnt := 0;
END_IF;

GSM_LearnFault := (LearnFailCnt >= RwLearnFailMax)
          OR (StopFailCnt >= RwStopFailMax);      (* -> alarm 16 *)

(* v6e: learning status for the HMI. FALSE until learned.                *)

GSM_Learned_OK := (RwMat_K > 0.0)
              AND ((RwStopDecel > 0.0) OR NOT RwLearnStop)
              AND NOT GSM_LearnFault;

(* ---- 13. v6h: keep T-one in automatic tension control during the stop - *)
(* While running: BrakeHold = Brake_Cmd (no change in running behaviour).  *)
(* After the stop: stays ON until the web is below 5 m/min, after 15 s.    *)
(* or on E-stop. Already TRUE on the stop scan (Output_Manager runs before *)
(* this section), so O_Brake never drops for one scan.                     *)
(* Output_Manager rung 2:  O_Brake := Brake_Cmd OR BrakeHold               *)

IF O_MotorStart THEN
    BrakeHold := Brake_Cmd AND ControlVoltage;
    BrakeHoldTicks := 0;
ELSIF BrakeHold THEN

    IF Tick THEN
        BrakeHoldTicks := BrakeHoldTicks + 1;
    END_IF;

    (* v6v: signed web speed: a web pulled BACK (negative speed) also     *)
    (* releases the T-one (trace 10/28 15:00: -6.6 m/min, about 20 cm).   *)

    IF NOT ControlVoltage OR (Velocity_M_min < 5.0) OR (BrakeHoldTicks >= 150)
THEN
        BrakeHold := FALSE;
    END_IF;
END_IF;

(* ---- 14. v6j: TEACH CORE - automatic offset with an empty core -------- *)
(* HMI: set RwTeachCore (button, Set ON). Conditions: motor stopped, web   *)
(* not moving, DT35 OK. After 2 s stable (filter settled):                 *)
(*    RwAxisDist := RwDist + RwCoreDiameter / 2   -> RwDiam = RwCoreDiameter
*)
(* RwTeachState: 0 idle, 1 teaching (wait 2 s), 2 done OK,                 *)
(*               3 refused (motor running / web moving / DT35 fault).      *)
(* Only the offset is taught: the scale is RwDist10V - RwDist0V            *)
(* (= real distance between the two DT35 taught points).                   *)
```

```
668 IF RwTeachCore THEN
669    IF O_MotorStart OR (RwSpeed > 1.0) OR RwSensFault
670       OR NOT RwFiltInit OR (RwCoreDiameter <= 0.0) THEN
671       RwTeachState := 3;                    (* refused *)
672       RwTeachCore := FALSE;
673       TeachTicks := 0;
674    ELSE
675       RwTeachState := 1;                    (* teaching *)
676
677       IF Tick THEN
678          TeachTicks := TeachTicks + 1;
679       END_IF;
680
681       IF TeachTicks >= 20 THEN              (* 2 s stable *)
682          RwAxisDist_Prev := RwAxisDist;     (* v6l: keep previous *)
683          RwAxisDist := RwDist + RwCoreDiameter / 2.0;
684          RwTeachState := 2;                 (* done *)
685          RwTeachCore := FALSE;
686          TeachTicks := 0;
687       END_IF;
688    END_IF;
689 ELSE
690    TeachTicks := 0;
691
692    IF O_MotorStart THEN
693       RwTeachState := 0;                    (* clear status on next run *)
694    END_IF;
695 END_IF;
696
697 (* ---- 14b. v6k: unwinder TEACH CORE (HMI button Core_Measure) ---------- *)
698 (* Same behaviour as section 14. Sonar is not filtered, so the 2 s are   *)
699 (* averaged:  UwCoreDistance := average(UwDiam_Measure)  (sonar -> core   *)
700 (* surface). UwDiam_Act = (UwCoreDistance - UwDiam_Measure) * 2          *)
701 (*                      + UwCoreDiameter   (General rung 14, unchanged). *)
702 (* UwTeachState: 0 idle, 1 teaching, 2 done OK, 3 refused.               *)
703
704 IF Core_Measure THEN
705
706    IF O_MotorStart OR (RwSpeed > 1.0) OR (UwCoreDiameter <= 0.0)
707       OR (UwDiam_Measure <= 0.0) THEN
708       UwTeachState := 3;                    (* refused *)
709       Core_Measure := FALSE;
710       UwTeachTicks := 0;
711       UwTeachSum := 0.0;
712    ELSE
713       UwTeachState := 1;                    (* teaching *)
714
715       IF Tick THEN
716          UwTeachTicks := UwTeachTicks + 1;
717          UwTeachSum := UwTeachSum + UwDiam_Measure;
718       END_IF;
719
720       IF UwTeachTicks >= 20 THEN            (* 2 s, 20 samples *)
721          UwCoreDist_Prev := UwCoreDistance;     (* v6l: keep previous *)
722          UwCoreDistance := UwTeachSum / INT_TO_REAL(UwTeachTicks);
723          UwTeachState := 2;                 (* done *)
724          Core_Measure := FALSE;
725          UwTeachTicks := 0;
```

```
726             UwTeachSum := 0.0;
727         END_IF;
728     END_IF;
729 ELSE
730     UwTeachTicks := 0;
731     UwTeachSum := 0.0;
732
733     IF O_MotorStart THEN
734         UwTeachState := 0;                  (* clear status on next run *)
735     END_IF;
736 END_IF;
737
738 (* ---- 14c. v6l: UNDO teach = swap current <-> previous (machine stopped) *)
739 (* Pressing UNDO twice goes back to the taught value. State 4 = RESTORED. *)
740
741 IF RwTeachUndo THEN
742
743     IF NOT O_MotorStart AND (RwAxisDist_Prev > 0.0) THEN
744         TeachSwap := RwAxisDist;
745         RwAxisDist := RwAxisDist_Prev;
746         RwAxisDist_Prev := TeachSwap;
747         RwTeachState := 4;                  (* restored *)
748     ELSE
749         RwTeachState := 3;                  (* refused *)
750     END_IF;
751     RwTeachUndo := FALSE;
752 END_IF;
753
754 IF UwTeachUndo THEN
755
756     IF NOT O_MotorStart AND (UwCoreDist_Prev > 0.0) THEN
757         TeachSwap := UwCoreDistance;
758         UwCoreDistance := UwCoreDist_Prev;
759         UwCoreDist_Prev := TeachSwap;
760         UwTeachState := 4;                  (* restored *)
761     ELSE
762         UwTeachState := 3;                  (* refused *)
763     END_IF;
764     UwTeachUndo := FALSE;
765 END_IF;
```

1-8-1-1-9.Tension_Helper

0

```
// TENSION HELPER: recommended T-one tension from paper GSM and roll width.
// Rule of thumb (paper): 0.54 N per mm of width per 100 g/m2 (TensFactor).
// Button TENSION UNITS (momentary) cycles kg -> lb -> N.

IF PaperGSM < REAL#10 THEN
    PaperGSM := REAL#10;
ELSIF PaperGSM > REAL#500 THEN
    PaperGSM := REAL#500;
END_IF;

IF WebWidth_mm < REAL#100 THEN
    WebWidth_mm := REAL#100;
ELSIF WebWidth_mm > REAL#3000 THEN
    WebWidth_mm := REAL#3000;
END_IF;

IF (TensFactor < REAL#0.1) OR (TensFactor > REAL#2) THEN
    TensFactor := REAL#0.54;
END_IF;

// Unit button: one press = next unit

IF TensUnitPB AND NOT TensUnitMem THEN
    TensUnit := TensUnit + 1;
    IF TensUnit > 2 THEN
        TensUnit := 0;
    END_IF;
END_IF;

TensUnitMem := TensUnitPB;

IF (TensUnit < 0) OR (TensUnit > 2) THEN
    TensUnit := 0;
END_IF;

TensIdeal_N := PaperGSM / REAL#100 * WebWidth_mm * TensFactor;

CASE TensUnit OF
    1:  TensIdeal_Disp := TensIdeal_N / REAL#9.81 * REAL#2.20462;   // lb
    2:  TensIdeal_Disp := TensIdeal_N;                              // N
ELSE
        TensIdeal_Disp := TensIdeal_N / REAL#9.81;                  // kg
END_CASE;

TensShowKg := (TensUnit = 0);
TensShowLb := (TensUnit = 1);
TensShowN  := (TensUnit = 2);
```

1-9.Data

1-9-1.Data Types

Name	Type	Comment

1-9-2.Global Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Network Publish	Comment
VAR_GLOBAL							
Pot1_Work_Velocity	INT		BuiltInIO://opt/#0/Ch1_Analog_Input_Value	False	False	Publish Only	Potentiometer 1 value - Work speed -
InvRef	INT		BuiltInIO://opt/#0/Ch2_Analog_Output_Value	False	False	Publish Only	Inverter speed reference
Fix10Vdc	INT		BuiltInIO://opt/#0/Ch1_Analog_Output_Value	False	False	Publish Only	Fixed 10 Vdc generator
VelMax	REAL	800		True	False	Publish Only	Maximum speed
VelSet	REAL			False	False	Publish Only	Set speed
RwSlowDecelK	REAL	0.15		True	False	Publish Only	Learned slow-down decel per mm of rewinder diameter (mm/s2 per mm)
RwMaxSlowFallback	REAL	1285.0		True	False	Publish Only	Rewinder max: slow-down diameter while K not learned (mm)
PaperDensity	REAL			False	False	Publish Only	
Language	DINT			True	False	Publish Only	Language
RwThickMin_mm	REAL	0.02		True	False	Publish Only	Learning: min plausible paper thickness (mm)
PulseRate	UDINT		IOBus://unit#3/Ch1 Pulse Rate	False	False	Publish Only	Pulses per 100 ms
RwThickMax_mm	REAL	2.0		True	False	Publish Only	Learning: max plausible paper thickness (mm)
RwDecelMin_mm	REAL	20.0		True	False	Publish Only	Learning: min plausible stop decel (mm/s2)
RwDecelMax_mm	REAL	5000.0		True	False	Publish Only	Learning: max plausible stop decel (mm/s2)
RwLearnFailMax	INT	12		True	False	Publish Only	Learning: failed windows before alarm (12 = 2 min)
RwStopFailMax	INT	3		True	False	Publish Only	Learning: failed stops before alarm
RwThick_mm	REAL			True	False	Publish Only	Learned paper thickness (mm)
OverSpeed	BOOL			False	False	Publish Only	Speed too high
WorkSpd_Max	REAL	500.0		True	False	Publish Only	Work speed pot full scale (m/min), 50..VelMax
WorkSpd_Max_Set	REAL	400.0		True	False	Publish Only	HMI request: work pot full scale (m/min), applied only with motor stopped
UwRawMin	INT	10		True	False	Publish Only	Unwinder sonar raw min (below = sensor fault, alarm 17)
UwRawMax	INT	3990		True	False	Publish Only	Unwinder sonar raw max (above = sensor fault / no echo, alarm 17)
TensCheckEnable	BOOL	False		True	False	Publish Only	Tension check ON (HMI switch); OFF = start as before
TensWaitSpd	REAL	10.0		True	False	Publish Only	Speed held after START until tension established (m/min)
TensionOK	BOOL	False		False	False	Publish Only	Tension established (deboun

							ced T-one Alarm 0)
TensWait	BOOL	False		False	False	Publish Only	Waiting for tension (speed held at TensWaitSpd, HMI lamp)
SonarAsk	BOOL	False		False	False	Publish Only	V10 HMI popup: sonar is OFF, turn it on? (HMI NO resets it)
UwSlow_at_D_Req	BOOL	True		True	False	Publish Only	V7c: unwinder diameter slow-down follows the speed (FALSE = fixed UwSlowDown_Diam)
UwSlowDiamAct	REAL	0.0		False	False	Publish Only	V7c: unwinder slows down at this diameter (mm), HMI display
FinalStopDecel	REAL	500.0		True	False	Publish Only	V12: end-of-roll stop ramp, web decel (mm/s2), 0 = drive DEC ramp as before, 50..2000
FinalStopRamp	BOOL	False		False	False	Publish Only	V12: end-of-roll stop ramp running (HMI lamp)
RwJogMaxDiam	REAL	400.0		True	False	Publish Only	V7d: jog blocked above this rewinder diameter (mm), 0 = no limit
JogInhibit	BOOL	False		False	False	Publish Only	V7d: jog blocked (rewinder roll too big), HMI lamp
SCurveTime	REAL	0.5		True	False	Publish Only	V7f: S-curve rounding of the AUTO ramps (s), 0 = off, max 2
TOneDecel	BOOL	False		False	False	Publish Only	V7f: line decelerating (for a T-one stop/boost input)
HornStartBeep_s	REAL	1.0		True	False	Publish Only	V9 horn: beep when the slow-down starts (s), 0 = off, max 5
HornBeepLen_s	REAL	0.2		True	False	Publish Only	V9 horn: length of the short beeps during slow-down (s), 0 = off, max 2
HornBeepPeriod	REAL	5.0		True	False	Publish Only	V9 horn: one short beep every ... s during slow-down, 0 = off, max 30
HornStopBeep_s	REAL	3.0		True	False	Publish Only	V9 horn: beep when stopped after a slow-down (s), 0 = off, max 10
HornTest	BOOL	False		False	False	Publish Only	V9 horn: HMI test button (momentary), machine stopped only
RwLockDiam	REAL	300.0		True	False	Publish Only	V9: paper thickness LOCKED at this rewinder diameter (mm), 0 = never lock
RwLockMinGrow_mm	REAL	50.0		True	False	Publish Only	V9: min diameter growth since the start of learning before locking (mm)
RwMatLocked	BOOL	False		True	False	Publish Only	V9: paper thickness locked (HMI lamp), NEW PAPER / GSM YES unlocks
LongThick_mm	REAL	0.0		False	False	Publish Only	V9: thickness from the long baseline (mm), HMI display
RwDiamRev	REAL	0.0		False	False	Publish Only	V9: rewinder diameter averaged over one revolution (mm)
RwWave_mm	REAL	0.0		False	False	Publish Only	V9: diameter wave peak-pe

							ak in the last revolution (mm), HMI
UwStopLenShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder STOP LENGTH (AUTO STOP UNWINDER active)
UwSlowDiamShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder SLOW AT diameter (normal operation, sonar ON)
SoftCancelAccel	REAL	100.0		True	False	Publish Only	V9: then ramp up at this web accel (mm/s2), 0 = AccelWeb
RwMatReset	BOOL	False		False	False	Publish Only	HMI NEW PAPER: forget the learned paper thickness RwMat_K (PLC resets it)
FinalSpd_Max_Set	REAL	80.0		True	False	Publish Only	HMI request: final pot full scale (m/min), applied only with motor stopped
FinalSpd_Max	REAL	80.0		True	False	Publish Only	Final speed pot full scale (m/min), 5..WorkSpd_Max
UnderSpeed	BOOL			False	False	Publish Only	Speed too low
RwLearnReset	BOOL	False		False	False	Publish Only	HMI / Watch: clear learned decels RwStopDecelK, RwStopDecel, RwSlowDecelK (PLC resets it)
RwCoreDiameter	REAL	100.0		True	False	Publish Only	Rewinder core outside diameter for TEACH CORE (mm)
RwTeachCore	BOOL	False		False	False	Publish Only	HMI: TEACH CORE request (PLC resets it)
RwTeachState	INT	0		False	False	Publish Only	TEACH CORE: 0 idle 1 teaching 2 done 3 refused
SpdDelta_Percent	REAL	12.0		True	False	Publish Only	Speed difference percentage
Encoder_Pos	DINT		IOBus://unit#3/Ch1 Encoder Present Position	False	False	Publish Only	Encoder counter
UwTeachState	INT	0		False	False	Publish Only	Unwinder TEACH CORE: 0 idle 1 teaching 2 done 3 refused
Encoder_On	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Counter Enable	False	False	Publish Only	Encoder enable
Encoder_Reset	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Internal Reset Execution	False	False	Publish Only	Encoder reset
HMI_Screen	ARRAY[0..1] OF DINT			False	False	Publish Only	HMI panel Screen
Velocity_M_	REAL			True	False	Publish Only	Speed in m/min

min							
Current_Metre	REAL			True	False	Publish Only	Current metres
I_RsetCnt	BOOL			False	False	Publish Only	Metre counter reset button
Speed_Display	REAL			True	False	Publish Only	Set speed (displayed)
I_ControlVoltage	BOOL		BuiltInIO://cpu/#0/Input_Bit_00	False	False	Publish Only	In_0 ControlVoltageON
I_Start_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_01	False	False	Publish Only	In_1 Start button
I_Stop_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_02	False	False	Publish Only	In_2 Stop button
I_Motor_Thermal	BOOL		BuiltInIO://cpu/#0/Input_Bit_03	False	False	Publish Only	In_3 Motor thermal switch
I_Prot_ByPass	BOOL		BuiltInIO://cpu/#0/Input_Bit_04	False	False	Publish Only	In_4 Guards-bypassed selector switch - Under Panel
I_Hub_Position	BOOL		BuiltInIO://cpu/#0/Input_Bit_05	False	False	Publish Only	In_5 Hub/chuck in extraction position
I_FrontGuard_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_06	False	False	Publish Only	In_6 Front guard closed
I_Jog_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_07	False	False	Publish Only	In_7 Jog Push button
I_VFD_Alarm	BOOL		BuiltInIO://cpu/#0/Input_Bit_08	False	False	Publish Only	In_8 Inverter in alarm
I_Side_Door_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_13	False	False	Publish Only	In_Side Door is Closed
I_OR_Position_Sens	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN01 (Pin 1.1)_6000_01	False	False	Publish Only	OverRider position sens
I_OR_Hoover_Mode	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN02 (Pin 2.1)_6000_02	False	False	Publish Only	OverRider Hoover Mode
I_OR_Auto_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN03 (Pin 1.2)_6000_03	False	False	Publish Only	OverRider in automatic
I_OR_Cont_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN04 (Pin 2.2)_6000_0	False	False	Publish Only	OverRider in continuous

			4				
O_MotorStart	BOOL		BuiltInIO://cpu/#0/Output_Bit_00	False	False	Publish Only	Q_0 Motor start
I_FanTemp_Ok	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN06 (Pin 2.3) _6000_06	False	False	Publish Only	Motor (fan) thermal OK
O_Alarm_Array	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 01 (Pin 1.1) _7000_01	False	False	Publish Only	Alarm lamp
O_Prot_ByPass_Lamp	BOOL			False	False	Publish Only	Guards bypassed lamp
O_Motor_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_01	False	False	Publish Only	Q_1 Motor ON lamp
O_RwUnload	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 03 (Pin 1.2) _7000_03	False	False	Publish Only	Unloaders enable - opening
O_FrontGuard_ByPass	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT 04 (Pin 2.2) _7000_04	False	False	Publish Only	Front guard bypass
O_Brake	BOOL		BuiltInIO://cpu/#0/Output_Bit_02	False	False	Publish Only	Q_2 Brake command
O_OR_Auto_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_06	False	False	Publish Only	Q_6 OverRider auto/continuous lamp
O_OverRider_Up	BOOL		BuiltInIO://cpu/#0/Output_Bit_09	False	False	Publish Only	Q_9 Raise OverRider ~10mm Over Round Rewinder roll
O_OverRider_Hold	BOOL		BuiltInIO://cpu/#0/Output_Bit_04	False	False	Publish Only	Q_4 OverRider Lock
O_CoolingFan	BOOL		BuiltInIO://cpu/#0/Output_Bit_03	False	False	Publish Only	Q_3 Brake and motor fans
Fan_Timeout	REAL	0.5		True	False	Publish Only	Fans stop timeout [min]
OverRider_TimeOFF	REAL	0.01		True	False	Publish Only	OverRider photocell OFF timeout (panel)
OverRider_TimeON	REAL	0.05		True	False	Publish Only	OverRider photocell ON timeout (panel)
AlarmOn	BOOL			False	False	Publish Only	Alarm active
BuzzerOn	BOOL			False	False	Publish Only	Panel buzzer command
Msg	INT			False	False	Publish Only	Message: Warnings / Alarms
Index_B	INT			False	False	Publish Only	Alarm index

MotStop	BOOL			False	False	Publish Only	Motor stopped
Clock1s	BOOL			False	False	Publish Only	1 s clock
ManMode	BOOL			True	False	Publish Only	Manual mode
Counter_Reset	BOOL			False	False	Publish Only	Counter reset from panel
I_FrontGuard_Open	BOOL		BuiltInIO://cpu/#0/Input_Bit_09	False	False	Publish Only	In_9 Front guard open
O_RwLoading_Enable	BOOL		BuiltInIO://cpu/#0/Output_Bit_07	False	False	Publish Only	Q_7 Unloaders Load enable - closing
JogSpd_SP	REAL	10.0		True	False	Publish Only	Speed in manual
O_AccDec2	BOOL		BuiltInIO://cpu/#0/Output_Bit_05	False	False	Publish Only	Q_5 Second acc / dec ramp
AccelTrim	REAL	0.08		True	False	Publish Only	Acceleration coefficient in run
DecelTrim	REAL	0.08		True	False	Publish Only	Deceleration coefficient in run
DecelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests decel)
AccelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests accel)
UwStopDiamShow	BOOL	False		False	False	Publish Only	V11 HMI: show unwinder STOP AT diameter (slow-down reached, decel / final speed)
O_AcousticSignal	BOOL		BuiltInIO://cpu/#0/Output_Bit_08	False	False	Publish Only	Q_8 Horn / Acoustic signal
Pot2_Slow_Velocity	INT		BuiltInIO://opt/#0/Ch2_Analog_Input_Value	False	False	Publish Only	Potentiometer 2 value - Final speed -
I_AutoStop_OS	BOOL		BuiltInIO://cpu/#0/Input_Bit_10	False	False	Publish Only	In_10 Stop on metre counter
I_Input_11_Spare_1255	BOOL		BuiltInIO://cpu/#0/Input_Bit_11	False	False	Do not publish	In_11 SPARE (wire 12.55) Under Panel
I_WebBreak_Monitor	BOOL		BuiltInIO://cpu/#0/Input_Bit_12	False	False	Publish Only	In_12 Paper break monitoring ON
I_Vacuum_OK	BOOL		IOBus://unit#3/Ch1 Reset/External Input Status/Ch1 External Input 0 Status	False	False	Publish Only	Extractor fan thermal OK
I_T_one_TensionOK	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN05 (Pin 1.3) _6000_05	False	False	Publish Only	T-one Alarm 0 (15-16): tension established (replaces I_SlowSens_PH)
I_WebBreak_PH	BOOL		ECAT://node#[1,0]/IB IL	False	False	Publish Only	Paper break photocell

			24 DI 8/HD-ECO Input-_IN07 (Pin 1.4) _6000_07				
UwStop_at_M_Val	REAL	5100		True	False	Publish Only	Unwinder Stop at metres
UwSlowMargin_m	REAL	10		True	False	Publish Only	Unwinder Slow-down metres
LowSpd_Display	REAL			True	False	Publish Only	Slow-down speed (displayed)
RwMaxDiam	REAL	1315.0		True	False	Publish Only	Rewinder max diameter protection (mm)
RwMaxStop	BOOL			False	False	Publish Only	Rewinder max diameter STOP (latched)
Encoder_Pulse	INT	1000		True	False	Publish Only	Encoder pulses/Rotation
RwSlowDecelLrn	REAL	0.0		True	False	Publish Only	Learned real slow-down decel (mm/s2) - 0 = not learned
CountReset_Req	BOOL			False	False	Publish Only	Counter reset requested
Psw	INT			False	False	Publish Only	Password Entry
I_RsetAlarm	BOOL		ECAT://node #[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN08 (Pin 2.4) _6000_08	False	False	Publish Only	Alarm reset
EncoderWheel_Circumf	REAL	402		True	False	Publish Only	Encoder wheel (Cylinder / Roller) Circumference [mm]
AlarmCodeSV	ARRAY[0..32] OF BOOL			False	False	Publish Only	Panel alarms
SafeGuard_ByPass	BOOL			False	False	Publish Only	Guards bypassed
FrontGuard_Closed	BOOL			False	False	Publish Only	Front guard closed
SoftStop	BOOL	False		False	False	Publish Only	Soft stop in progress (HMI)
SoftStopEnable	BOOL	True		True	False	Publish Only	STOP button = soft stop (False = immediate as before)
SoftStopEndSpd	REAL	8.0		True	False	Publish Only	Soft stop: run command drops below this speed (m/min)
UwDiam_Measure	REAL			True	False	Publish Only	Measured distance
UwCoreDiameter	REAL	100		True	False	Publish Only	Core Outside diameter [mm]
Core_Measure	BOOL	False		True	False	Publish Only	Stored core value
UwCoreDistance	REAL	785.56		True	False	Publish Only	Sonar distance to empty core. [mm]
AI_Diam_Unwinder	INT		IOBus://unit#2/Ch1 Analog Input Value	False	False	Publish Only	Analog value - UnWinder Diameter
Measure_ON	BOOL	True		True	False	Publish Only	Unwinder Diameter measurement enabled
UwDiam_Act	REAL			True	False	Publish Only	Measured roll diameter
UwSlow_at_	REAL	200		True	False	Publish Only	Unwinder Slow-down diam

D_Val							eter [mm]
UwStopDiam_Limit	REAL	110		True	False	Publish Only	Stop diameter
Alarm_Reset	BOOL			False	False	Publish Only	Alarm reset from panel
Psw22	BOOL			False	False	Publish Only	Password 22 for Popup 22
Psw44	BOOL			False	False	Publish Only	Password 44 for Popup 44
I_Roll_OK	BOOL			False	False	Publish Only	Roll OK / Broken Holder Joint
AI_Diam_Rewinder	INT		IOBus://unit#2/Ch2 Analog Input Value	False	False	Publish Only	Analog value - ReWinder Diameter
RwStopEnable	BOOL	False		True	False	Publish Only	Rewinder diameter stop enabled
RwTargetDiam	REAL	300.0		True	False	Publish Only	Rewinder final diameter (mm)
RwUseSlow	BOOL	True		True	False	Publish Only	Slow down before target
RwSlowMargin_m	REAL	10.0		True	False	Publish Only	Rewinder Metres at slow speed before stop
RwSlowDecel	REAL	0.0		True	False	Publish Only	Rewinder Slow-down decel mm/s2 - 0 = auto from DecStart
RwResetHyst	REAL	50.0		True	False	Publish Only	Reset requests below target - this [mm]
RwDist0V	REAL	0.0		True	False	Publish Only	DT35 distance at 0 V (mm)
RwDist10V	REAL	782.0		True	False	Publish Only	DT35 distance at 10 V (mm)
RwRawMin	INT	10		True	False	Publish Only	Raw below = sensor fault
RwRawMax	INT	3900		True	False	Publish Only	Raw above = sensor fault
RwStopOnFault	BOOL	True		False	False	Publish Only	Stop on sensor fault
RwAxisDist	REAL	727.0		True	False	Publish Only	Sensor to shaft centre (mm) - teach
RwFilter_K	REAL	0.2		True	False	Publish Only	Distance filter per 100 ms (0..1)
RwMinPulses	DINT	1000		True	False	Publish Only	Min pulses in window to learn
RwLearn_K	REAL	0.3		True	False	Publish Only	Material factor filter weight
RwMat_K	REAL	0.0		True	False	Publish Only	Learned material factor
RwLearnStop	BOOL	True		True	False	Publish Only	Learn stop decel on every stop
RwMinLearnSpd	REAL	15.0		True	False	Publish Only	Min speed to learn a stop (m/min)
RwStopDecel	REAL	0.0		True	False	Publish Only	Learned stop decel (pulses/s2)
RwTreact	REAL	0.03		True	False	Publish Only	Stop reaction time (s)
RwTstopDef	REAL	3.0		True	False	Publish Only	Stop time until decel learned (s)
RwTrim	REAL	1.0		True	False	Publish Only	Stop length trim (>1 = earlier)
RwDist	REAL			False	False	Publish Only	Filtered distance (mm)
RwDiam	REAL			False	False	Publish Only	Rewinder diameter (mm)
RwSpeed	REAL			False	False	Publish Only	Line speed from encoder (m/min)
RwRemain_m	REAL			False	False	Publish Only	Metres to target diameter
RwStopLen_m	REAL			False	False	Publish Only	Stopping length now (m)
RwSlowLen_m	REAL			False	False	Publish Only	Rewinder Slow-down length [m]
RwStop_at_D	BOOL			False	False	Publish Only	Rewinder diameter STOP re

_Req							quest
RwSlow_at_D_Req	BOOL			False	False	Publish Only	Rewinder diameter SLOW request
RwSensFault	BOOL			False	False	Publish Only	Rewinder DT35 sensor fault
UwStopEnable	BOOL	False		True	False	Publish Only	Unwinder Auto Stop - Enable
SlowAtDisp	REAL			False	False	Publish Only	HMI: slow-down starts at (m or mm, see SlowAtMode)
SlowAtMode	INT			False	False	Publish Only	HMI: 0=none 1=unwinder [m] 2=rewinder [mm]
SlowAtShow	BOOL			False	False	Publish Only	HMI: show slow-down display
UwSlow_at_M_Val	REAL			False	False	Publish Only	Unwinder: slow-down starts at [m]
UwSlow_at_M_Req	BOOL			False	False	Publish Only	Unwinder metre slow-down request (latched)
RwSlow_at_D_Val	REAL			False	False	Publish Only	Rewinder slow-down starts at diameter [mm]
RwStop_at_D_Val	REAL			False	False	Publish Only	Rewinder: stop command at diameter (mm)
RwMaxSlow	BOOL			False	False	Publish Only	Rewinder max diameter SLOW (latched)
UwEmptyHyst	REAL	30.0		True	False	Publish Only	Unwinder Empty Hysteresis
GSM_Learned_OK	BOOL			False	False	Publish Only	Paper [Grams / Square Meter] FALSE until GSM learned
RwMaxSlowAtDiam	REAL	1370.0		False	False	Publish Only	Rewinder max: slow-down starts at diameter (mm)
RwStopDecelK	REAL	0.0		True	False	Publish Only	Learned stop decel per mm of rewinder diameter (mm/s2 per mm)
SoftStopDecel	REAL	500.0		True	False	Publish Only	Soft stop decel (mm/s2), 0 = DecelCoarse
SpdMax_Pending	BOOL	False		False	False	Publish Only	New pot range waiting for the motor to stop
RwAxisDist_Prev	REAL	727.0		True	False	Publish Only	Rewinder: RwAxisDist before the last TEACH (for UNDO)
UwCoreDist_Prev	REAL	785.56		True	False	Publish Only	Unwinder: UwCoreDistance before the last TEACH (for UNDO)
RwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO rewinder TEACH (PLC resets it)
UwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO unwinder TEACH (PLC resets it)
Psw33	BOOL			False	False	Publish Only	Password 33 for Popup 33
RwMatChangePct	REAL	35.0		True	False	Publish Only	Paper change detection: window deviation (%), 2 windows in a row = new paper, 0 = off
RwSlowFallbackMm	REAL	30.0		True	False	Publish Only	Rewinder: slow-down this many mm before the target while the paper thickness is not learned (0 = off)
RwAutoMaxSpd	REAL	500.0		True	False	Publish Only	Rewinder AUTO stop ON: maximum work speed (m/min), 0 = no limit
RwWinLen_m	REAL	60.0		True	False	Publish Only	Paper thickness learning: MAX window length (m of

							paper)
RwWinGrow_mm	REAL	5		True	False	Publish Only	Paper thickness learning: window = this diameter growth (mm), 4 m .. RwWinLen_m
GsmAsk	BOOL	False		False	False	Publish Only	HMI popup: new parent roll, did you change the paper? (HMI NO resets it)
GsmAskYes	BOOL	False		False	False	Publish Only	HMI popup YES: forget the learned thickness (PLC resets it)
DecelWeb	REAL	250.0		True	False	Publish Only	V7: AUTO decel ramp, real web mm/s2 (0 = DecelCoarse)
AccelWeb	REAL	200.0		True	False	Publish Only	AUTO accel ramp, real web mm/s2 (0 = AccelCoarse)
RwRatioD	REAL	3366.0		True	False	Publish Only	V7: drive counts per m/min x rewinder diameter (mm), adapted while running
FFEnable	BOOL	True		True	False	Publish Only	V7: feed-forward speed loop ON (FALSE = old loop of rung 16)
SpdHold	BOOL	False		False	False	Publish Only	V9: speed held after START during a soft stop, START again = ramp up (HMI lamp)
SonarAskYes	BOOL	False		False	False	Publish Only	V10 HMI popup YES: sonar ON (PLC resets it)
UwStop_at_D_Val	REAL	0.0		False	False	Publish Only	V11: unwinder stop command diameter (mm) so the web stops at UwDiamStop, HMI "STOP AT"
Clock10ms	BOOL			False	False	Publish Only	10mS clock
Clock100ms	bool			False	False	Publish Only	100mS clock
Sec_to_Min	REAL			False	False	Do not publish	
Psw48	BOOL			False	False	Publish Only	Password 48 for Popup 48
JogRun	BOOL			False	False	Publish Only	V10: JOG really running (button, not a normal run, not a soft stop)
GuardStopOK	BOOL			False	False	Publish Only	V11: stopped for real -> guard bypass allowed (Output 13)
SlowDown	BOOL			False	False	Publish Only	Slow down
WarningOn	bool			False	False	Publish Only	HMI Warning Display
AlarmSilenced	BOOL			False	False	Publish Only	
PaperGSM	REAL			True	False	Publish Only	
WebWidth_mm	REAL			True	False	Publish Only	
TensFactor	REAL			True	False	Publish Only	Tension Helper
TensShowKg	BOOL			True	False	Publish Only	Tension Helper
TensShowLb	BOOL			True	False	Publish Only	Tension Helper
TensShowN	BOOL			True	False	Publish Only	Tension Helper
TensIdeal_Disp	REAL			True	False	Publish Only	Tension Helper
TensUnit	INT			True	False	Publish Only	Tension Helper
TensUnitPB	BOOL			False	False	Publish Only	Tension Helper
TensLostWarn	BOOL	False		False	False	Publish Only	Tension lost while running (warning)
UwSensTest	BOOL	False		False	False	Publish Only	Unwinder sonar fault (raw o

								ut of range 1 s)
	UwSensFault	BOOL			False	False	Publish Only	UnWinder Sensor Fault
	Thickness_Fault	BOOL			False	False	Publish Only	Thickness Not Learned !
	AutoStopWhy	INT			False	False	Publish Only	
	AutoStopDone	BOOL			False	False	Publish Only	
	AutoEnRefused	BOOL			False	False	Publish Only	
	AutoEnWhy	INT			False	False	Publish Only	
	TensWaitHz	REAL	1.0		True	False	Publish Only	
	StartBlocked	BOOL	True		False	False	Publish Only	
	GuardOpen	BOOL	False		False	False	Publish Only	
	SideDoorOpen	BOOL	False		False	False	Publish Only	